

Price, value, valuation

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Approaches in science

Prescriptive (normative) science

- „That has to be like this.” (mathematics)

Descriptive science

- „That is like this.” (botany, sociology)

This semester is a compromise.

How are those different?

Project valuation

Today

Limited horizon

One single risk

Cross-effects

Accept or refuse?

Business valuation

At a given date

Endless operation

Multitude of risks

Financial statements

How much?

Equity valuation

In 1 year
(target price)

Endless operation

Risk of
market pricing

Technical analysis

Sell or buy?

What creates value (utility)?

Money

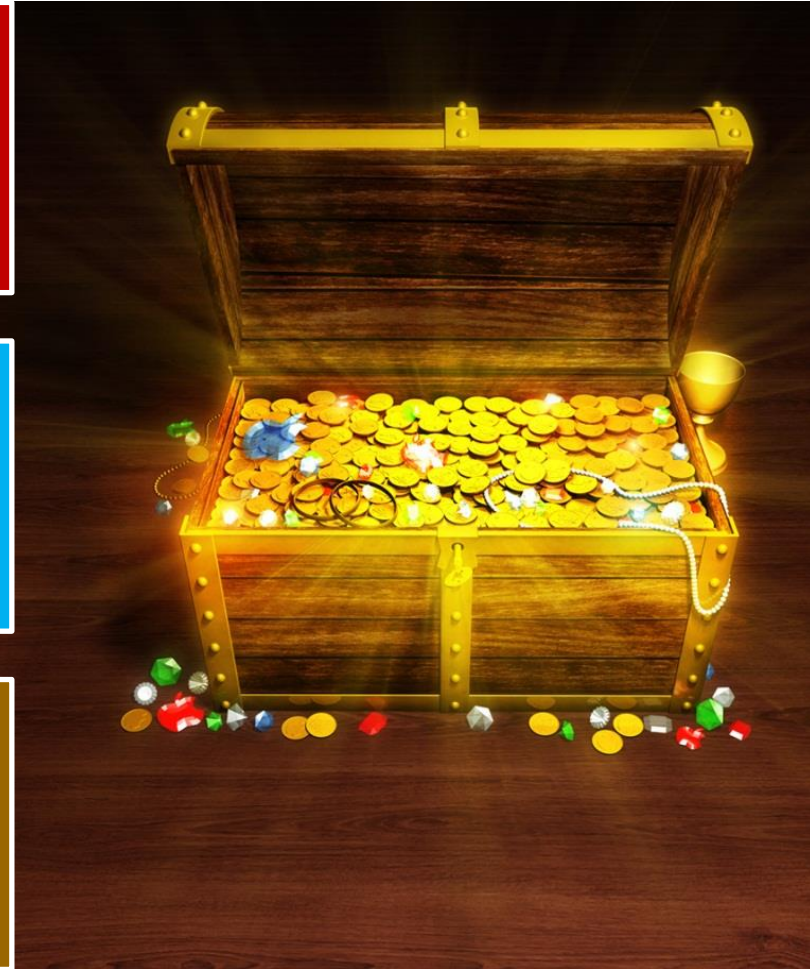
Problem-solving

Joy of work

Respecting traditions

Recognition

Power



Dual value creation

Value for the customer

- Pays the **price**, receives **utility**.
- Alternatives: competitive solutions.

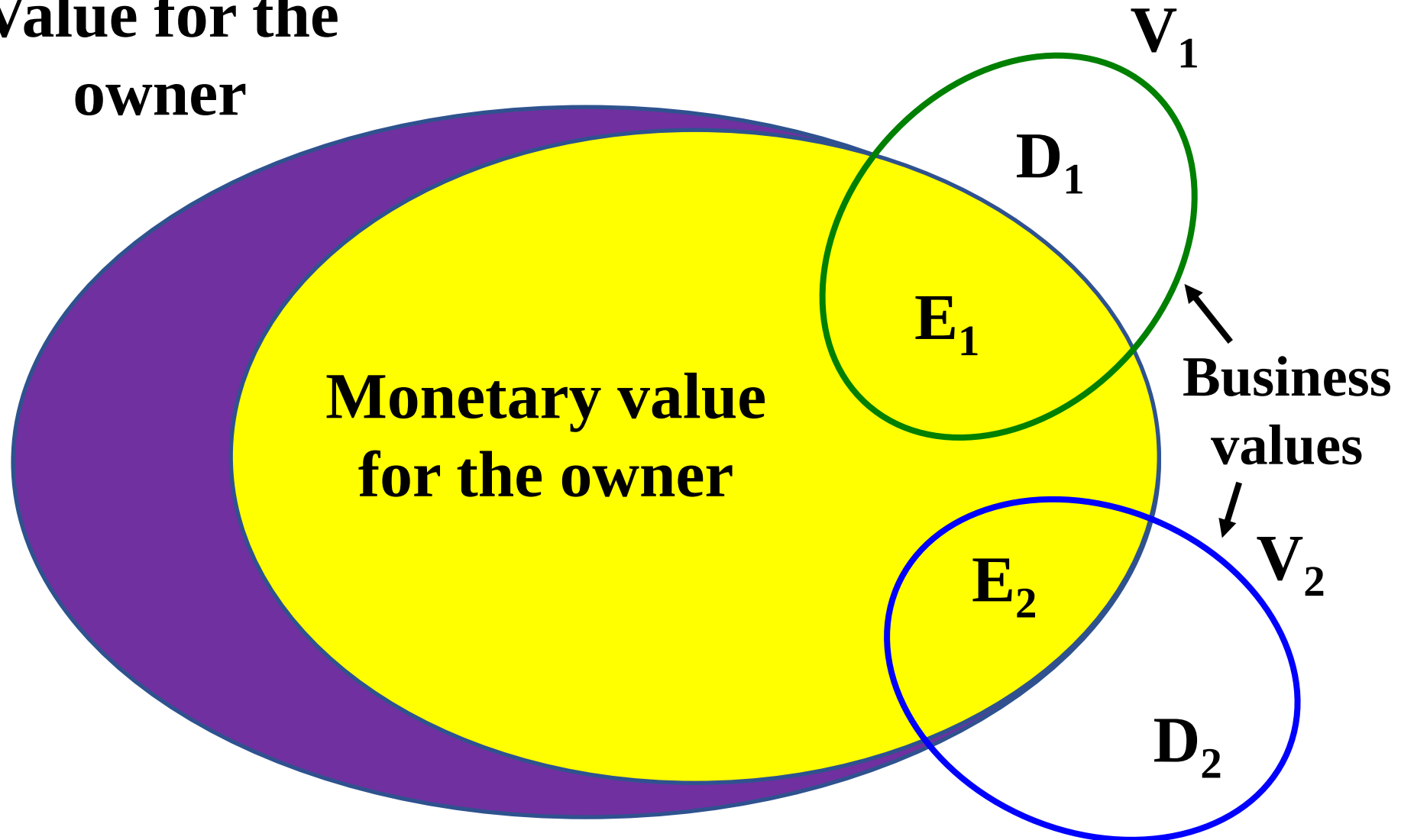
Value for the shareholders?

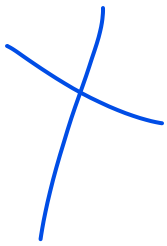
- Provides **capital**, receives **return**.
- Alternatives: competitive investment opportunities.

Value for the stakeholders?

Value dimensions

Value for the
owner





Why do we need the value?

Share trades

IPO

Privatisation

**Strategic
decisions**

**Management
performance**

M&A

Taxation

**Inheritance,
divorce**

Who needs to understand business valuation?

Equity analysts

Investment bankers

Private equity professionals

Fund managers

Privatisation officers

Business owners

All top managers



What is worth to study?

The process
of valuation

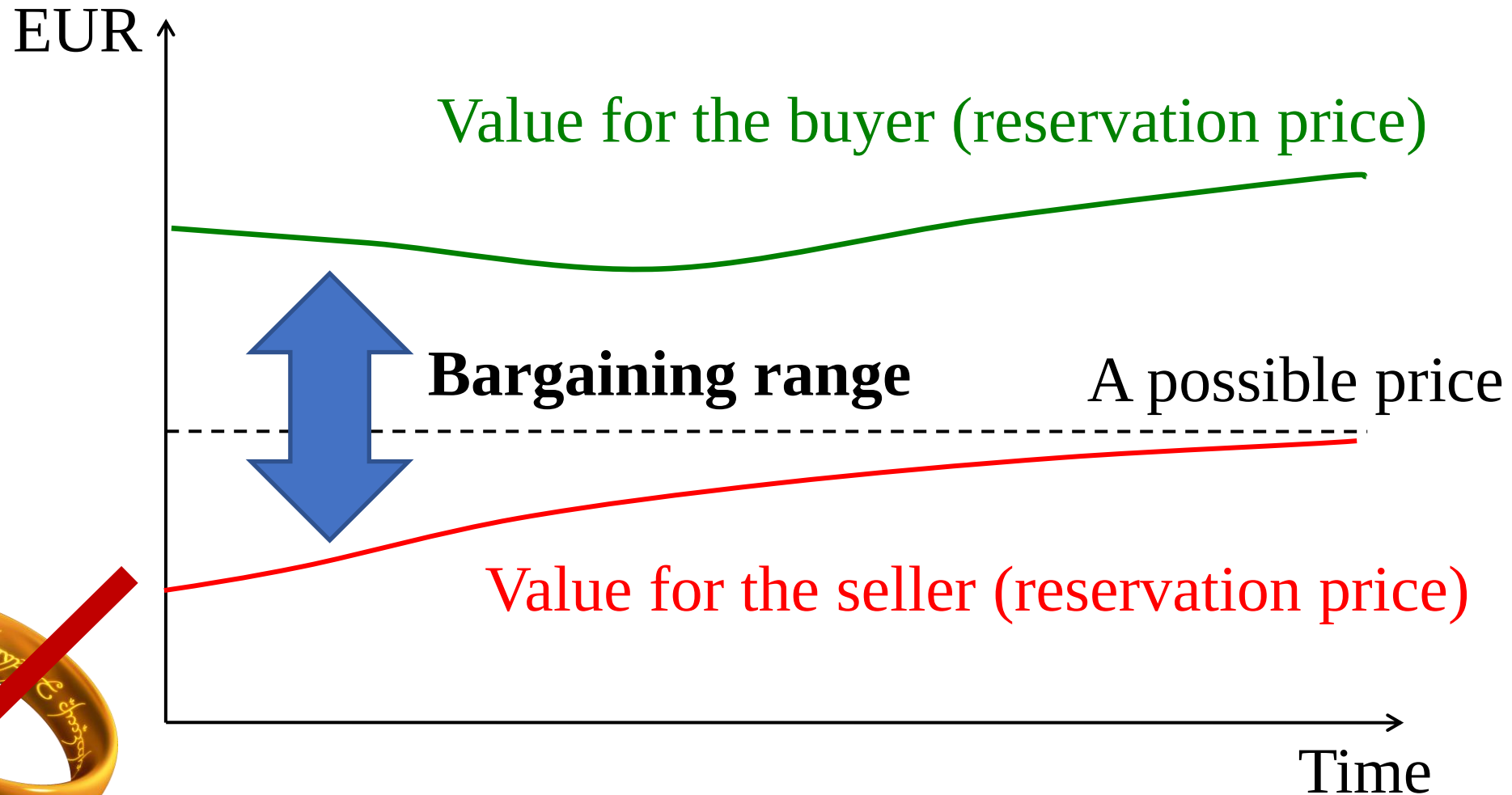
Interpreting
financial
statements

Basics of
financial
planning

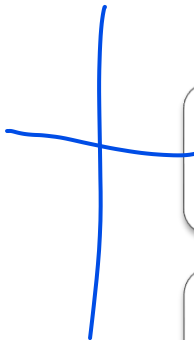
Value based
management

But, of course, valuation is also useful ...

X **There is no „one value above them all”**



Why would the values differ?



Expectations (macro assumptions)

Makings (knowledge, connections, synergies) PERSONAL CHARACTERISTICS

Time horizon (private businesses)

Investment alternatives (not everything is public, family issues)

Risk averseness (required return, level of diversification)

Compromises

Reality...	Classroom...
Investors with unique traits (special ownership and interest links)	„Typical” investor (small shareholder, professional and financial investor)
Imperfect markets (no fresh capital)	Efficient markets (CAPM)
Information asymmetry	Perfect information
Personal expectations (rationality?)	„Market” expectations, expert estimations
That is what is being offered.	That is the result of the formula.

The aim and the result of the valuation

Buyout
(firm unchanged)

Strategic decisions
(alternative future
paths)

M&A
(synergies)

Liquidation
(operation ends)



Valuation approaches

Income-based (dividend, DCF, capitalisation)

Multiple-based (comparative transactions, market multiples)

Asset-based (recreation, liquidation, forced liquidation)

Real options? (not a valuation approach, rather a conversion tool)

Connected fields of science

Corporate finance

Business economics

Managerial accounting

Value chain management

Financial accounting

Strategic management

Investments

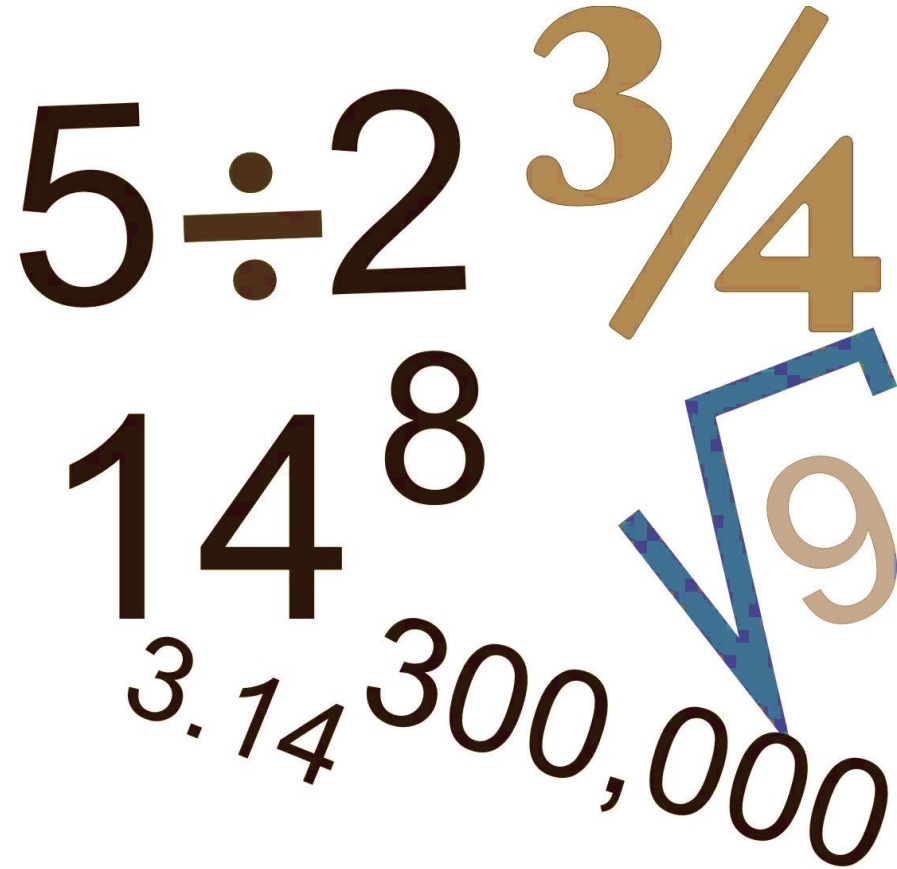
Organisation and leadership

Microeconomics

Performance measurement

Economic law

1. As valuation is quantitative,
it has to be **objective** too.



The price is objective, the value not.

2. The result of a well-prepared valuation is **timeless.**



3. A good valuation leads to an **exact result**.



**4. The more the quantified information,
the more exact the valuation result is.**



5. You may **earn money by valuation** only on an inefficient market .



The **value has not to be paid.**
The **price reflects the value for none.**

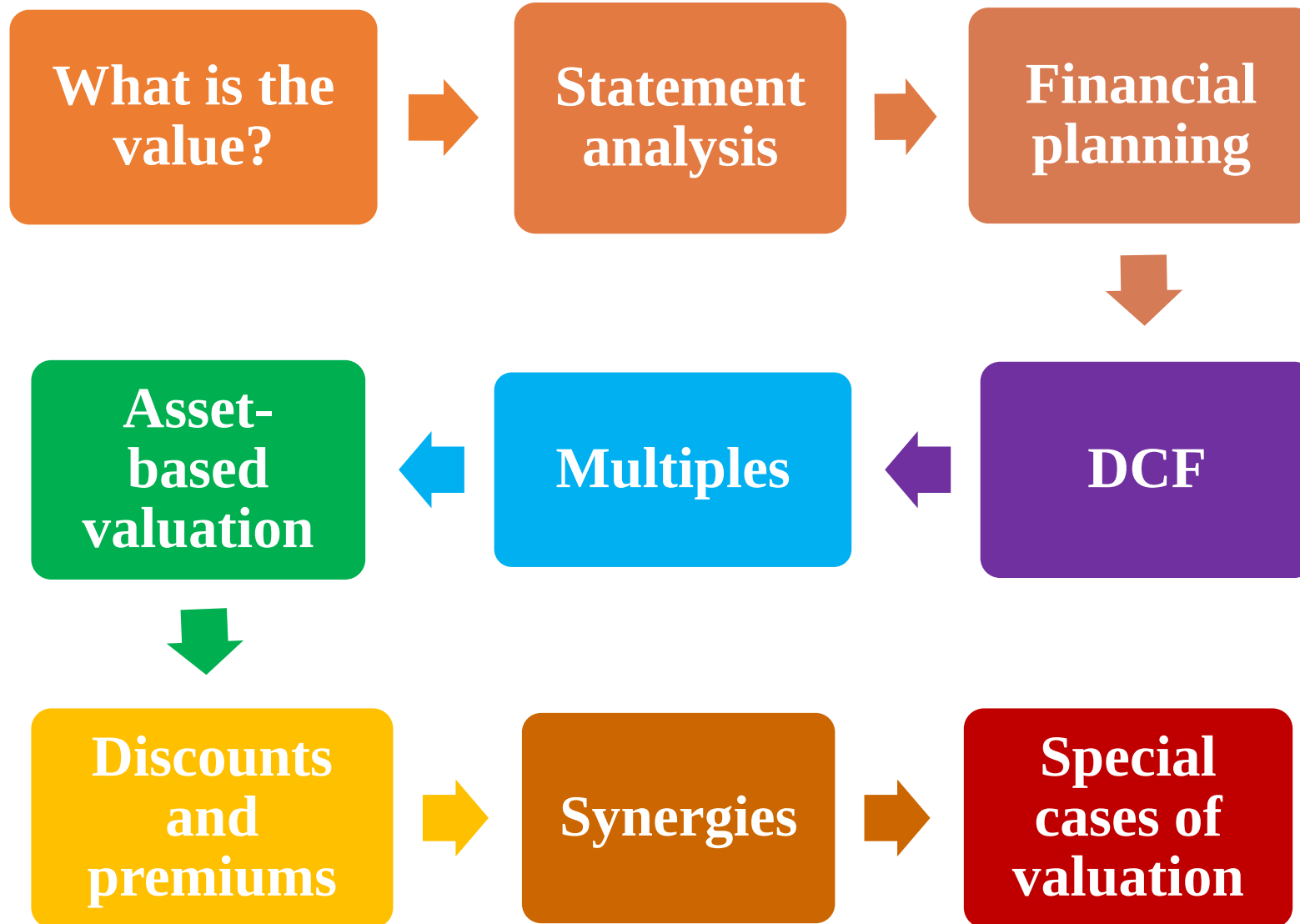
**6. It is only the result of the valuation
that counts.**

Methods are secondary.



Even a broken clock show you the right time twice a day.

Our schedule for the semester



Analysing Financial Statements

Business Cash Flow

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Why do we need **historical analysis**?

Planning

Former performance maintainable?

Reviewing plans

Are assumptions realistic?

Performance measurement

Did we do better than required?

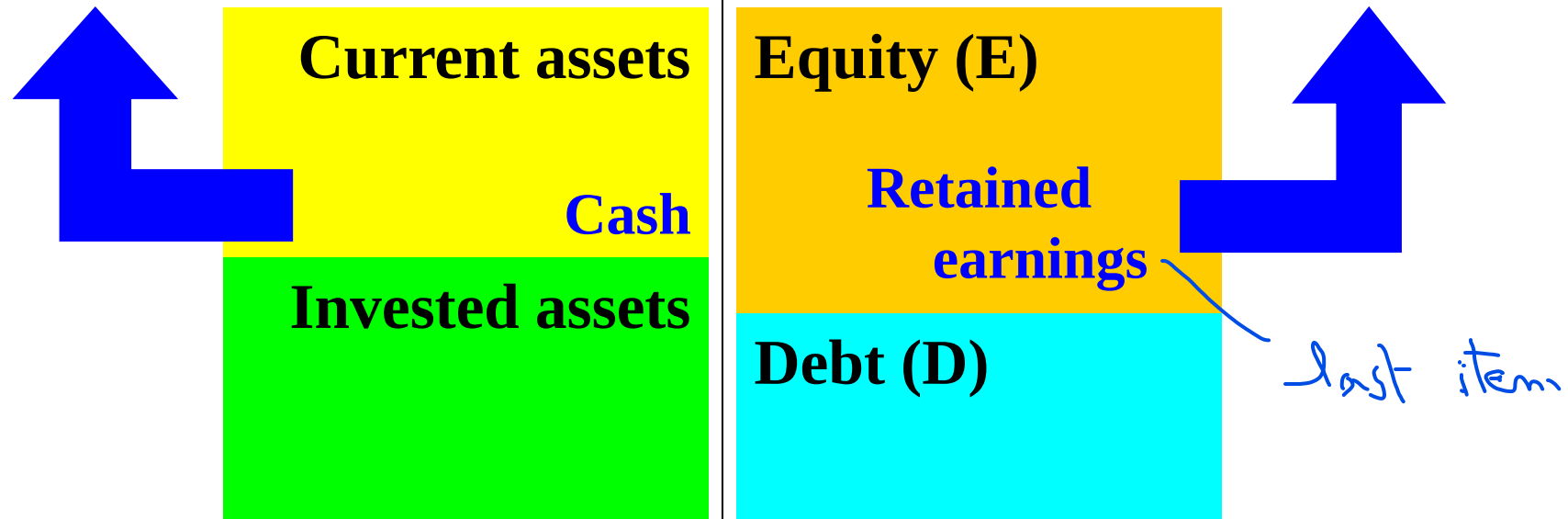


Financial statements

Cash flow

Balance sheet

P/L statement



**What is the last item in
the P/L statement?**

Δ Retained earnings

The three corrections of the statements

Main operation vs non-operative items

- Rented office building

Standard vs Extraordinary items

- Once in a lifetime grant

Operational vs Financing items

- Management fees, exchange gain on interest

„Magic” in accounting

Building reserves

Recognition of profit

Postponed investments

Tricks at inventory review

D&A, unexpected loss in value

Investment valuation

Extraordinary items

Transfer prices

Investment or operational expense (R&D, marketing)



Standard operation in the future



Long term
outlook

Comparative
advantage
New entrants

Outstanding
liabilities

Investments
Regulation

Ongoing
issues

Litigations
Changes in
regulations

The balance sheet in a business view

Assets	Liabilities & Equity
Operational cash	Payables
Receivables	Short term operational liab.
Inventories	Financing debt (D)
Other operational current assets	
Invested assets needed for operation	Equity (E)
Non-operational assets	

P/L in a business view

Operational items	Non-operational items
Sales - Op. expenses (logic) = EBITDA - D&A = EBIT (<i>business view</i>)  + Non-operative profit (logic) + „Extraordinary” profit (logic) = Operational profit (accounting) Business cash flow	+ Profit of Financial Investments + Profit of Financing + Financial effects of operation = Profit from financial activities (accounting) = Gross profit - Corporate tax = Net profit - Dividend = Retained earnings

Surprise Ltd. 3.

1. Assets in personal use: 600, D&A linked: 60, Debt linked: 300, Interest linked: 30.
2. Invoiced personal costs: 1000.
3. Payment gap for CEO: 1200+600 taxes.
4. One buyer filed quality issues (2019): sales lost 100, additional loss 50.

Surprise Ltd. 4.

	2019	Corr	Final		2019	Corr	Final
Invested assets	2000	-600	1400	Equity	2500	-450	2050
Inventory	500		500	Debt	800	-300	500
Receivables	1000	-100	900	Payables	300		300
Cash	100		100	Other short t. liab.	0	50	50
Assets	3600	-700	2900	Liab. & Equity	3600	-700	2900

Surprise Ltd. 5.

	2019	Corr	Final
Sales	12000	-100	11900
Material exp.	8000	-1000+50	7050
Labour exp.	2500	+1800	4300
D&A	250	-60	190
Operating profit	1250	-840	360
Interest payment	80	-30	50
Gross profit	1170	-810	310
Tax	234	-162	72
Net profit	936	-648	248
Dividend	500		500
Retained earnings	436	-648	-252

A cash flow

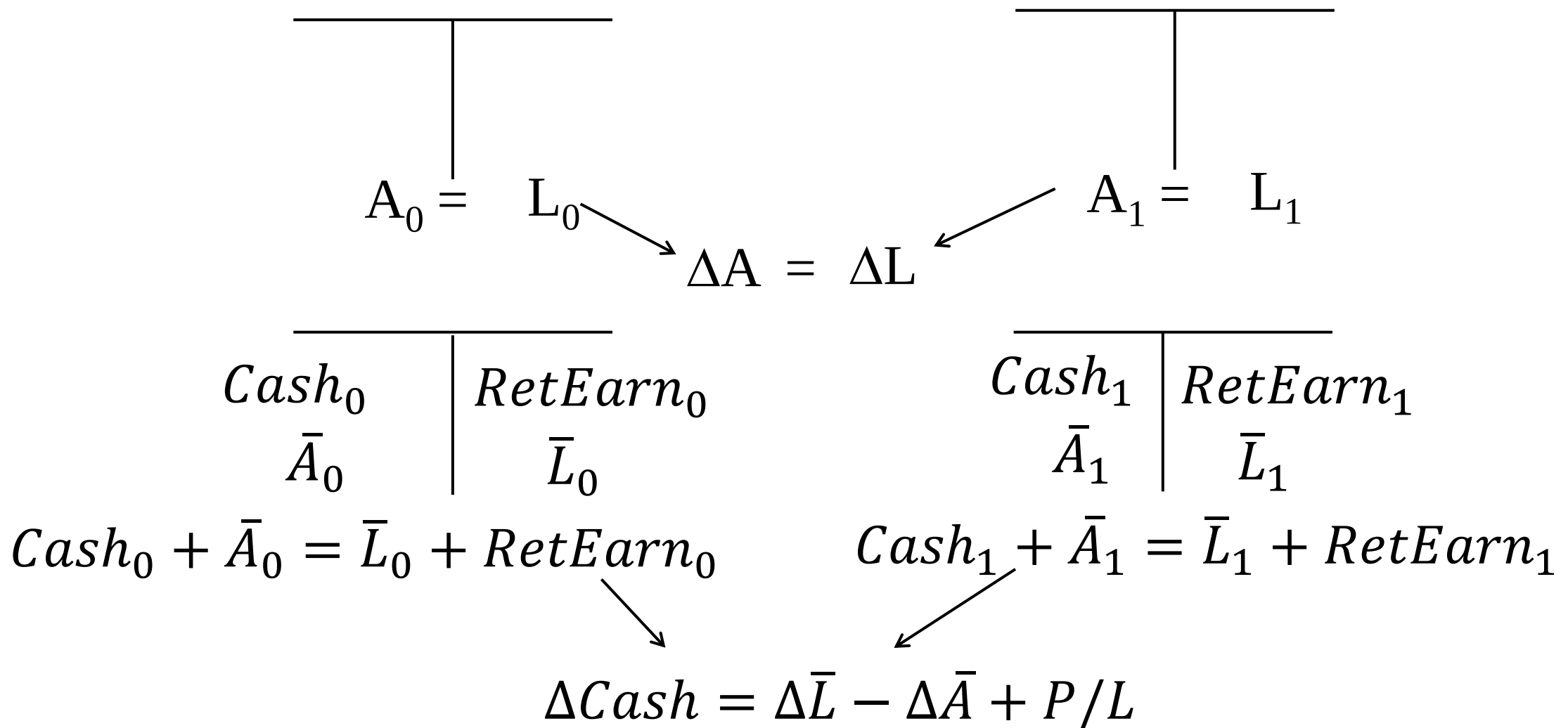
Direct

Directly tracing cash movements
Liquidity planning, short-term

Indirect

Builds on changes in Balance sheet items
Business planning, long-term

The logic behind the indirect cash flow



Business cash flow

Operational items	Non-operational items
EBIT	+ Interest paid
- EBIT*t	- Tax shield
= NOPLAT	+ Change in Financing Debt
+ D&A	= FCFD
= Gross CF	FCFF – FCFD = FCFE
- Change in working capital	- Dividend payment
= Operating CF	+ (-) Share issue/buyback
+ Investment CF	= Change in Non-operating wealth
= FCFF	(Not Change in Cash.)

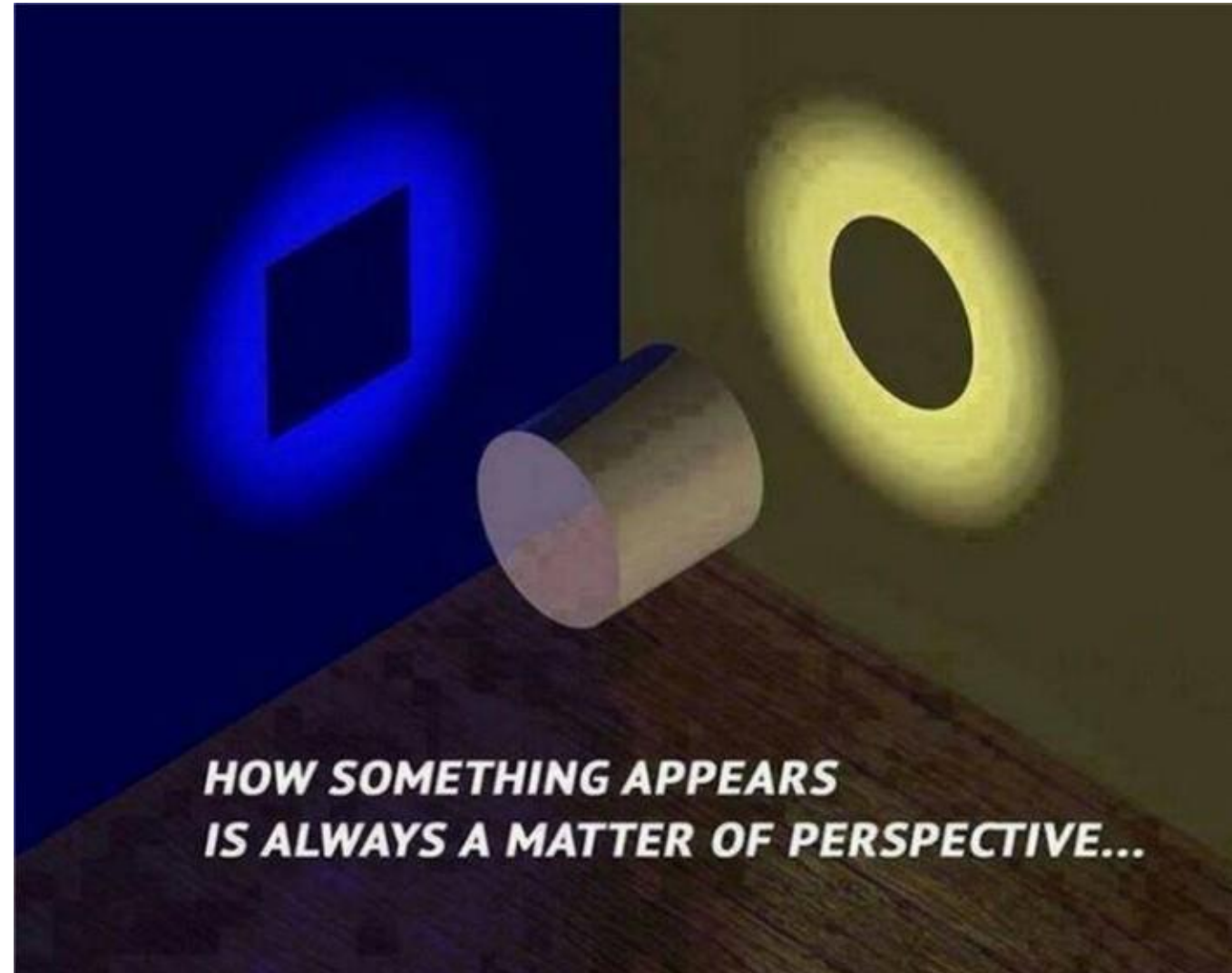
Difference between cash flows

	Accounting	Business
Application	Historical analysis, checking calculations	Predicting, supporting business decisions
Explained quantity	Change in cash	Change in non-operational wealth
Breakdown	Ordinary / Investment / Financial	Operational / Investment / Financing
Aim	Checking on change in cash	Quantifying sub-totals (FCFF, FCFE)

Analysis financial data

From whose point of view?

- management
- owner
- creditor
- competitor



Understand the ratios

Firm	X	Y
Equity	500	1000
Debt	500	0
EBIT	200	200
Interest (10%)	50	0
Gross profit	150	200
Tax (40%)	60	80
Net profit	90	120
ROE [Net Profit/E]	18 %	12 %
ROA [Net profit/(E+D)]	9 %	12 %
ROI [EBIT/(E+D)]	20 %	20 %

Contrasting historical data

If no change in:

strategy, targeted markets, pricing, competition, regulation, macro conditions, product structure, life cycle, market technology, competitive advantage



There was no

Resturcturing,
M&A,
Sale of
business unit,
New market,
New activity

Contrasting data of different firms

Only if those have similar

Accounting and D&A rules,
Asset valuation methodology,
Production processes,
Business policies,
Financing,
Business years.



The content of the analysis

What
happened?

How do you
know?

Is that good
for us?

Why did it
evolved so?

What should
we do?

Value drivers, financial planning

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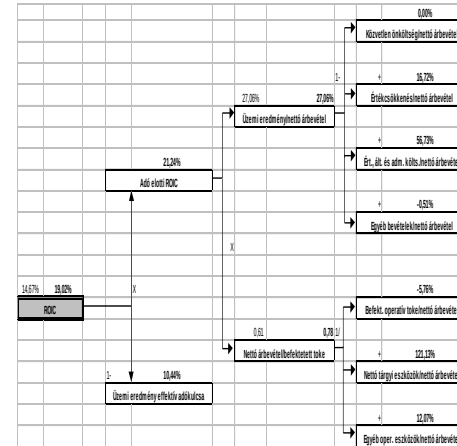
What are value drivers?

Operating parameters



Production,
Logistics,
Sales

Value drivers



Managerial
Accounting,
Controlling,
Business planning

Intrinsic value



Corporate
Finance

Share price



Investments

Value drivers 1

Quantities

1. ... linked directly to shareholder value creation,
2. ... representing both operation and financing,
3. ... covering the long term growth potential.

Value drivers 2

General use

- Any quantity in direct connection with business performance (CF/employee).

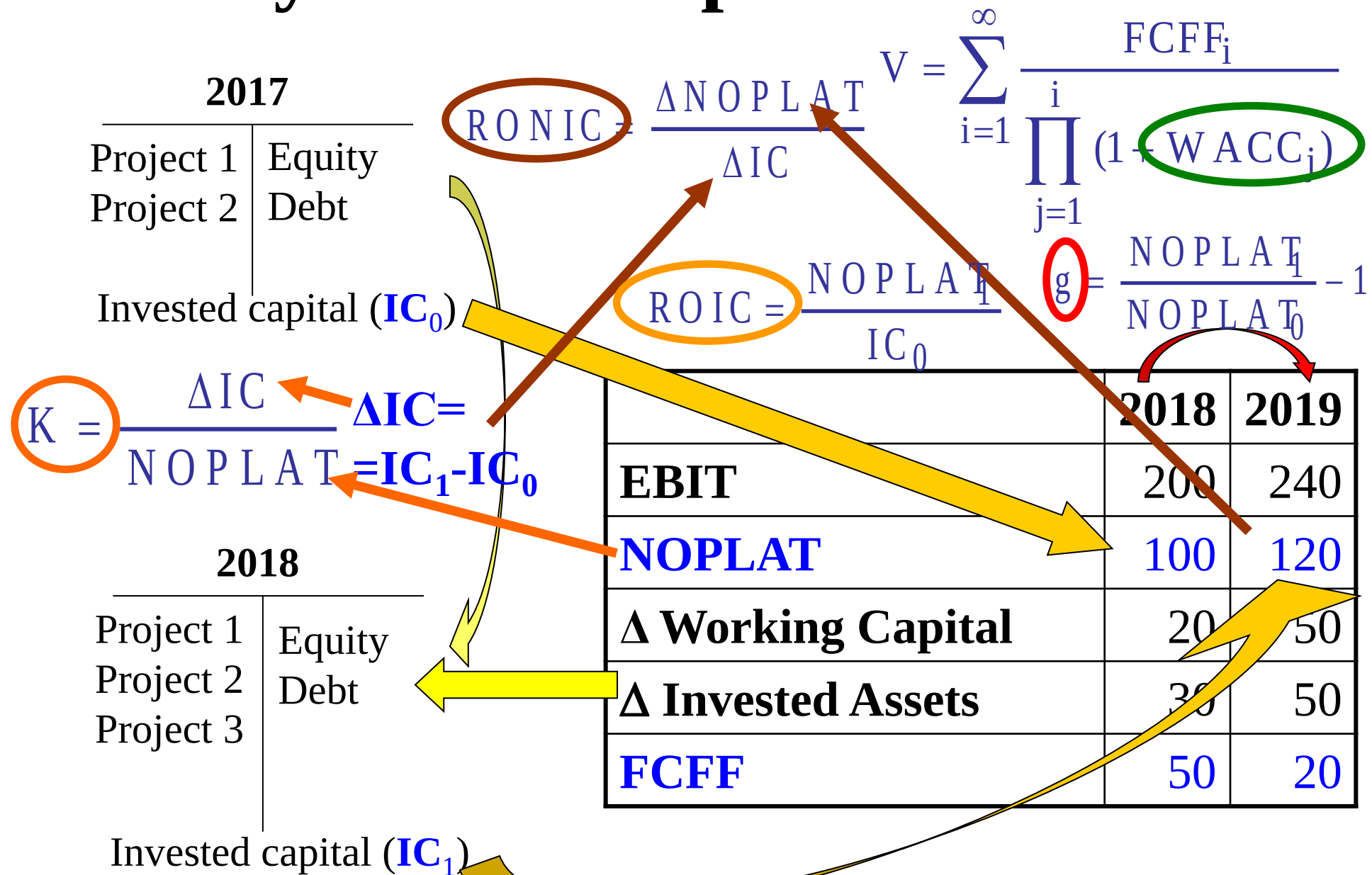
Specific use

- Quantities linked to the output a certain business area (customer satisfaction).

Valuation interpretation

- Financial quantities used to describe the financial plan.

Key drivers of performance 1



Key drivers of performance 2

NOPLAT = Net Operating Profit Less Adjusted Taxes

g = annual growth rate of Noplat

IC = Invested Capital

ΔIC = annual change of IC, $\Delta IC_T = IC_{T+1} - IC_T$

Capital reinvested into the operation

FCFF = Free Cash Flow to Firm, available to distribute to financiers

$$FCFF_T = NOPLAT_T - \Delta IC_T$$

WACC = Weighted Average Cost of Capital,

artificial quantity to reflect average required rate of capital employed

Key drivers of performance 3

K = reinvestment rate

the theoretical proportion of NOPLAT reinvested into the operation:

$$K_T = \Delta IC_T / NOPLAT_T$$

ROIC = Return on Invested Capital :

$$ROIC_T = NOPLAT_T / IC_{T-1}$$

RONIC = Return on Newly Invested Capital,

(assuming the ROIC of the old project be unchanged):

$$RONIC_T = \Delta NOPLAT_T / \Delta IC_{T-1}$$

Links among value drivers

$$\begin{aligned}\text{FCFF}_T &= \text{NOPLAT}_T - \Delta \text{IC}_T = \\ &= \text{NOPLAT}_T - \text{NOPLAT}_T * K_T = \\ &= \text{NOPLAT}_T * (1 - K_T)\end{aligned}$$

$$\begin{aligned}\text{RONIC}_{T+1} &= \Delta \text{NOPLAT}_{T+1} / \Delta \text{IC}_T \\ K_T &= \Delta \text{IC}_T / \text{NOPLAT}_T \\ g_{T+1} &= \Delta \text{NOPLAT}_{T+1} / \text{NOPLAT}_T = \\ &= \text{RONIC}_{T+1} * K_T \\ K_T &= g_{T+1} / \text{RONIC}_{T+1}\end{aligned}$$

$$\text{FCFF}_T = \text{NOPLAT}_T * (1 - g_{T+1} / \text{RONIC}_{T+1})$$

$$\text{FCFF}_T = \text{IC}_{T-1} * \text{ROIC}_T * (1 - g_{T+1} / \text{RONIC}_{T+1})$$

Value drivers across the lifecycle

	Launch	Growth	Maturity	Decline
g_{Sales}	High	High	None	Negative
Operative profit margin	Low/High	High	Medium	Low
Corporate tax burden	Low	Low	Normal	Depends
Working capital need	High	High	Low	Decreasing
Invested asset need	High	High	Low/Medium	Low/Decreasing
r_A	Very high	High	Medium	Low
Planning horizon	Short	Medium/ Long	Long	Short/Medium
Proportion of terminal value	High	High	Medium	Medium

Business value by value drivers

$$V_{OP} = \sum_{t=1}^{\infty} \frac{IC_{t-1} * ROIC_t * \left(1 - \frac{g_{t+1}}{RONIC_{t+1}}\right)}{\prod_{i=1}^t (1 + WACC_i)}$$

Assuming a growing perpetuity

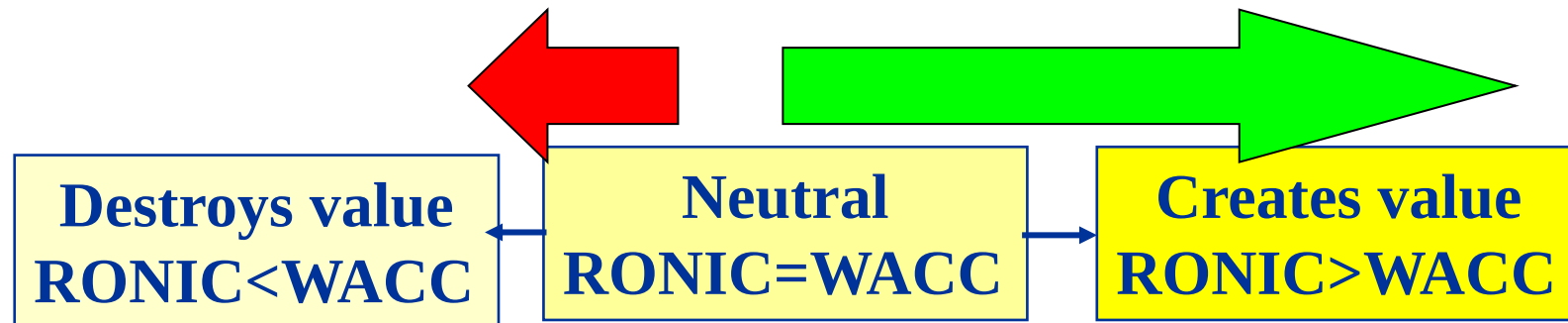
$$V_{OP} = \frac{NOPLAT * \left(1 - \frac{g}{RONIC}\right)}{(WACC - g)}$$

g, ROIC, WACC and the business value

		RONIC				
		6%	8%	10%	12%	14%
g	4%	709	891	1 000	1 073	1 125
	6%	495	811	1 000	1 126	1 217
	8%	215	706	1 000	1 196	1 336

$IC_0=1000$, $ROIC_1=RONIC$, $WACC=10\%$, after 20 years $RONIC=WACC$

Growth:



Measuring value added

Economic Value Added (EVA)

$$EVA = NOPLAT - WACC * IC = (ROIC - WACC) * IC$$

Residual Income (RI)

$$RI = \text{Net profit} - \text{Equity} * r_E$$

Market Value Added (MVA)

$$MVA = V - IC$$

Introducing value based management

1. Identifying.

What drives value locally?

2. Setting priorities.

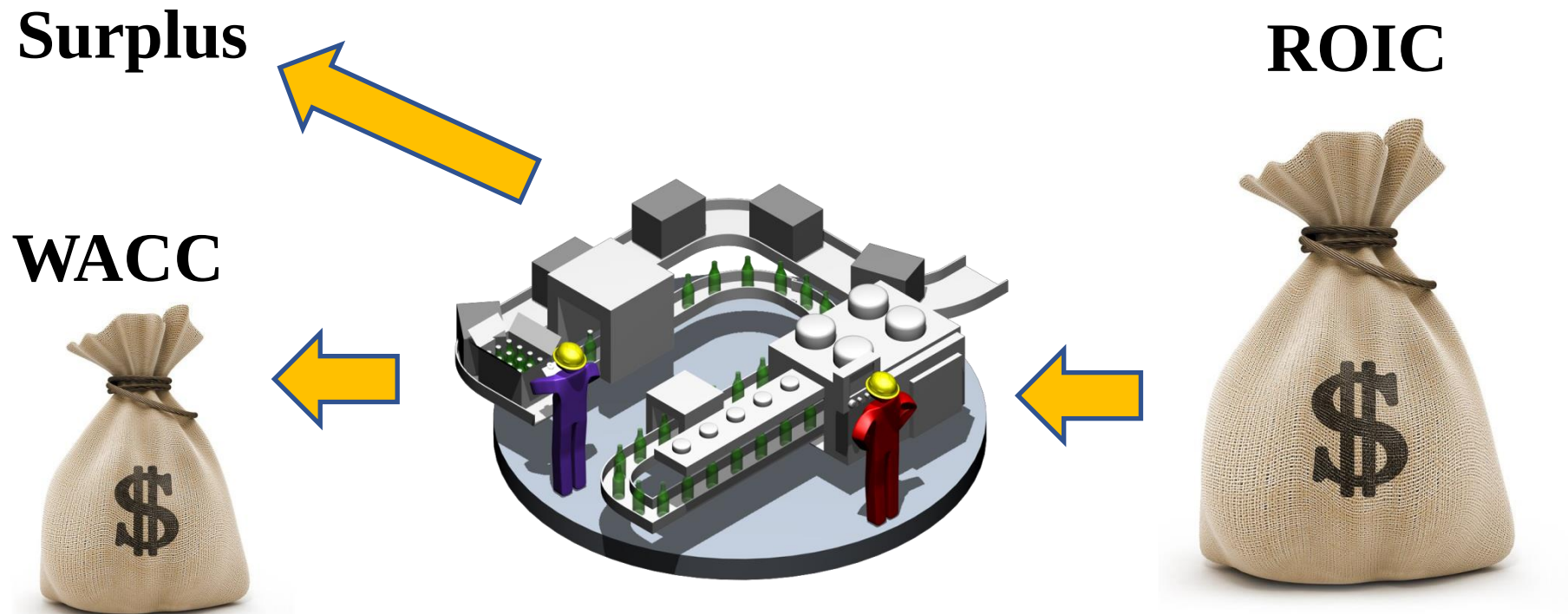
Importance of factors can be set based on sensitivity tests.

3. Integrating into the culture.

Employees have to understand the concept.

Lessons of VBM 1

Value is generated by achieving return on investment **on the top of the required level.**



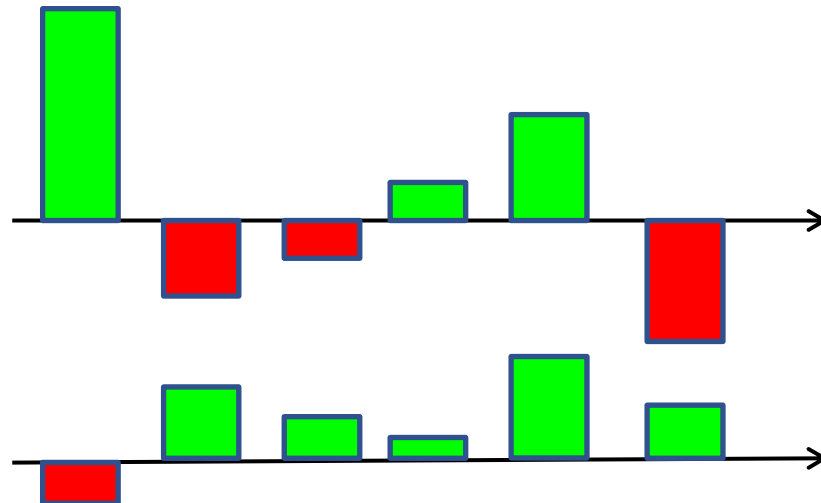
Lessons of VBM 2

The more capital we invest, and the longer we invest it, the more value can be created. This goes hand in hand with higher CF growth and longer period of growth.



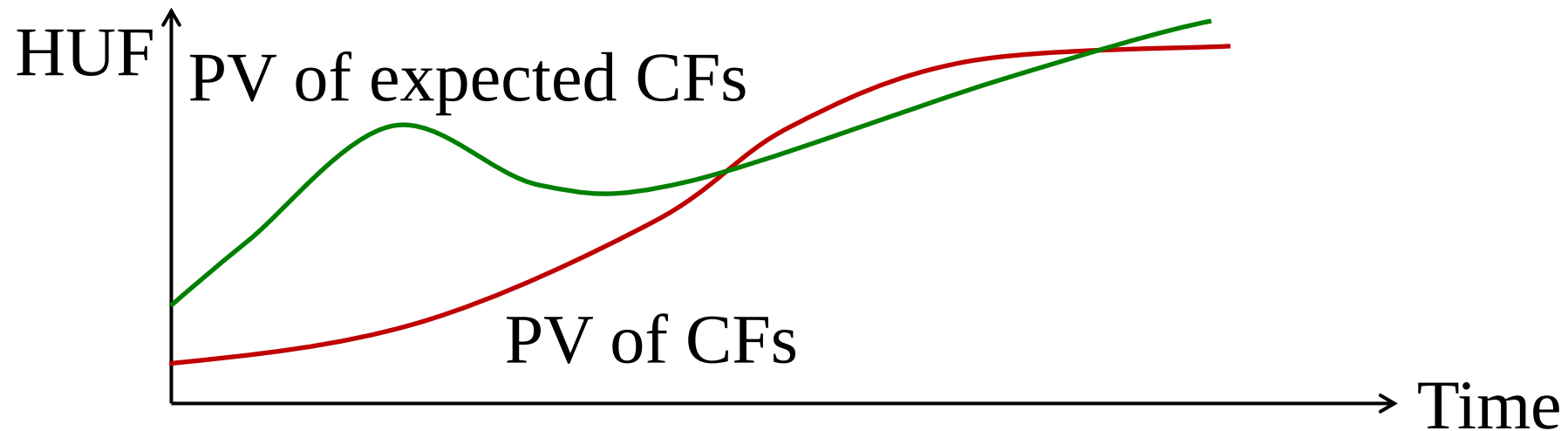
Lessons of VBM 3

We should prefer the strategy that maximises the present value of future FCFs or EVAs.

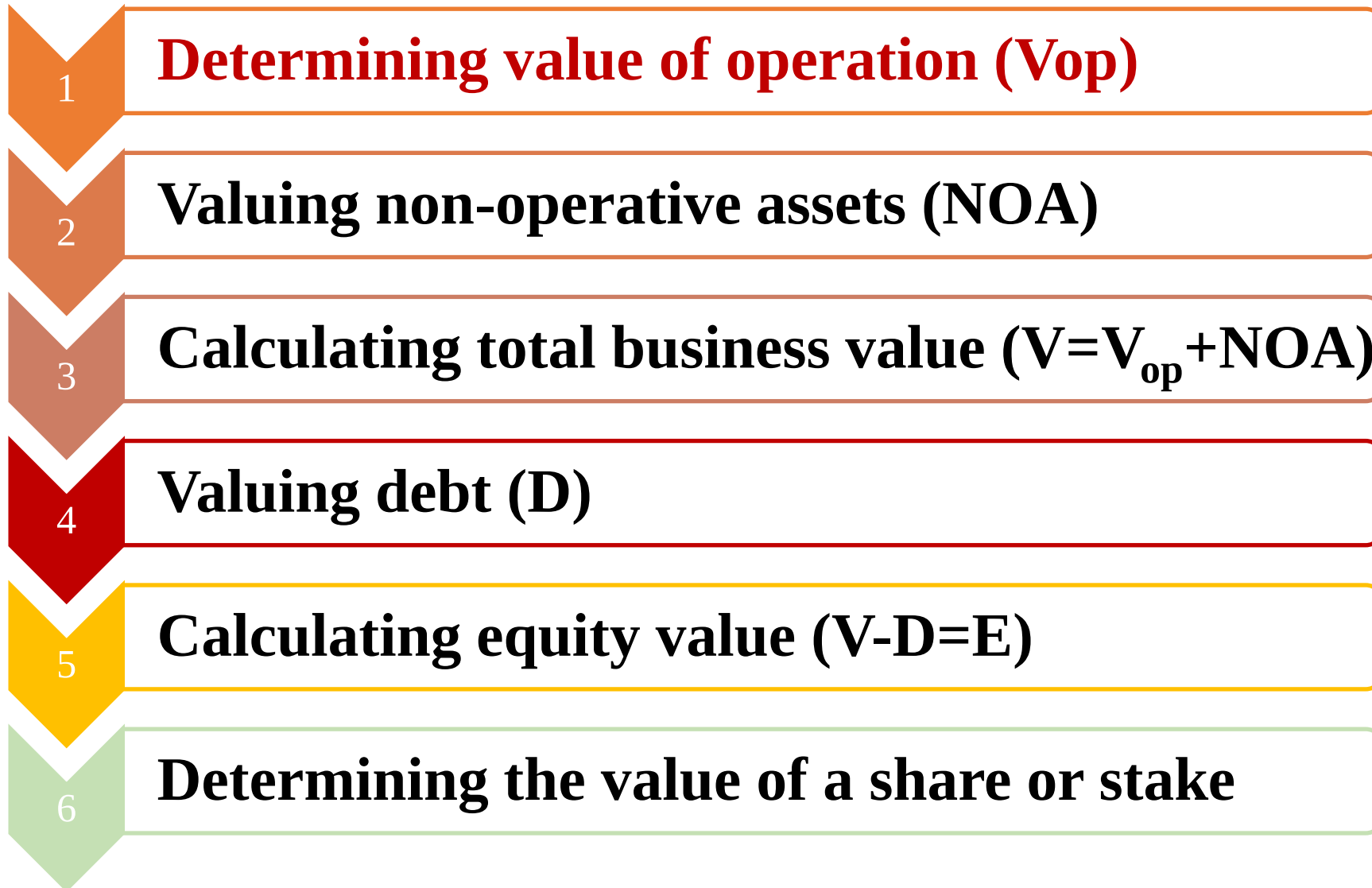


Lessons of VBM 4

The business *value* is determined by the expected *performance*, while the *quotes* of shares and bonds are rather linked to the changes in *expectations* about the future performance.



The process of valuation



Determining value of operation 1

Business plan

1. **Macro and industry analysis.** Trends, outlooks, new entrants, regulations.
2. **Competitors.** Strategies, fields of business, organisation, financial performance, comparable ratios, competitive advantages, market power.
3. **The business.** Foundation, owners, mission, strategy, stakeholder policy, markets, operation, organisation.

Determining value of operation 2

4. **Historical data.** Firm data, financial reports, ratios.
5. **Predictions.** Future market conditions, growth in sales, cost structure, cash flow. Consultation!
6. **Valuation.** What do we value? Choice of method, contrasting results, premiums & discounts.
7. **Checking results.** Assumptions used, market pricing vs result received, M&A data.

Which project/firm is better?

$$r = 10\%$$

$$NPV_A = 73.55$$

$$NPV_B = 54.13$$

$$IRR_A = 173.50\%$$

$$IRR_B = 208.26\%$$

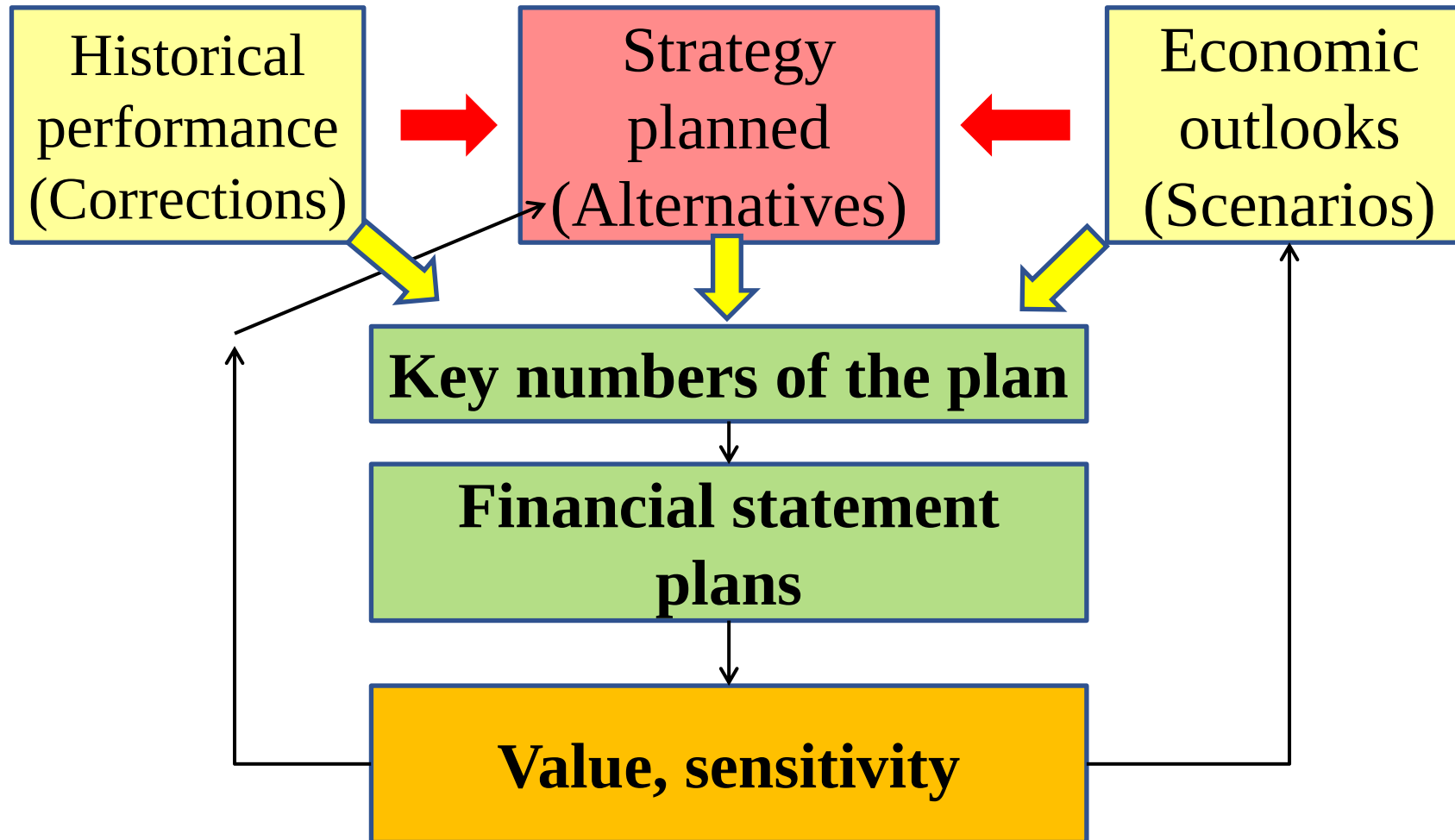
$$PI_A = 73.55\%$$

$$PI_B = 108.26\%$$

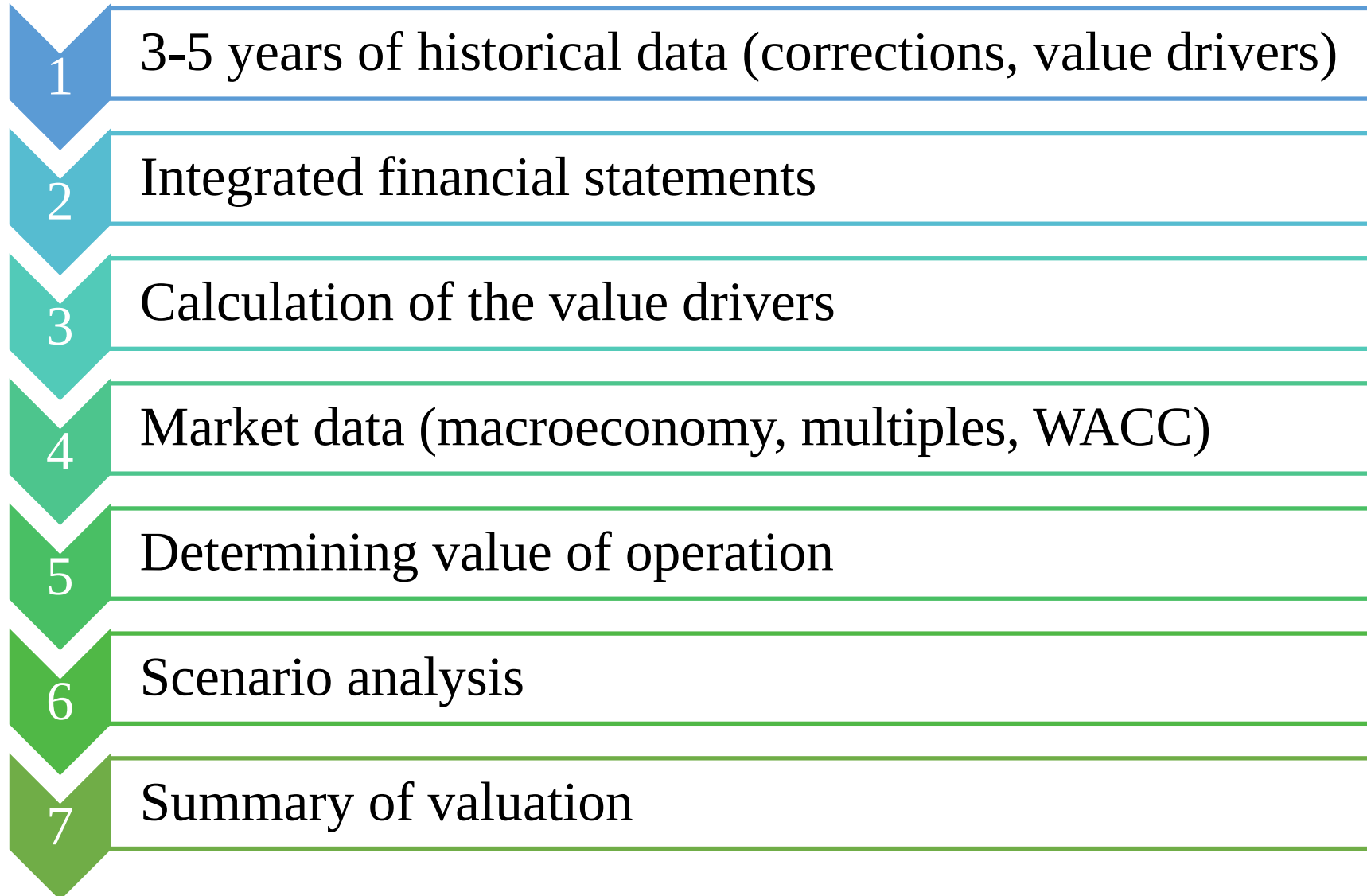
t	A	B
0	-100	-50
1	100	60
2	100	60

Better $\neq$ More valuable

The process of planning



Parts of the valuation model



Technical requirements

All inputs at one place

One directional logical connections

References instead of fix numbers

Key results are highlighted

Forecasting economic conditions

Porter five
forces

SWOT

PESTEL

Predicting
inflation

Yield curve

Market
outlooks

Development
of the
technology

Product life
cycle

Strategy

Steps of financial planning

1. Market performance

Quantity, price, conditions, timing $\Rightarrow$ Operational profit

2. Asset need

Invested and current assets, operational liabilities $\Rightarrow$ Total of Assets

3. Quantifying IC

Total of Assets - Operational sources $\Rightarrow$ IC need

4. Financing

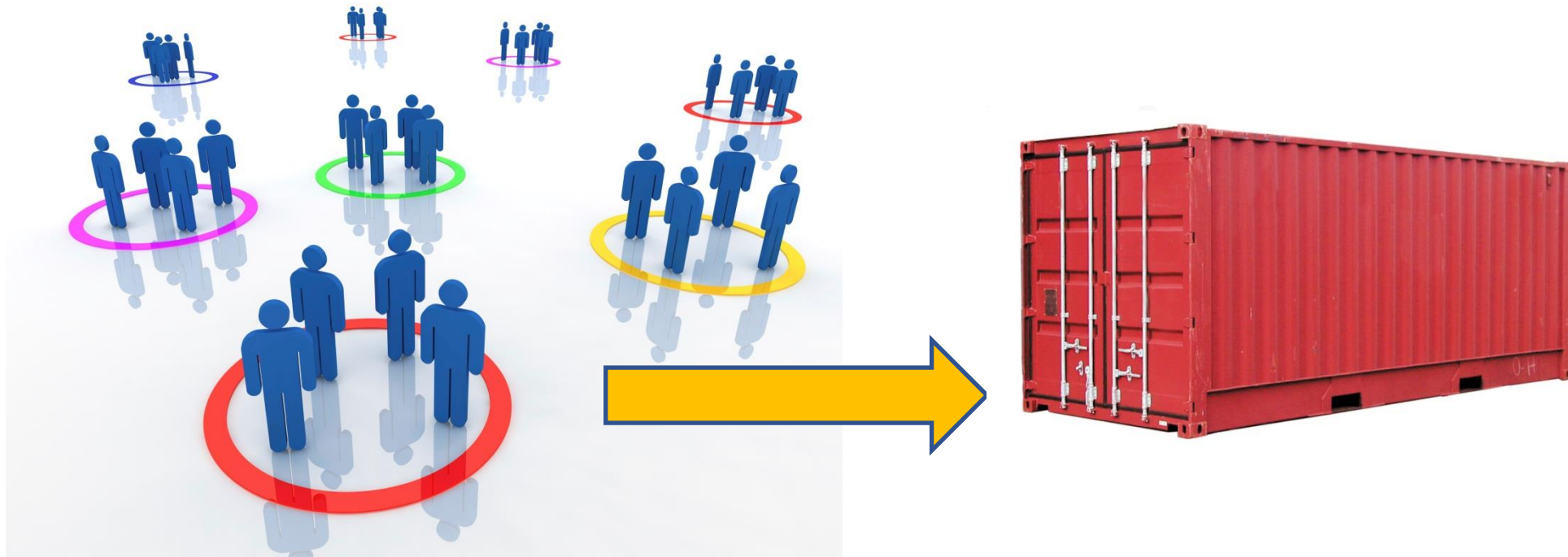
Optimal sourcing of IC $\Rightarrow$ Total of Equity & Liabilities, finalising P/L

Predicting sales 1

Top-down approach: We derive the sales from the changes in the market conditions

logic: total market size, own share, development of prices

application: mature industries, big players, reliable market data available

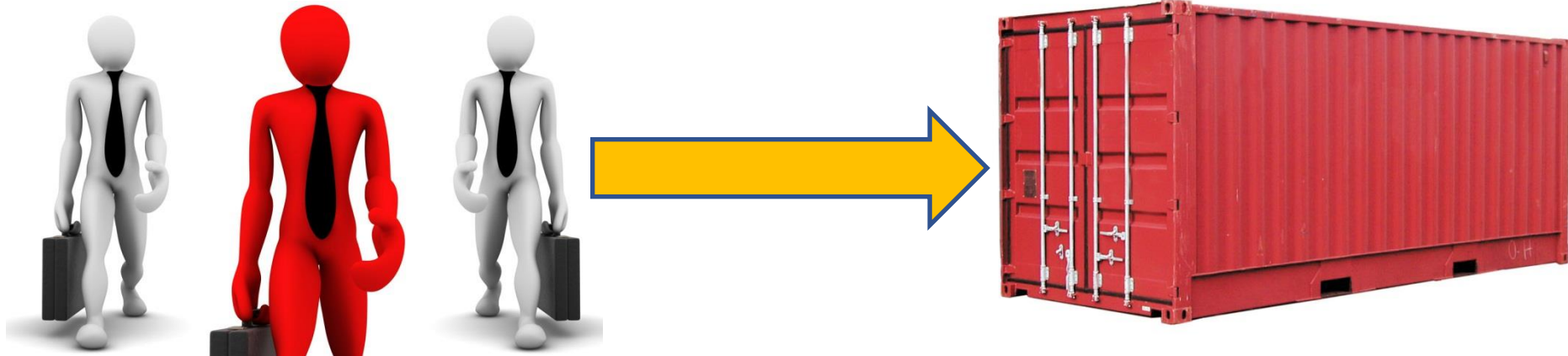


Predicting sales 2

Bottom-up approach: Prediction of demand for the current and potential buyers.

logic: demand data from customers, churn rates, acquisition possibilities

application: new industries, small players, application of data from comparable sectors



Predicting P/L

Type	P/L lines	Basis	Rate proposed
Operative	COGS	Sales (Q)	COGS / Sales
	SG&A	Sales (Q)	SG&A / Sales
	Material expenses	Sales (Q)	Material expenses / Sales
	D&A	Invested assets at the end of the last year	D&A / Invested assets
Non-operative	Non-operative incomes	Linked non-operative assets	Non-op income / non-op assets
	Interest paid	Debt at the end of the last year	Interest paid / opening debt
	Interest received	Free cash at the end of the last year	Interest received / opening free cash

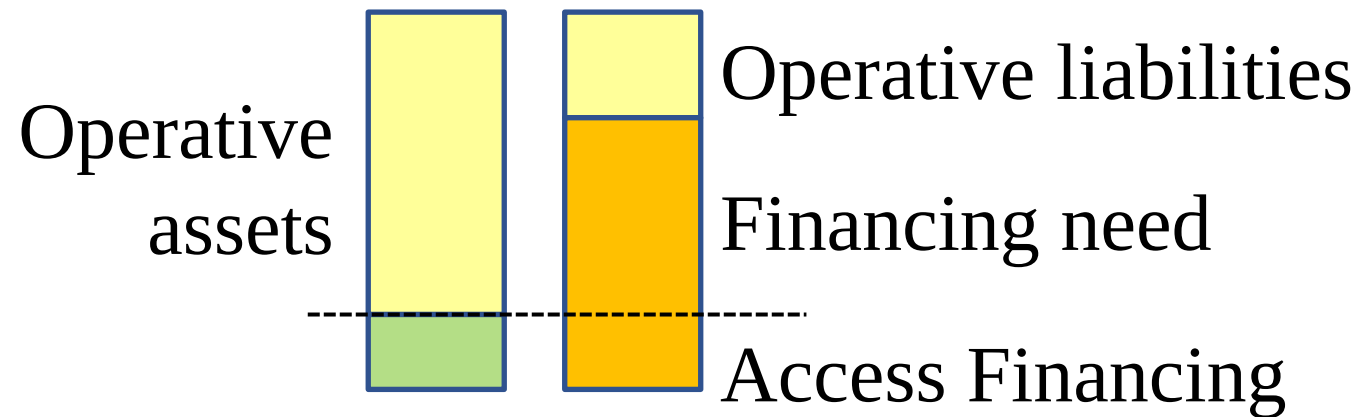
Predicting the balance sheet

Type	Balance sheet lines	Basis	Rate proposed
Operative	Operative cash	Sales	Operative cash / Sales
	Receivables	Sales	Average turnover days
	Inventory	COGS or Material + Labour expenses	Average turnover days
	Invested Assets	Sales (or quantity)	Invested assets / Sales
	Active accruals	Sales	Active accruals / Sales
	Payables	COGS or Material expenses	Average turnover days
Non-operative	Non-operative assets	None	Growth rate

Planning for financing

IC need is the difference of Assets and Operational liabilities:

$$\text{IC needed} = \text{Equity} + \text{Debt}$$



Assumptions on financing

Foreign capital

1. Remainder (balancing line),
2. Fix proportion: market average, percent of IC or V, own target value

Dividend

1. Remainder (based on reinvestment need),
2. Fixed proportion,
3. Nominal target (steady growth)

Which is the balancing line?

Debt

Operation,
Dividend, Equity
→ Debt (**Will we
get enough?**)

Cash

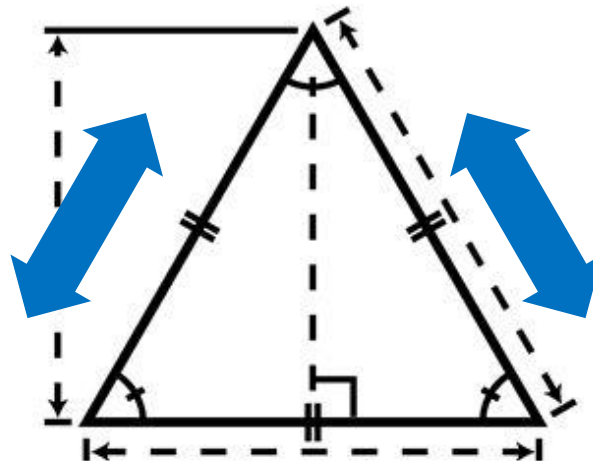
Operation,
Dividend, Debt →
Cash (**Too much?
Negative?**)

Dividend + Equity
increase/decrease

Operation,
Financing, Equity
→ Dividend +
Capital
in/decrease

Two factors are enough

Operation and
investment policy



Dividend
policy



Financing
policy

Iteration for financing

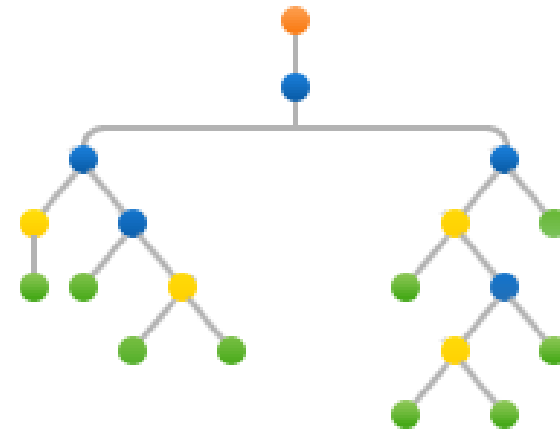
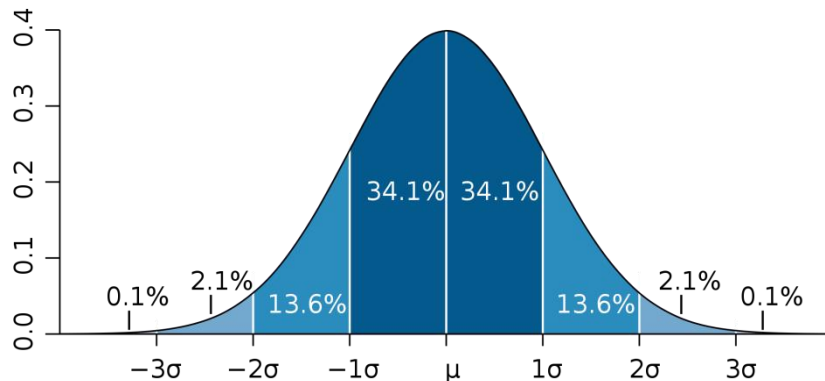
Assuming start of the year debt adjustment may lead to circular referencing:

Interests paid/received → Net Profit → Dividend →
Reinvestment → Equity → Debt need/Extra cash →
Interests paid/received



Why and how?

1. **Sensitivity analysis** (improving precision)
2. **Scenario analysis** (uncertain paths)
3. **Decision trees** (future choices)
4. **Monte Carlo simulation** (using distributions instead of expected values)



Discounted Cash Flow based (DCF) methods

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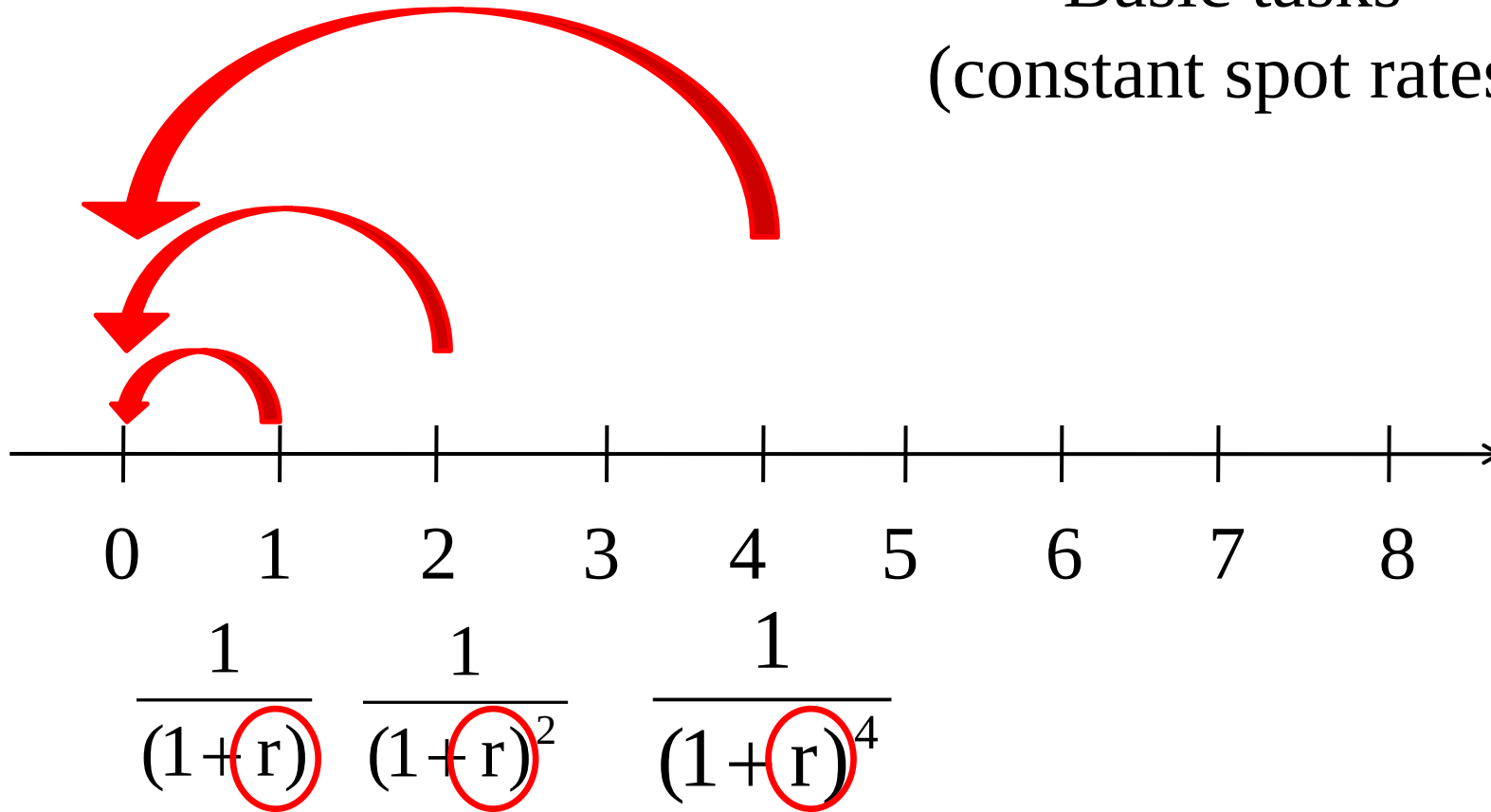
Corporate finance and reality

Corporate finance: flat yield curve, spot rates, required rates given

Reality: variable yield curve, forward rates, required rates are to estimated

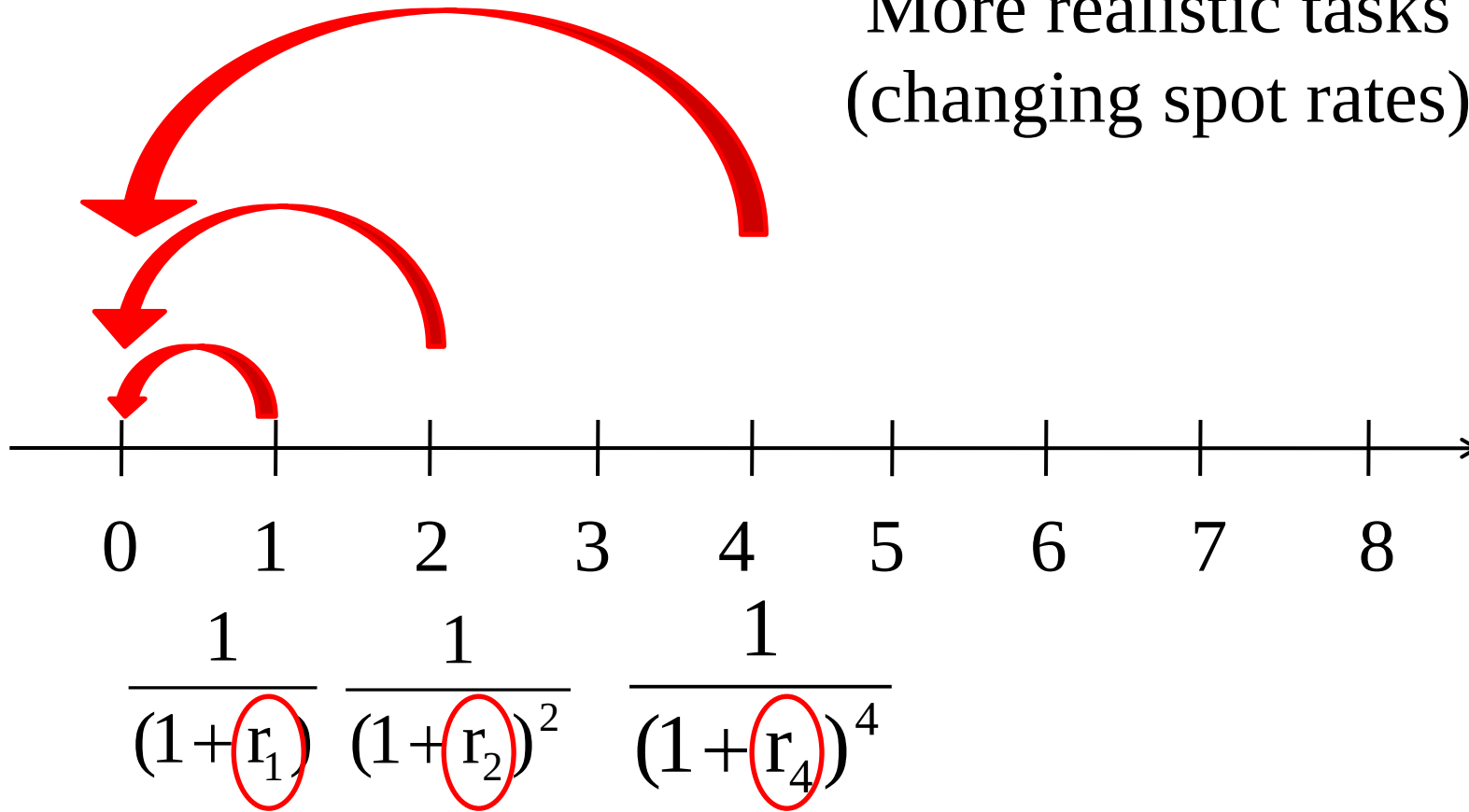
Calculating the present value 1

Basic tasks
(constant spot rates)



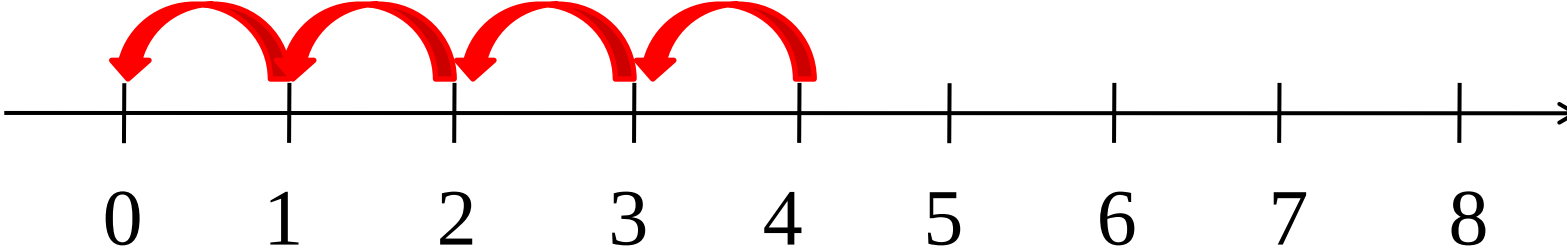
Calculating the present value 2

More realistic tasks
(changing spot rates)



Calculating the present value 3

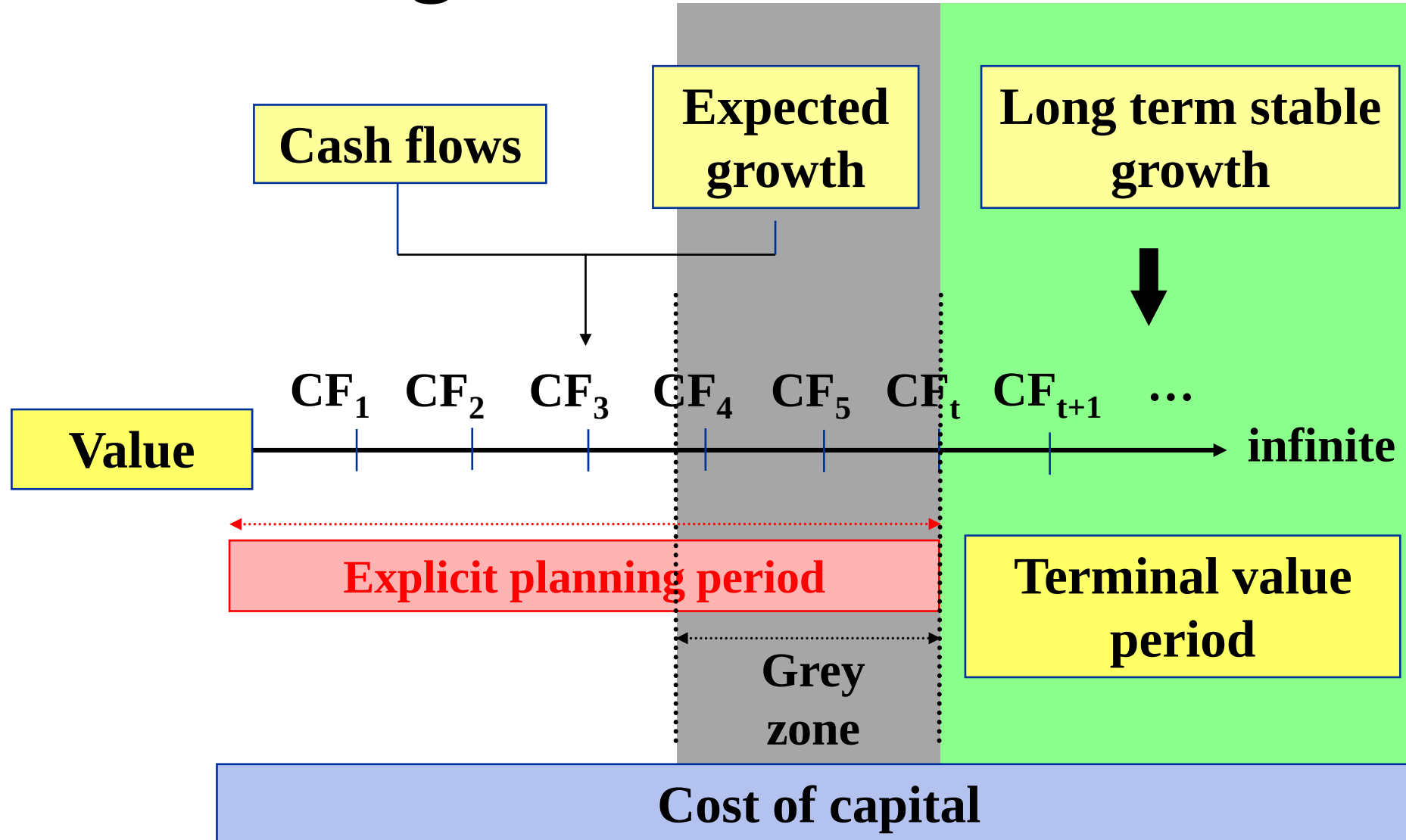
Reality
(forward rates)

$$\frac{1}{(1 + {}_0f_1)} * \frac{1}{(1 + {}_1f_2)} * \dots * \frac{1}{(1 + {}_3f_4)}$$


The general DCF model 1

$$NPV = \sum_{i=0}^{\infty} \frac{CF_i}{\prod_{j=1}^i (1 + r_j)}$$

The general DCF model 2



Durations

Explicit period

The longer the better, min 5-7 years

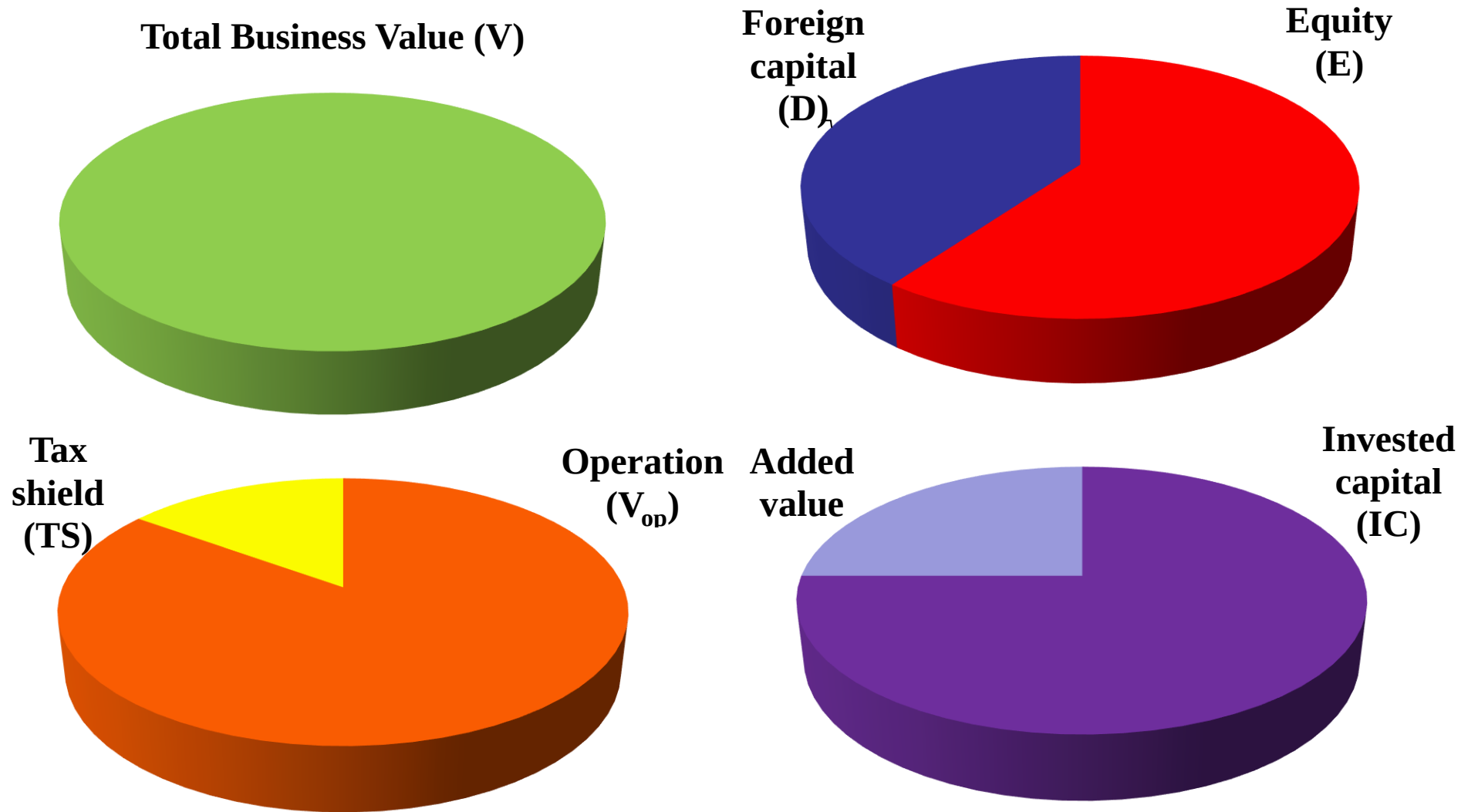
Terminal Value period

Only to start when everything will be stable

Grey (twilight) zone

In-between the earlier two, may use a simplified (value driver only) model

The division of the business value



DCF models

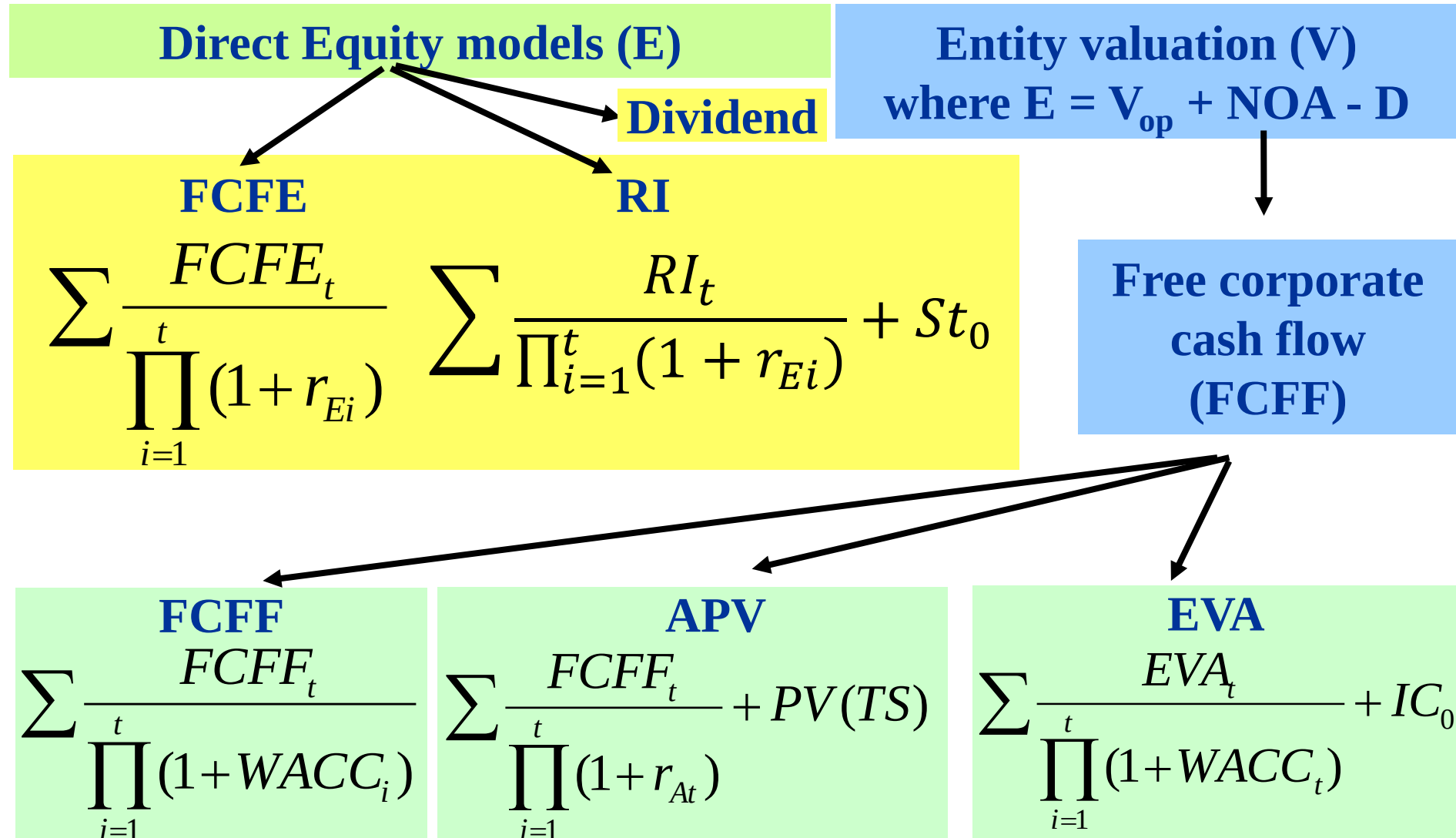
Entity models

- FCFF
- APV
- EVA

Equity models

- FCFE (Dividend models)
- RI (Residual Income)

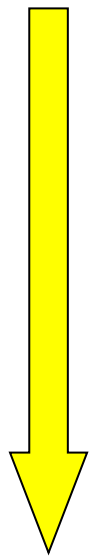
DCF valuation methods



Entity or Equity models?

All DCF models lead to the same result.

(This is the essence of Miller-Modigliani theorems.)



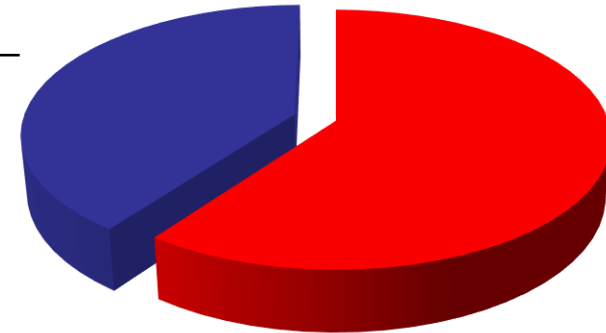
Value of operation (V_{op})

+ Value of non-operative assets

Total firm value (V)

- Value of Foreign capital (D)

= Value of equity (E)



Use rather entity models

Because:

we usually focus on the value creation process,

those can be used for Value Based Management (motivation),

multi-business firms can be analysed,

offer more freedom to model financing alternatives.

Steps of DCF valuations

1.

Estimating free cash flows

2.

Estimating cost of capital

3.

Calculating terminal value

4.

Calculating current value

5.

Sensitivity tests, Scenario analysis

6.

Considering other circumstances

Rules of Cash flow „matching”

Expiry

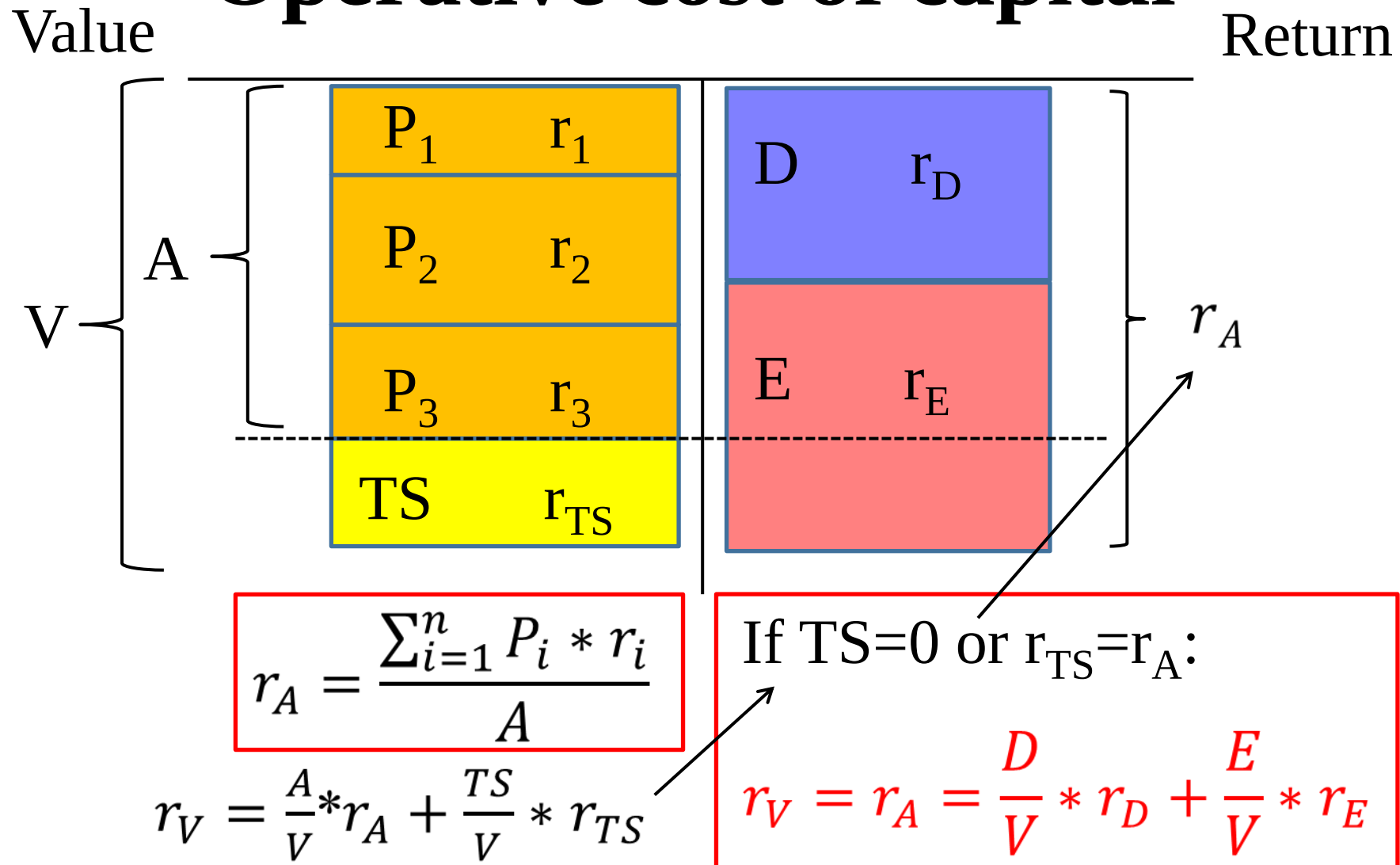
Currency

Nominal/real

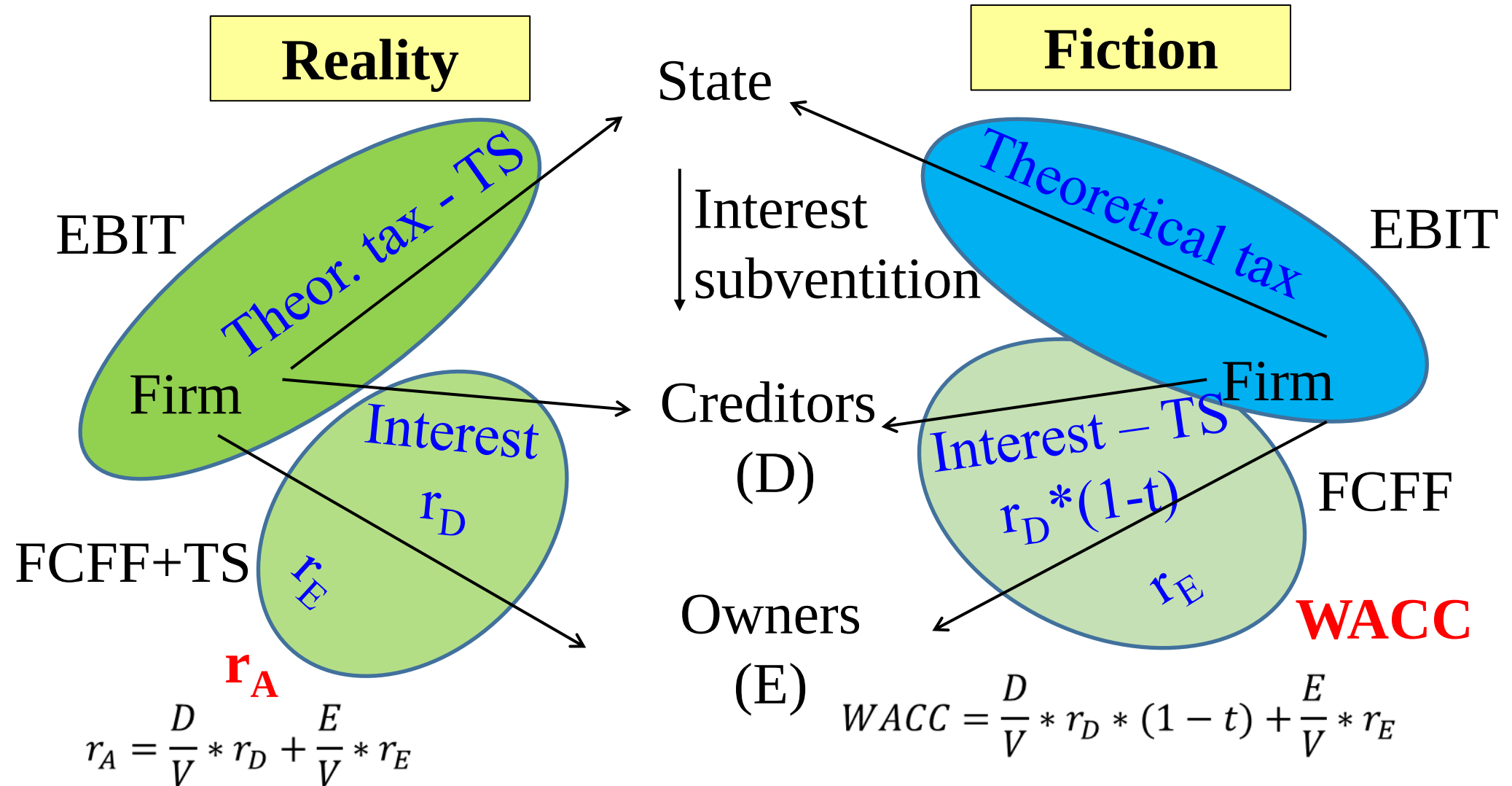
Effects of financing (TS)

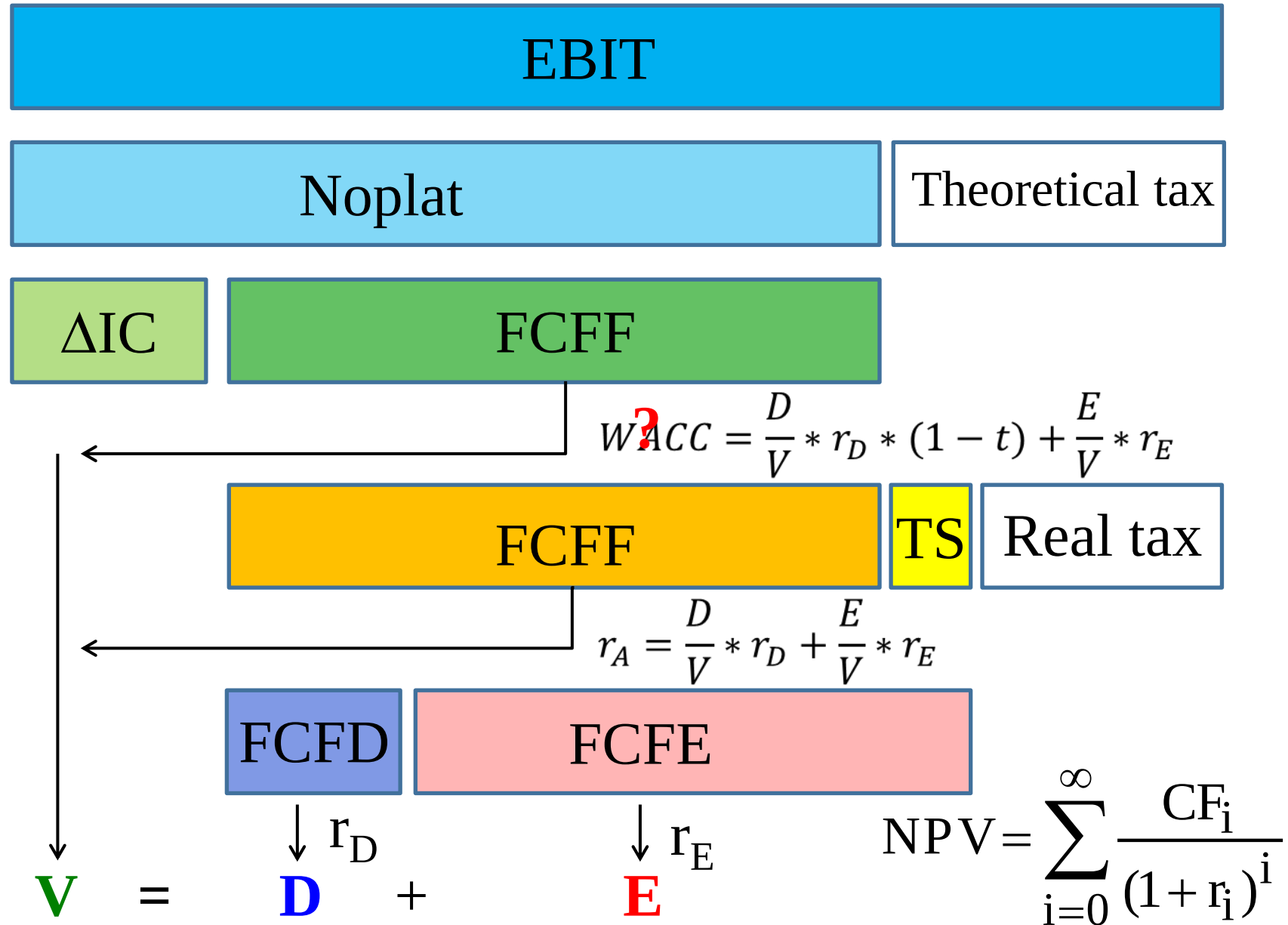
Risk

Operative cost of capital



Weighted Average Cost of Capital (WACC)





FCFF valuation

$$V = \sum_t \frac{FCFF_t}{\prod_{i=1}^t (1 + WACC_i)}$$

FCFF: CF available to all financiers. (CF independent of financing)

$$\mathbf{V - D = E}$$

FCFE valuation

$$E = \sum_t \frac{FCFE_t}{\prod_{i=1}^t (1 + r_{Ei})}$$

FCFE: CF available to shareholders. (CF depends on the financing structure, can be redrawn freely.)

$$\mathbf{E + D = V}$$

EVA/RI valuation

Limitations of FCFF valuation:

- 1) The seemingly high proportion of terminal value may create the impression that value is created after the explicit period.
- 2) does not show when the value is created.

Economic profit = added profit over the required level

$$\text{EVA}_i = \text{IC}_{i-1} * (\text{ROIC}_i - \text{WACC}_i) = \text{NOPLAT}_i - \text{IC}_{i-1} * \text{WACC}_i$$

$$\text{RI}_i = \text{BV(E)}_{i-1} * (\text{ROE}_i - r_{\text{Ei}}) = \text{Net Profit} - \text{BV(E)}_{i-1} * r_{\text{Ei}}$$

EVA valuation

$$V = \sum_t \frac{EVA_t}{\prod_{i=1}^t (1 + WACC_i)} + IC_0$$

In the long run ROIC = WACC (ROE = r_E), thus EVA = RI = 0.
The added value in TV in such a case is zero.

APV valuation

Limits of FCFF valuation:

- 1) shows the value for a given financing structure,
- 2) financing effects can not be separated.

$$\text{APV} = \text{V assuming equity only financing} + \text{Present value of tax shields [PV (TS)]}$$

$$V_{Op} = \sum_t \frac{FCFF_t}{\prod_{i=1}^t (1 + r_{Ai})}$$

What rate to use for discounting tax shield?

During this course we assume tax shield is linked to operation $\Rightarrow r_A$

DCF valuation methods

Method	To discount	Discount rate	Notes
Total firm DCF	FCFF	WACC	Useful for multiple business units and set D/V ratio firms
Economic profit	EVA / RI	WACC / r_E	Shows when the value will be created.
APV	FCFF and tax shield (TS)	Operative cost of capital (r_A)	Separates the effect of financing, handy for M&As.
Direct equity valuation	FCFE	Required rate on equity (r_E)	Detailed, hectic financial plan available, bottom-up CF, financial service industry

Which should I choose?

- 1. What to do if business units have no stand-alone financing?**
- 2. How risky is this business plan?**
- 3. What to do if it is very hard to estimate the cost of capital for this firm?**
- 4. What is the value of the shares given this outlook?**

Determining the terminal value

Péter Juhász, PhD, CFA

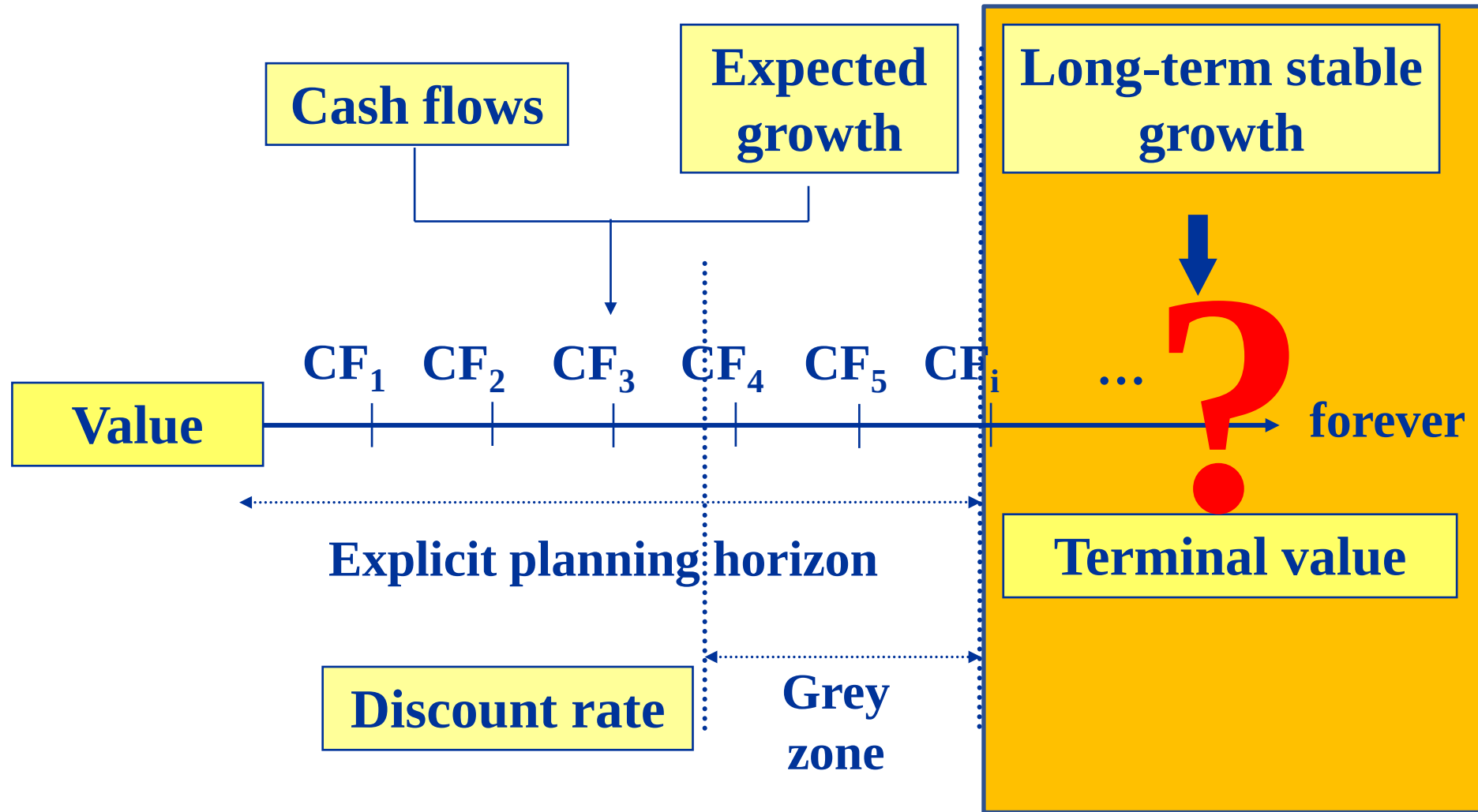


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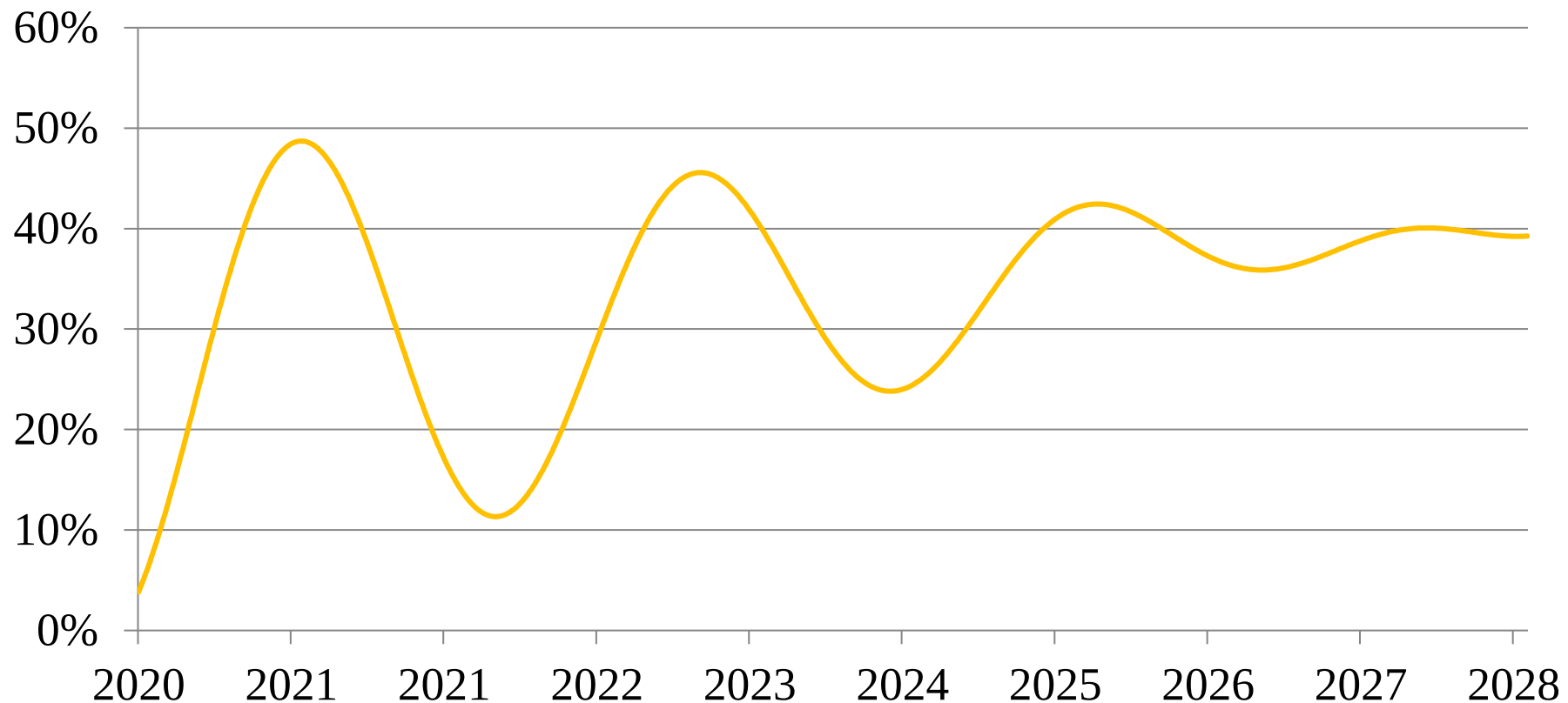


The general DCF model



How long should the explicit period last?

By the start of the terminal value period all inputs should stabilise.



The lifecycle of the ROIC and the g

Depend on the capabilities of the firm and the dynamics of the industry.

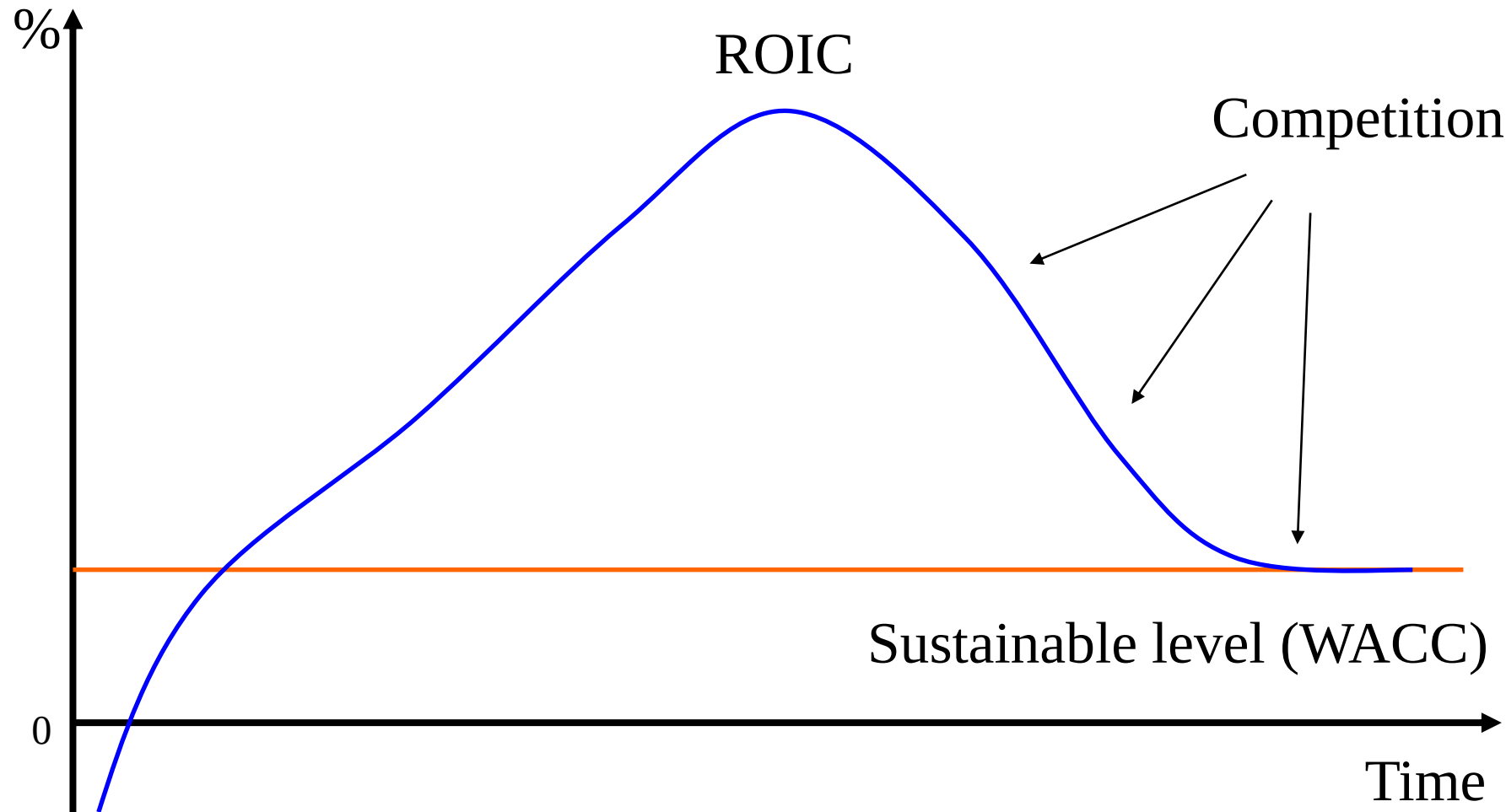
Those converge to the industry average.

Those mutually depend on each other (g generates value only if $ROIC > WACC$).

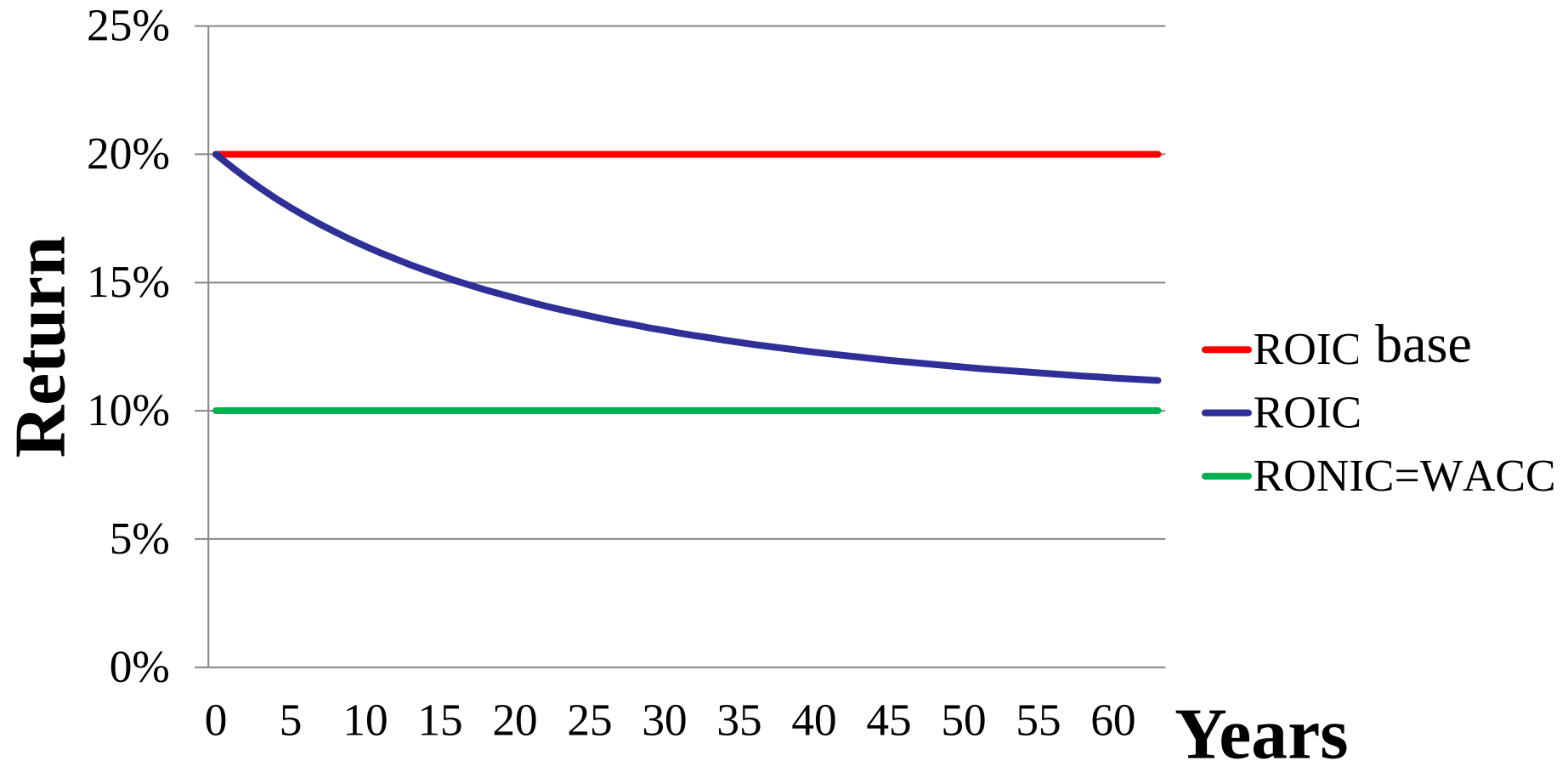
We plan for further details only if those improve the quality of the prediction.

(maintainability and precision of assumptions).

Changes in value creation over time



Connections among ROIC, RONIC, and WACC



ROIC remaining high 1

$$ROIC = (1 - t) * (\text{Unit price} - \text{Unit cost}) * \frac{\text{Quantity}}{IC}$$



Premium
price?



ROIC remaining high 2

$$ROIC = (1 - t) * (Unit\ price - Unit\ cost) * \frac{Quantity}{IC}$$

Lower costs?



ROIC remaining high 3

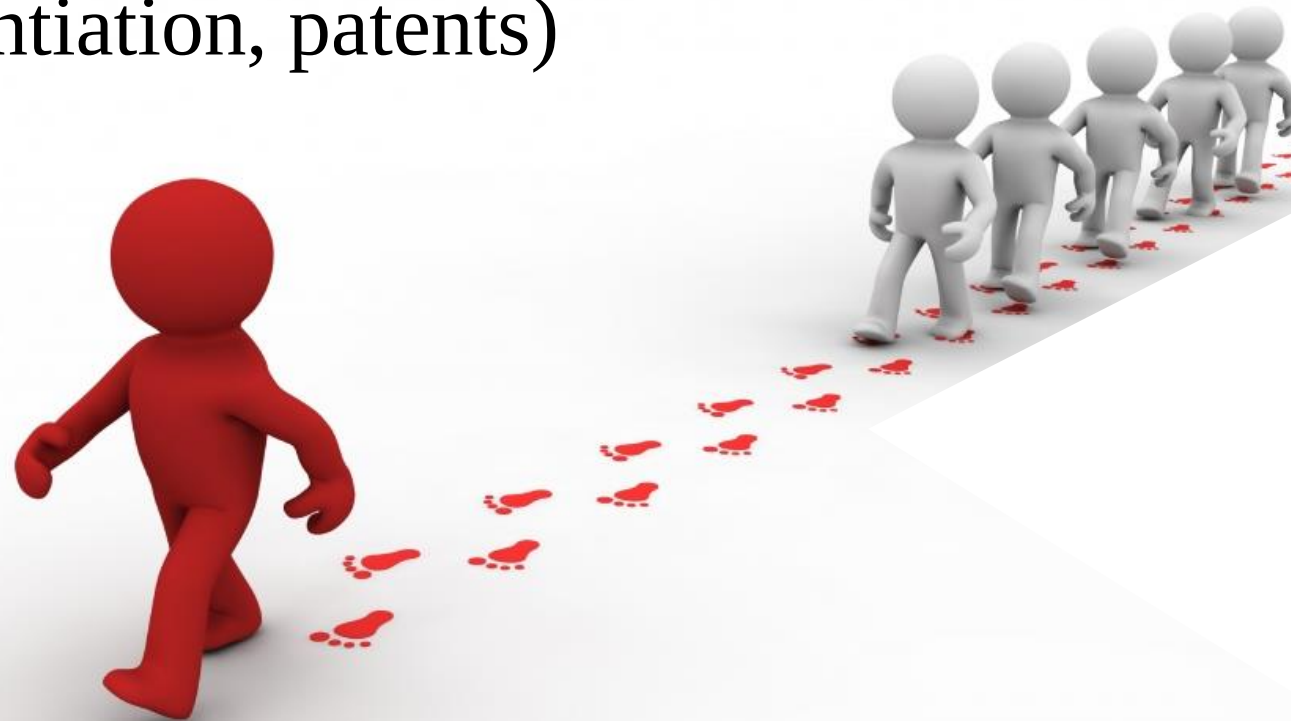
$$ROIC = (1 - t) * (Unit\ price - Unit\ cost) * \frac{Quantity}{IC}$$

More efficient capital
management?



ROIC remaining high 4

To maintain your competitive advantage in the long run you have to hinder copying. (state protection, exclusive contracts, licence, differentiation, patents)



Terminal value formulas

1. FCFF

$$TV = \sum_{t=1}^{\infty} \frac{FCFF_t}{\prod_{i=1}^t (1 + WACC_i)} = \frac{FCFF_{Standard}}{WACC^* - g_{FCFF}^*}$$

Assumption: $WACC^* > g_{FCFF}^*$!

Is that always true?

$$\mathbf{WACC^* < ? > g_{FCFF}^*}$$

If $r_f > g_{GDP}$:

$$\mathbf{WACC > r_D > r_f > g_{GDP} \geq g_{Firm}}$$

If $WACC < g_{Firm}$, then this is not the long-term stable status yet.

2. Value drivers

$$V_{TV} = \frac{NOPLAT_i - InvCF_i}{WACC^* - g_{NOPLAT}^*} = \frac{NOPLAT(1 - K)}{WACC^* - g_{NOPLAT}^*}$$

$$WACC^* > g_{NOPLAT}^* !$$

Inv CF = ?

$$= NOPLAT - FCFF = NOPLAT * K =$$

= Net capital investment (ΔIC)

$$V_{TV} = \frac{NOPLAT(1 - \frac{g_{NOPLAT}}{RONIC})}{WACC^* - g_{NOPLAT}^*}$$

WACC* > g_{NOPLAT}* < RONIC!

WACC* <?> RONIC

3. APV

$$TV = \frac{FCFF}{r_A - g} + \frac{D^* r_D^* t}{r_A - g}$$

4. FCFE

$$TV_E = \frac{FCFE}{r_E - g}$$

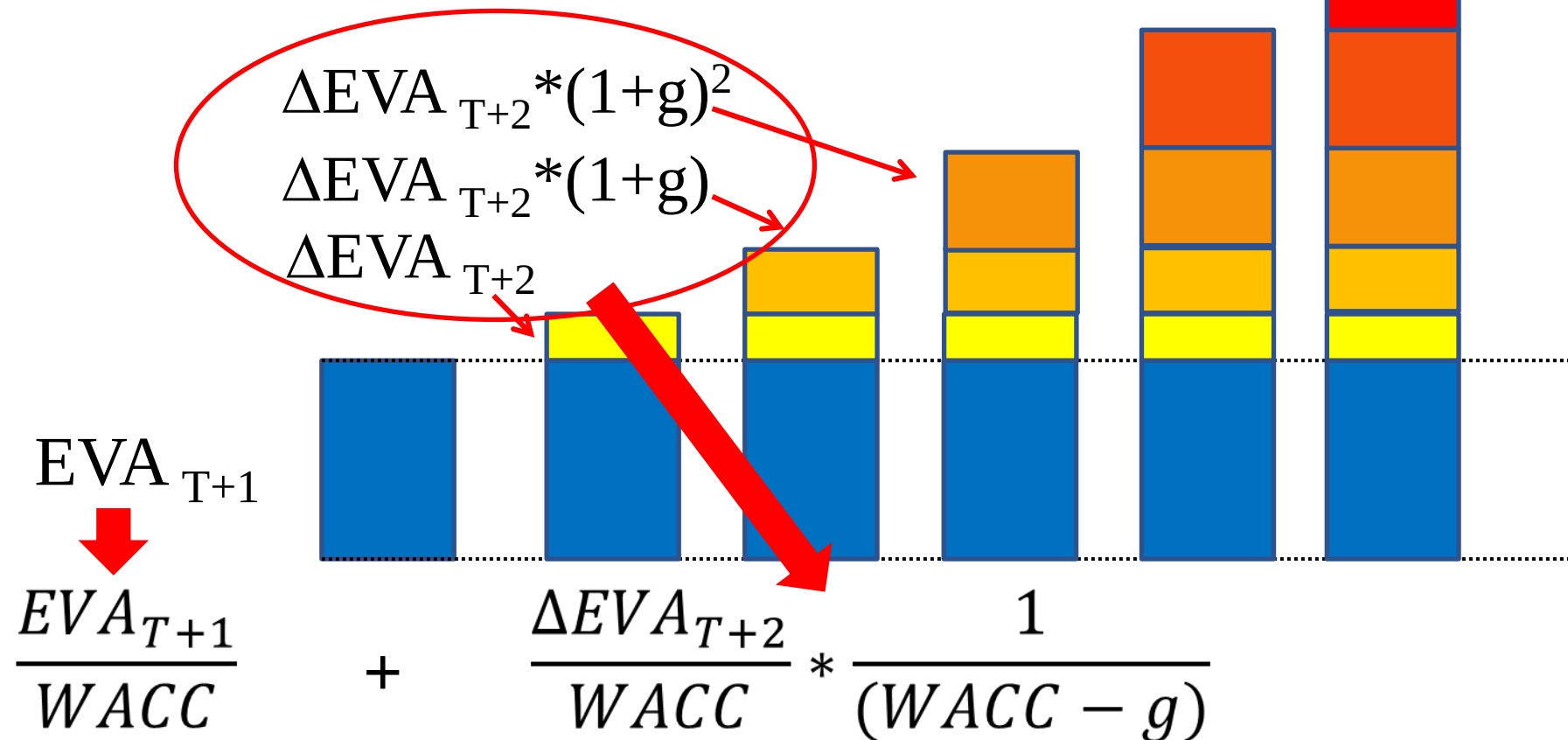
5. Economic profit

$$\begin{aligned} \text{EVA} &= \text{NOPLAT} - \text{IC} * \text{WACC} \\ &= \text{IC} * (\text{ROIC} - \text{WACC}) \end{aligned}$$

$$\text{EVA}_{TV} = \left(\frac{\text{EVA}_{T+1}}{\text{WACC}} + \frac{\text{NOPLAT}_{T+1} * \frac{g}{\text{RONIC}} * (\text{RONIC} - \text{WACC})}{\text{WACC} * (\text{WACC} - g)} \right)$$

How did we get that? ΔEVA_{T+2}

$$EVA_{TV_T} = \left(\frac{EVA_{T+1}}{WACC} + \frac{\Delta I C_{T+1} * (RONIC - WACC)}{WACC * (WACC - g)} \right)$$



Good news: convergence

Due to the competition in the long run: **WACC = RONIC**

$$V_{TV} = \frac{NOPLAT(1 - \frac{g}{RONIC})}{WACC - g} = \frac{NOPLAT(1 - \frac{g}{WACC})}{WACC - g} =$$

$$\frac{NOPLAT(\frac{WACC}{WACC} - \frac{g}{WACC})}{WACC - g} = \frac{NOPLAT(\frac{WACC - g}{WACC})}{WACC - g} =$$

$$= \frac{NOPLAT}{WACC}$$

Growth (g) plays no role.

What is the meaning of this formula?

$$TV = \frac{NOPLAT}{WACC - g}$$

Growth without investment.

4. Non-CF-based TV methods

Liquidation method

Liquidation or forced liquidation

Replacement methods

What has to and can be replaced from the market?

Multiple-based methods

Data on future market? Future comparable firms?

Typical mistakes in quantifying TV

Wrong choice of starting year

(Cycles, trends, standard operation, first TV year reflect maintainable performance)

Misprediction of ROIC and WACC

(Consider market growth, intensity of competition, and inflation.)

Mistake in discounting

(TV formulas provide the value for the beginning of the TV period.)

Using a real growth rate in a nominal model

(It is harder to believe but has no value effect if $RONIC = WACC$.)

Predicting growth

Historic data

Management expectations

Market/industry analysis

Macroanalysis



What do we predict?

**Sales, operational profitability, EBIT, Noplat,
operational CF, ..., net profit**

Averaging yearly growth rates

Arithmetical or geometric mean?

0	100	
1	120	+20%
2	96	-20%

Arithmetical: $(+20 + -20) / 2 = 0$

Geometrical:

$$\sqrt{(1.2 * 0.8)} = 0.9797 \longrightarrow \text{yearly } -2\%$$

Sustainable growth rate

$$\begin{aligned} g_{\text{Equity}} &= \Delta \text{Equity} / \text{Equity} = \\ &= (\text{Net profit} - \text{Dividend}) / \text{Equity} = \\ &= [\text{Net profit} * (1 - \text{div. payoff ratio})] / \text{Equity} = \\ &= (\text{Net profit} / \text{Equity}) * \text{reinvestment rate} = \\ &= \text{ROE} * \text{reinvestment rate} \end{aligned}$$

Equity and Debt are
book values here!

$$\text{ROE} = \text{ROIC} + \text{Debt/Equity} * [\text{ROIC} - r_D * (1-t)]$$

What changes together with Sales?

Quantity: COGS, variable costs

Receivables, if payment conditions unchanged

Suppliers, as procurement growth (price effect may differ!), if payment terms unchanged

Short term liabilities? Partly. Wages? Taxes?

Debt? Not automatic, new contract needed.

Equity: $g = \text{ROE} * \text{reinvestment rate}$

Effects of growth

Profit
and CF

Capital need
and structure

Operational
risk (r_A)

Limits of growth

Can be financed
from inner
sources?

Economic
(GDP/GNP)
growth?

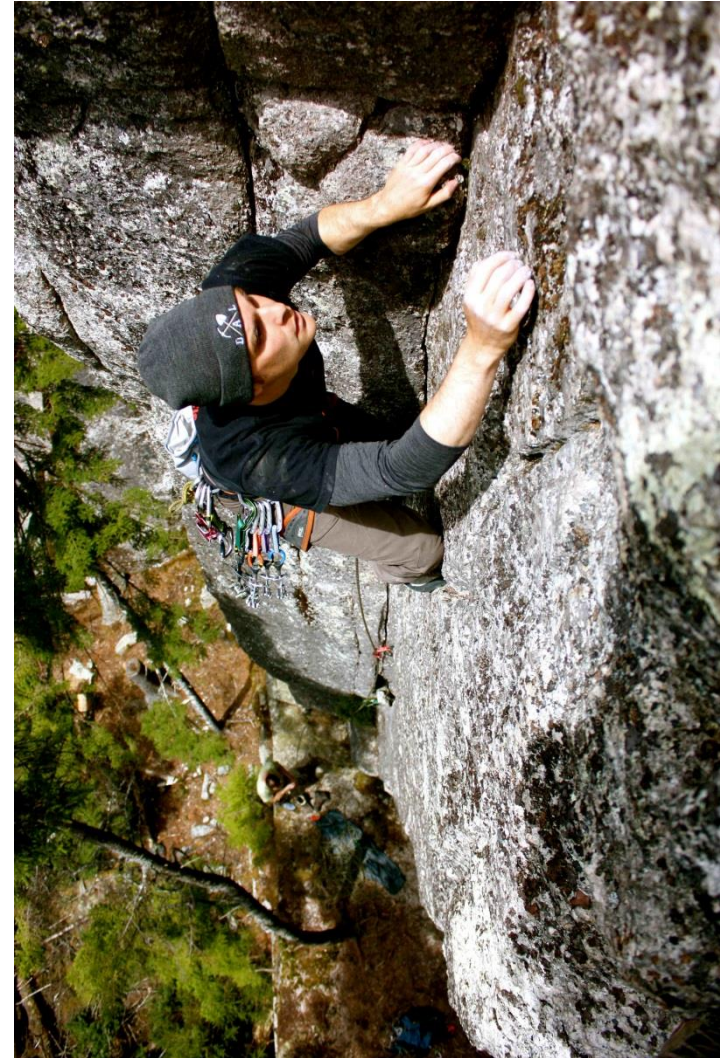
Growth of local
and international
markets?



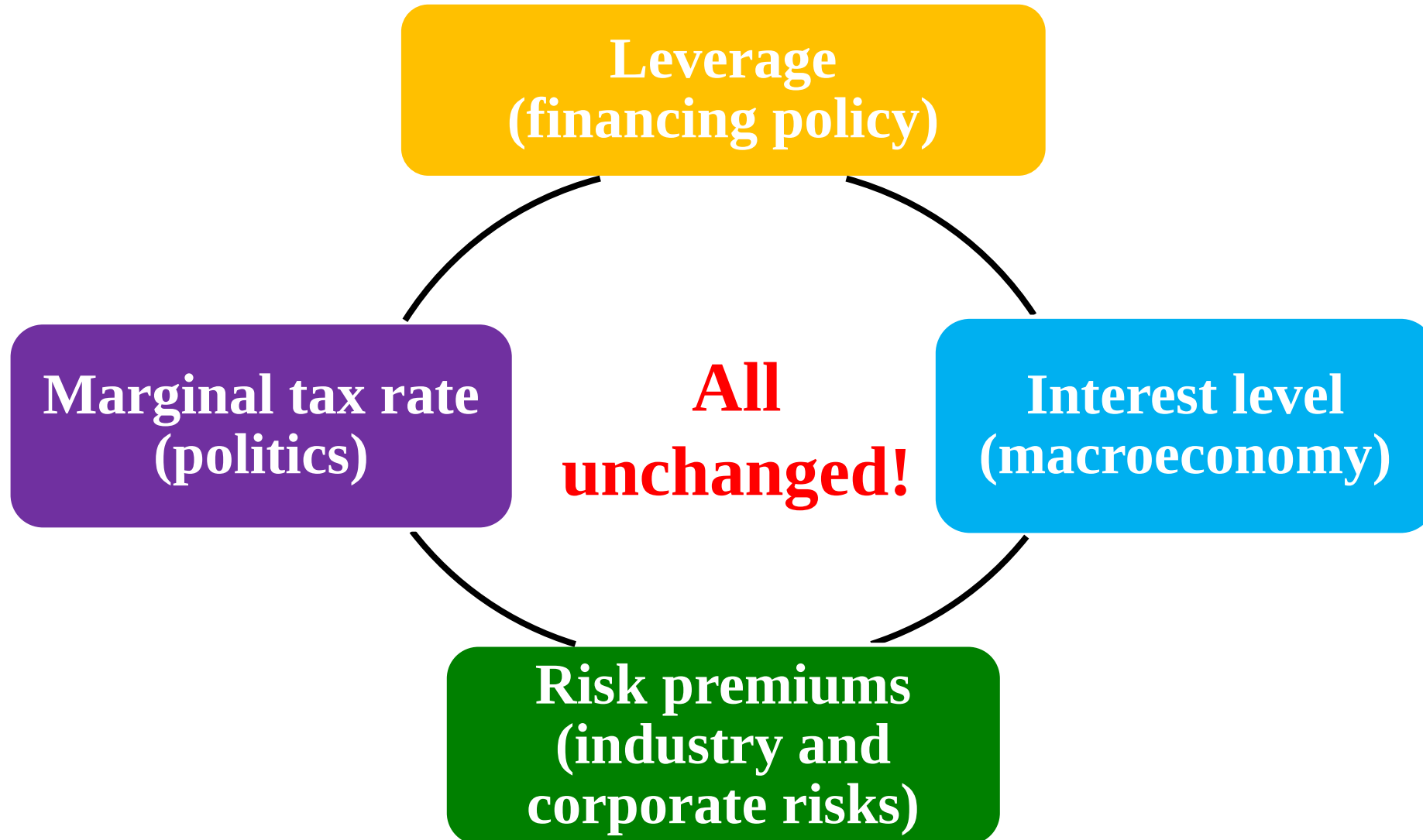
Issues linked to growth

- Nominal or real?
- Inflation stable?
- What currency?
- What supports it?

(market outlook, firm size,
current growth, competitive advantages,
competitors)



Assumptions of constant WACC



If WACC is constant ...

Leverage is stable and already optimal,

No competitive advantage or drawback,

Industry and regulation unchanged.

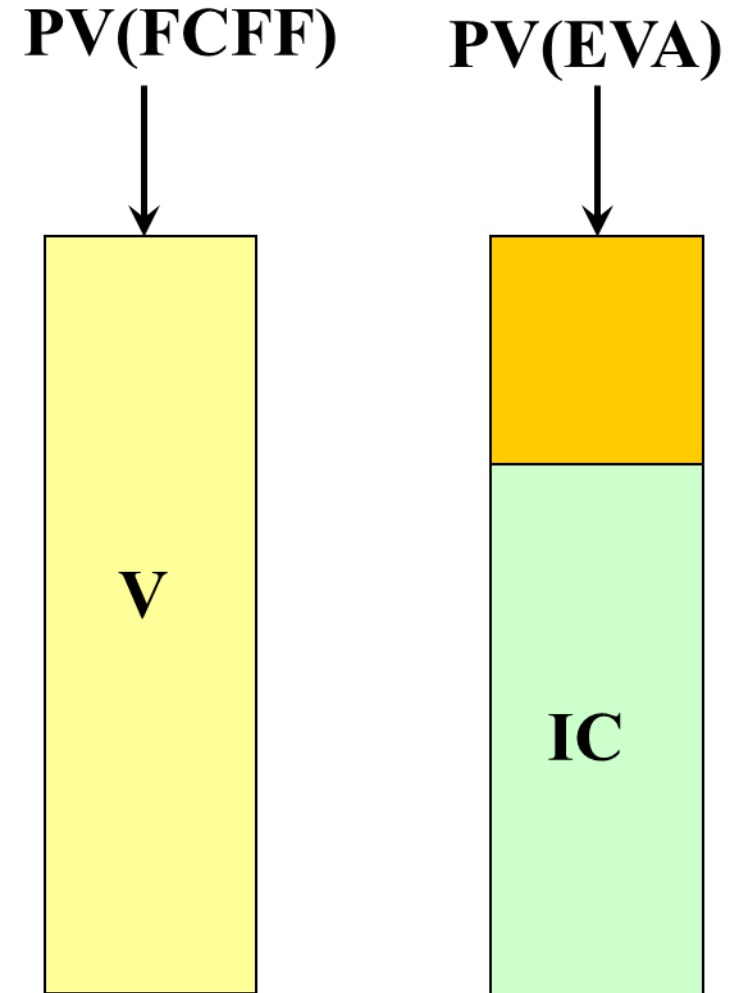


The meaning of TV for different methods

FCFF/FCFE: The value of firm/equity at the start of the TV period.

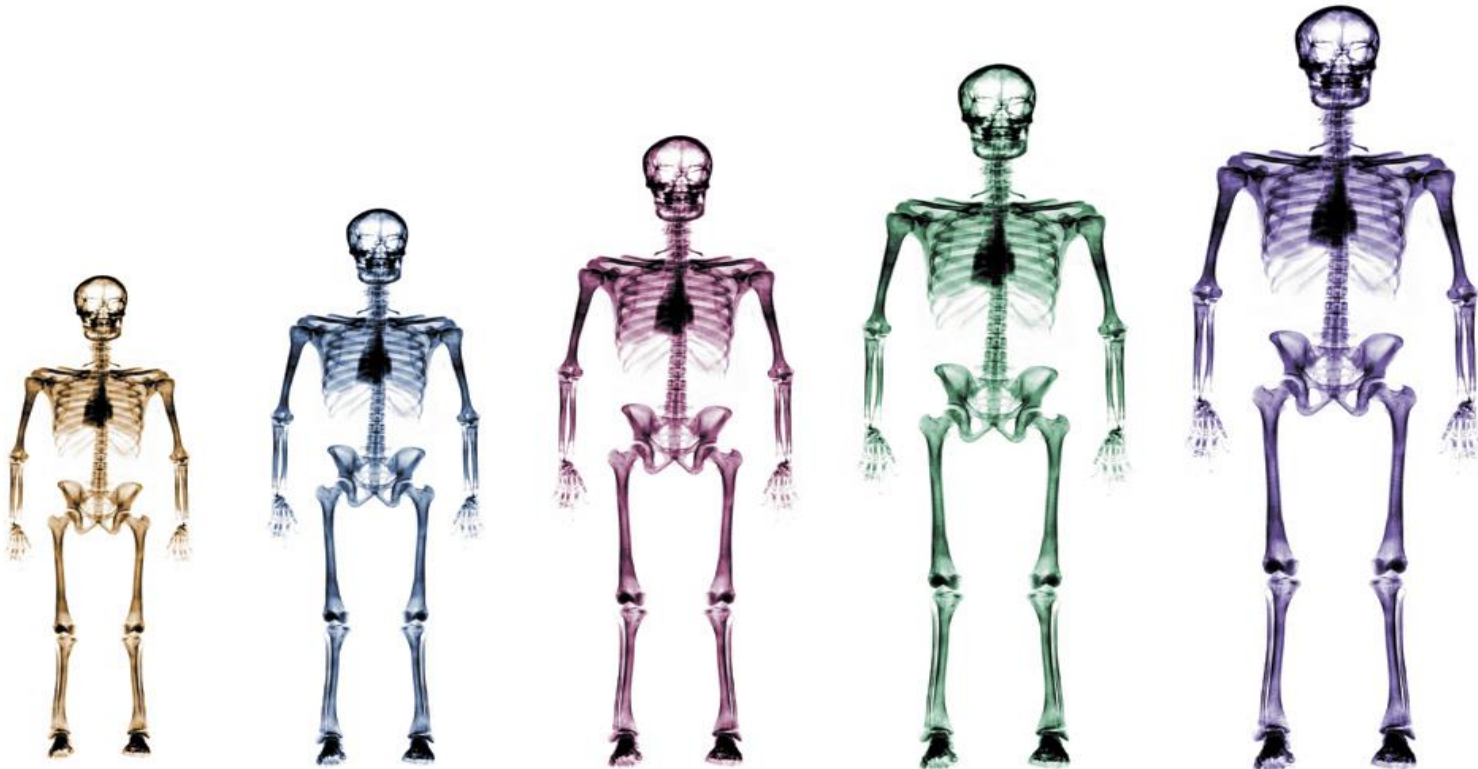
To get this value you have to invest IC/book Equity, so that is there without any risk.

EVA/RI: The added value created by the firm during the TV period.



Lessons of TV 1

The length of the explicit period should have no value effect.
We should extend the explicit period at least until all inputs reach their stable level.



Lessons of TV 2

Even in case of $RONIC=WACC$ it is not the same what is the ROIC at the beginning of the TV. If $ROIC \gg WACC$, there will be an economic profit generated for very long time.



Lessons of TV 3

Only the TVs of EVA/RI models show what part of the current value would be generated during the TV period.



Estimating the cost of capital

Péter Juhász, PhD, CFA



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Matching the CF and the cost of capital

Currency

**Effects of
financing
(TS)**

Inflation

Maturity

Risk

Estimating the required rate

Return of similar investments

How is that similar?

How do we know it is similar?

Measuring risk, calculating required rates

CAPM

Alternative methods (APT)



Whose risk is to be measured?

Investor	Risk	Metric	Requirement	Firm value
Private investor (one firm)	Total	Std deviation	40%	$100/0.40= 250$
Venture capital (one sector)	Sector portfolio +	Sector index beta	25%	$100/0.25= 400$
Hungarian investment found	Country portfolio +	Country index beta	15%	$100/0.15= 666$
Investment bank	Global portfolio +	Global index beta	10%	$100/0.10=1000$

Measuring the cost of capital

Cost of Debt

Historical risk premiums

Credit rating premiums

CAPM and debt beta

Cost of Equity

CAPM

APT

Implied return

Build-up method

Cost of capital inputs 1

Risk free rate – free only from repayment risk, other (e.g. market) risks still exist.

Real or nominal?

Which country?
(Switzerland vs Greece)

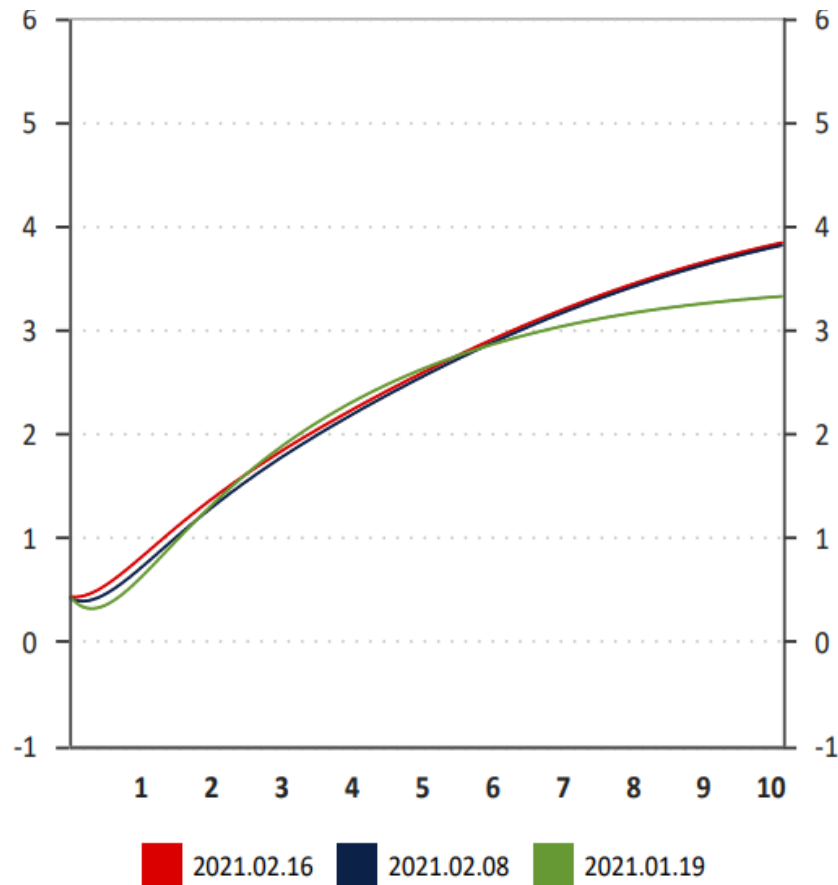
Yield curves
(spot/forward, never constant)

Exchange rate predictions
(parities)

Consistent with inflation predictions?

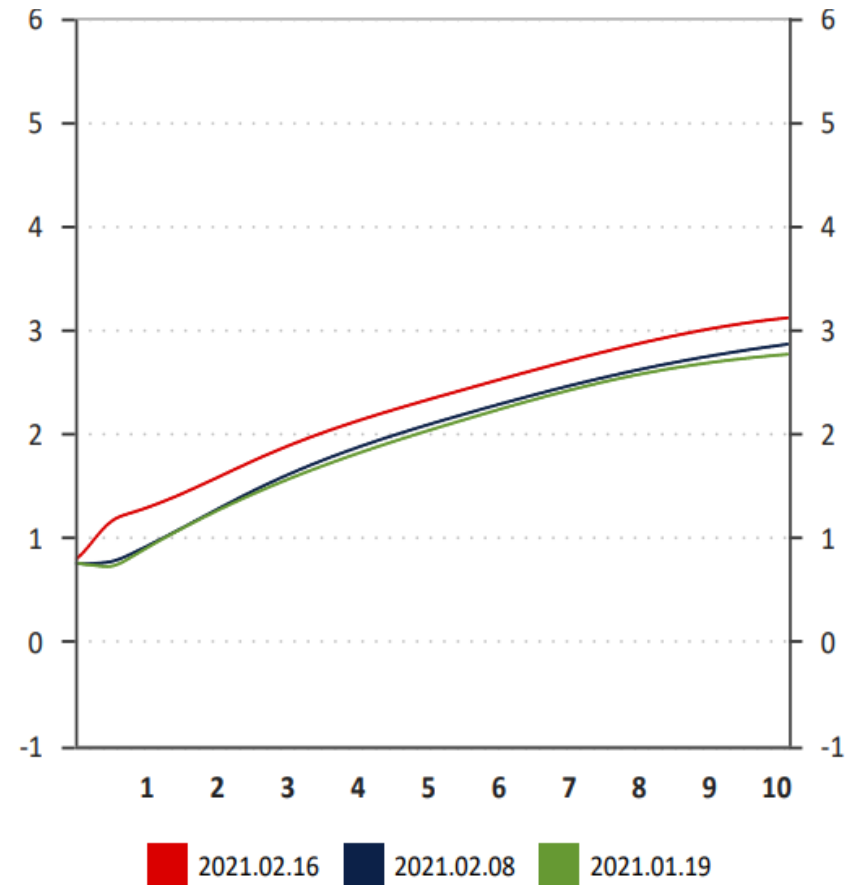
Estimating the forward rates

3-months gov. bonds



Forrás: MNB, MÁK
Megjegyzés: az MNB által Svensson módszerrel becsült kamatok. A vízszintes tengely lejáratokat mutat, években.

3-months interbank rates



Megjegyzés: az MNB által, bankközi pénzügyi (FRA) és kamatswap jegyzések felhasználásával, spline módszerrel becsült kamatok. A vízszintes tengely lejáratokat mutat, években.

Cost of capital inputs 2

Cost of **Foreign Capital**:

Former and
existing debt
contracts

Implied debt return
of own or similar
firms' bonds

Historical
regressions
(β_D)

Credit rating
(risk free rate +
premium)

Cost of capital inputs 3

Tax rate

Nominal (if no information available)

Effective (historic)

Marginal (extensional investments)

Combined (growing firm)

We need predictions.

Cost of capital inputs 4

Cost of Equity

Interview (What do owners want?)

Expert opinion (required rates for other firms)

Regression methods: CAPM, APT

Built-up method

Cost of capital inputs 5

Cost of equity contains two types of risk:

Risk of operation:

$$r_A = r_f + \beta_A(r_M - r_f)$$

Risk of financing:

$$\beta_E = \beta_A + \frac{D}{E}(\beta_A - \beta_D)$$

$$r_E = r_A + \frac{D}{E}(r_A - r_D)$$



$$r_E = r_f + \beta_E(r_M - r_f)$$

β estimation in practice 1

Historical regressions:

$$r_{\text{Asset}} = a + b * r_M$$
$$b \text{ (sloop)} = \text{covar}(r_{\text{Asset}}, r_M) / \sigma_M^2 = \text{„}\beta_E\text{”}$$

Market
portfolio?

Time frame?

Yield type?

Currency?

Exchange
rate
corrections?

β estimation in practice 2

Fundamental beta

- Industry (cycles, basic vs luxury) – „ β_{industry} ”
- Operative risks (cost structure: variable/fixed cost, $\Delta \text{ EBIT} / \Delta \text{ Sales}$ – DOL) – „ β_{strategy} ”
- $\Phi(\beta_{\text{industry}}, \beta_{\text{strategy}}) \Rightarrow$ „ β_A ”
- Financing risks (taxation, D/E) $\Rightarrow$ „ β_E ”

Cost of capital inputs 6

Leverage:

Historical
data

Predicted
trend

Target
(announced)
level

Iteration

Estimating the cost of equity using the build-up method

Estimating required rate on equity:

$$\epsilon(r_A) = r_f + \text{MRP} + \text{RP}_I + \text{RP}_S + \text{RP}_O$$

where

$E(r_E)$ = required rate of equity,

r_f = risk-free rate of return,

MRP = market risk premium,

RP_I = industry risk premium,

RP_S = size risk premium,

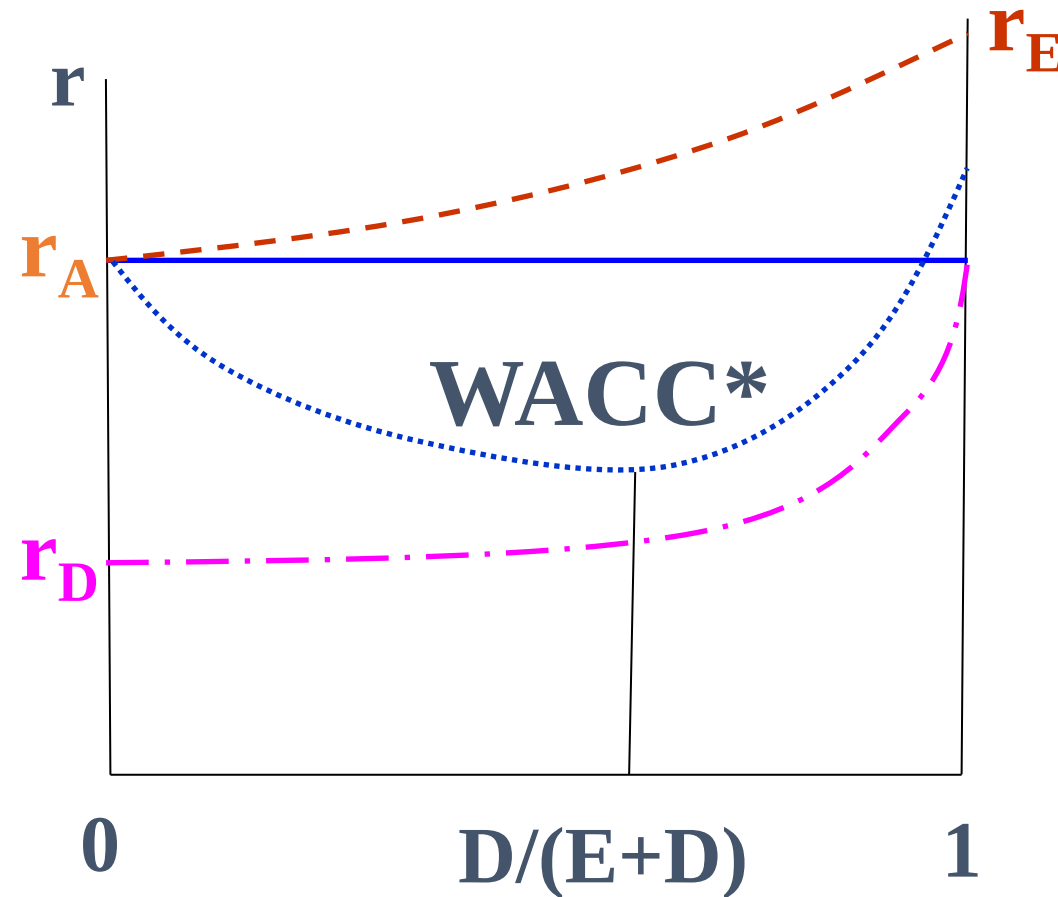
RP_O = other firm specific risk premium.

Government bond rates of developed countries	1.0%
Country risk premium of Hungary	1.5%
Inflation difference	1.0%
Risk free rate (r_f)	3.5%
Market risk premium (MRP: $r_M - r_f$)	5.5%
Industry premium	1.5%
Size and other firm specific premiums	2.0%
Required cost of assets (r_A)	12.5%

No need for betas.

$$WACC = \frac{D}{V} * r_D * (1 - t) + \frac{E}{V} * r_E$$

But this formula misses the effects of financial distress.



* With financial distress effects

Can you reach the optimum?

Curves are not continuous,

Only your direct neighbourhood can be tracked.

The amount of debt could be limited.

Curves may deviate across financiers.

Curves change over time.



Special challenges of DCF-based valuation

Péter Juhász, PhD, CFA

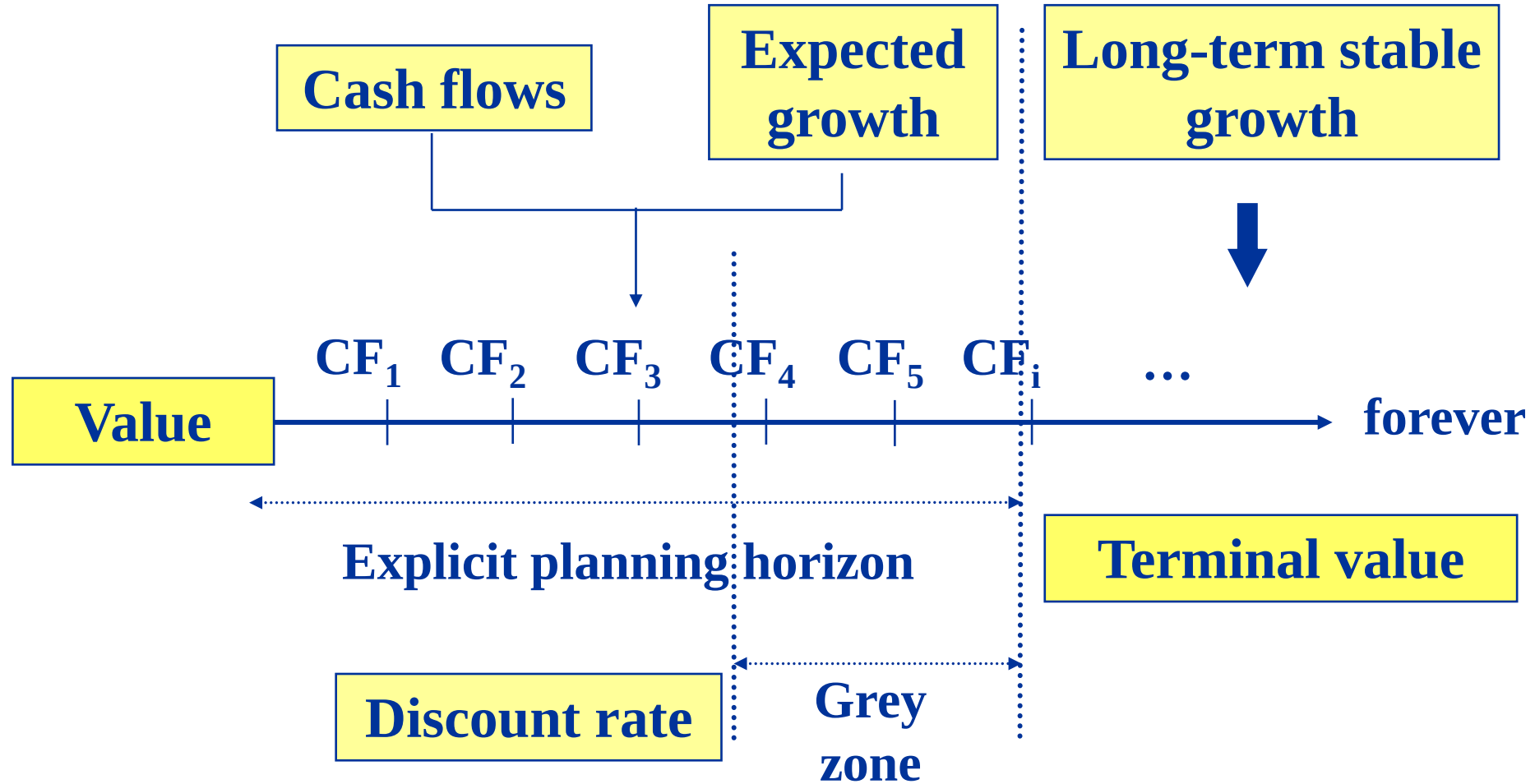


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The general DCF model



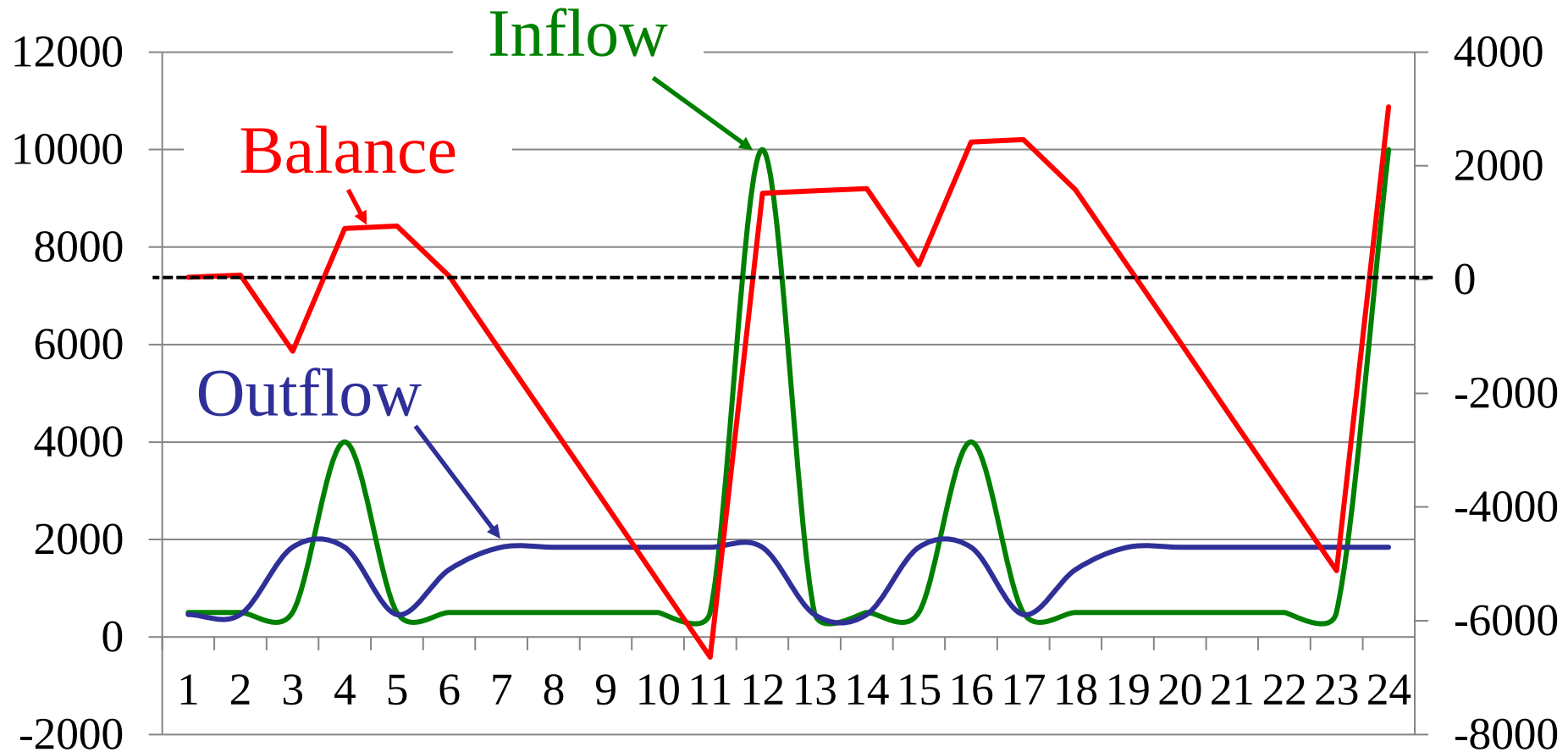
Seasonality

Are all CFs due
at the same
point in time?

Is that point the
end of the
year?



Seasonality effect on cash flows



At a rare of 1% the multiplier is 87.32%.

Dealing with choices

Decision trees: some alternative future paths, probabilities can be estimated, more realistic models

Real options: opportunities that gain value for an increase in risk.

Poor projects are often offered for sale as real options.

Monte Carlo simulation

Inputs are estimated as a distribution not as a single value.

The end result is a range (distribution) not a discrete value.

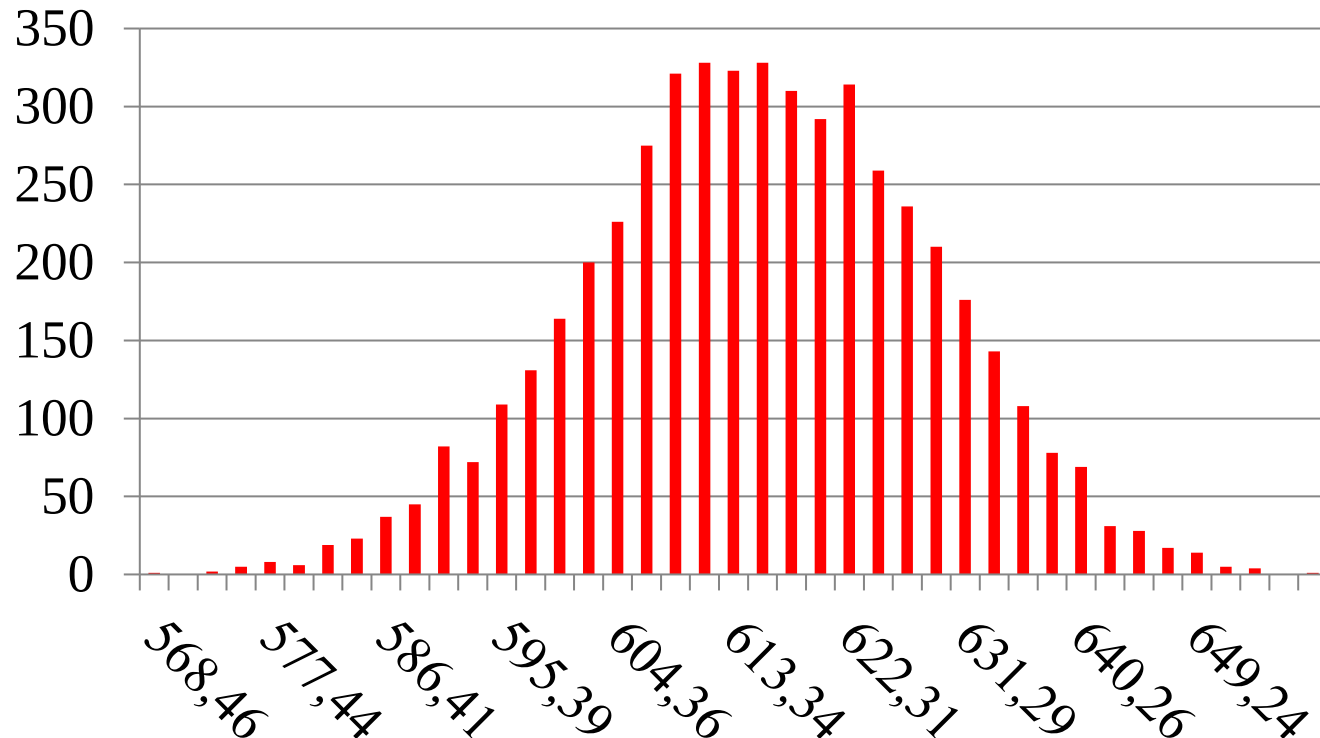


The value of an annuity

N=10 years CF: 90-110, unique distribution

R_{forward} : 10%, normal distribution, 0.5% standard deviation

Value calculated based on means: 614.46



Min	568.74
Max	662.34
Average	614.47
Range %	15.23%

Multi-business companies

Business units have different risks so CFs can not be merged

Invested capital and required rate for Bus can often be estimated only.

We have to address the transfer prices

What would be the market price?

HQ may have a negative value

Is there any synergy? Where does it come from?

Valuing a multi-BU firm 1

Perpetuity-like FCFF=100, $g=5\%$, $D/V=50\%$, $r_D=5\%$ $t=20\%$, $r_A=10,5\%$, thus: $WACC=10\%$.

Business value:

$$V = \frac{FCFF}{(WACC - g)} = \frac{100}{(10\% - 5\%)} = 2000$$

Valuing a multi-BU firm 2

BU A FCFF=30, BU B FCFF=90, HQ FCFF=-20.

A superficial estimation:

$$V_A = \frac{30}{(10\% - 5\%)} = 600$$

$$V_B = \frac{90}{(10\% - 5\%)} = 1800$$

$$V_K = \frac{-20}{(10\% - 5\%)} = -400$$



Valuing a multi-BU firm 3

Detailed analysis:

	A	B	HQ	Average*
FCFF	30.00	90.00	-20.00	100.00
g	7.00%	4.33%	5.00%	5.00%
r _A	10.00%	10.66%	10.50%	10.50%
WACC	9.50%	10.16%	10.00%	10.00%

*FCFFs used as weights.

Valuing a multi-BU firm 4

Enhanced valuation:

$$V_A = \frac{30}{(9,5\% - 7\%)} = 1200,00$$

$$V_B = \frac{90}{(10,16\% - 4,33\%)} = 1543,74$$

$$V_K = \frac{-20}{(10\% - 5\%)} = -400,00$$

$$\mathbf{V=2343.74}$$

Valuing a multi-BU firm 5

If BU A would be a stand-alone firm, its FCFF were 20, while that of the HWQ -15 (synergy lost):

$$V_A = \frac{20}{(9,5\% - 7\%)} = 800,00$$

$$V_B = \frac{90}{(10,16\% - 4,33\%)} = 1543,74$$

$$V_K = \frac{-15}{(10\% - 5\%)} = -300,00$$

Valuing a multi-BU firm 6

At what price would we sell BU A?

$$V_A = V_{A+B+K} - V_{B+K}$$

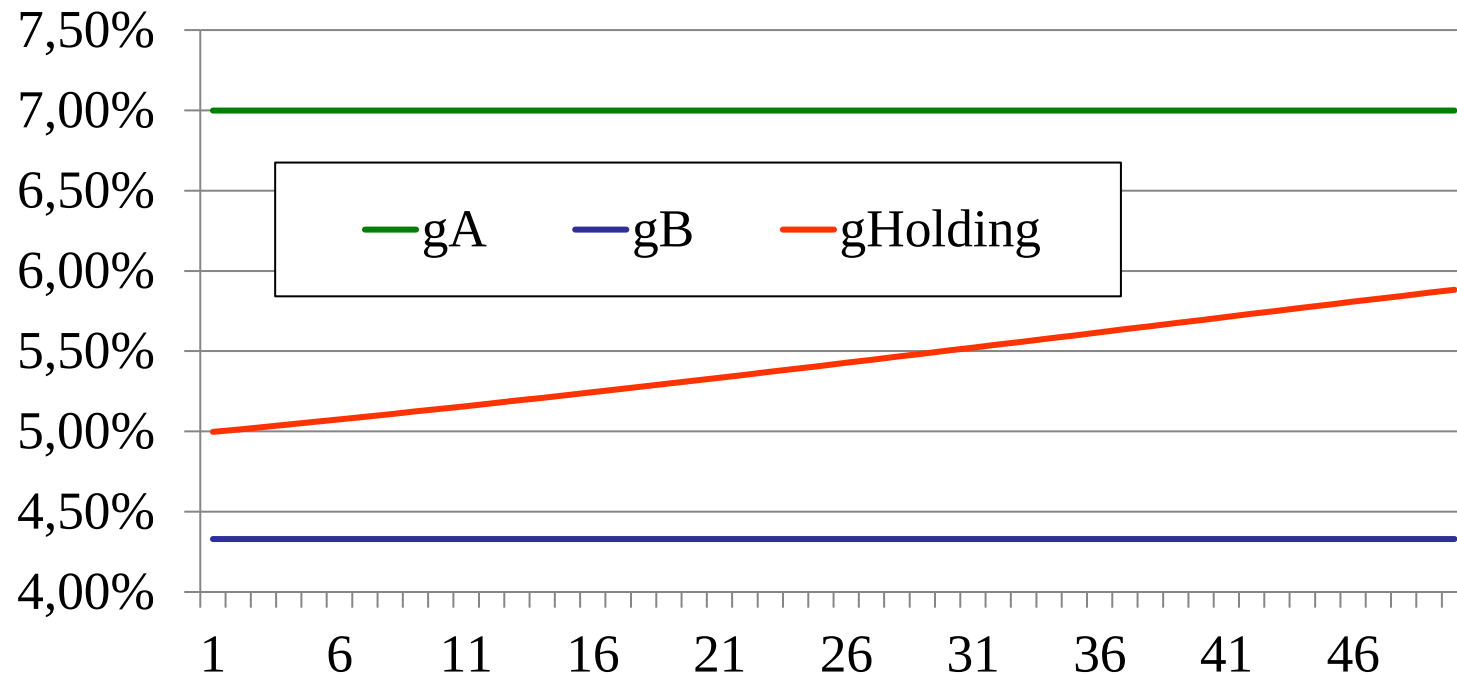
**We do not need the
value of BU A.**

$$V_A = 2343,74 - (1543,74 - 300) = 1100$$

This is the value of the BU decreased by the costs emerging at the HQ.

Valuing a multi-BU firm 7

What is the actual growth rate of the firm if there would be no change in the operation?



Perpetuity-like Bus do not necessary add up to a perpetuity.

Effects of the inflation

Inflation is not value neutral.

1. Not all items increase with inflation: D&A
2. Not all items are driven by the same inflation: selling price, procurement price, energy, wages, import, export
3. Effect of inflation may appear delayed: inventory



**Valuation may only be done
based on nominal plans.**

Exchange rate risk

1. Market hedging (forward, swap, option)
2. Changes in exchange rates due to inflation rate differences are not to be managed, only purchase parity deviations should be addressed
3. Corporate diversification across markets
4. Investor diversification

Other valuation issues 1

For what date do we value? What if we have to value for the middle of the year?

Firms are not bought only in January.

Valuing financial service companies

We can not separate supplier and creditor, operation and financing,
Only FCFE method can be used, Business planning and statements are completely different

Consistency and business rationality

Other valuation issues 2

Sensitivity

Change in value for possible input values (estimation error vs variability of the environment)

Value range

+/-15% range is standard. (USA: historical yearly volatility of share prices was 25%, thus monthly volatility is around 15%.)

The valuation process

1. Determining the value of operation (V_{op})
2. Valuing non-operative assets (NOA)
3. Calculating total value ($V = V_{op} + NOA$)
- 4. Valuing foreign capital (D)**
5. Determining the value of Equity ($E = V - D$)
6. Appraisal of stake or share

Valuing liabilities 1

This category here is wider than in accounting

Bank loans

Bonds

Rental agreements

Lease

Convertible bonds

Management
options

Minority interests

Preferred equity

Pension funds

Legal processes

Environmental
obligations



Valuing liabilities 2

Debt:

Generally at book value (if a variable rate debt)

Fixed rate debt and subsidised debt calls for a separate valuation (May influence WACC calculation!)

In case of financial difficulties, the value of debt is directly linked to the value of the firm

Non-operational debt should not be considered in WACC

Valuing liabilities 3

Other liabilities:

Off-balance sheet items

(e.g. legal processes) extent, probability, date

Dividend preferred shares

Similar to uncovered debt

Options

(management options, warrants, convertible debt) Option valuation, value may depend on the value of the firm (circular reference)

Minority interests

Needs valuation of affiliates

Limits of applying the DCF approach 1

Loss-making firms

(the future has to be different, so available information is not relevant)

- Negative CF
- Growth rate is hard to estimate
- Tax calculation is challenging
- Continuous operations is not realistic (bankruptcy)



Causes behind the loss

Firm specific issues

Litigation, fine, natural disaster, product recall

Industry specific issues

Fall in price, lack of raw material, decrease in demand

Business cycles

Cars, luxury items, home building, chemicals

Macro effects

Crises, wars, global trends

Sorting out problems

Poor strategy (product mix, pricing)

New management, training, consultants

Poor operational efficiency

Re-engineering, lean systems, new technology, training

Financing decisions

Refinancing, renegotiating debt

Life cycle issues

Creating new BUs, R&D, inheritance

Limits of applying the DCF approach 2

Young or new companies

(the necessary information is not existing)

Probabilities only $\Rightarrow$ Venture capital

Differences: frequency of information service,
required rate depends on the VC firm

Liquidity discount, extra risk due to fast growth



Limits of applying the DCF approach 3

Private firms

(the necessary information is not available)

More lax reporting rules than elsewhere
(comparability?)

Less information shared
(Auditor? Analysts?)

Exit is harder

No market data (competitors data, betas)

Management and ownership overlapping
(salary vs dividend, non-operational assets)



Further issues 1

Owners less diversified

Family businesses

Cost of debt

Less information → higher debt rate, no credit rating → bond issue difficult

Lack of comparable data

Industry risk is hard to measure

Cost of equity

No historical share price for CAPM, risk premium for less liquidity?

Further issues 2

CF estimation difficult

Labour cost should be contrasted to market data

Business and personal expenses may mix

Detailed review may be needed

Growth is hard to estimate

No analysis, historical data blurred, many pieces of information are not public, lack of data collected

Firms may not exist for ever

Expiring licences, changing market and personal needs, illness, getting older, inheritance

A DCF „alternatives”

1.

Use non CF data
(Dividend, net profit)

2.

Use data from other firms
(Multiple-based methods)

3.

Use alternative approaches
(going concern vs recreation or liquidation)

Multiple-based valuation methods

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Multiple-based methods



Comparable transactions

Based on **earlier actual take-over transactions**

Distortions

The markets change over time
Synergic premiums



Market multiples

Based on **current stock exchange quotes**



DOW J	15,075.18	- 43.31
NASDAQ	3,436.63	+ 0.05
AAPL	456.95	+ 3.98
S&P 500	1,632.57	- 1.13
GOOG	878.04	- 2.19
FTSE 100	6,631.76	+ 6.78

Apple Inc	
Open: 451.53	Mkt Cap: 428.3B
High: 457.90	52w High: 705.07
Low: 451.50	52w Low: 385.10
Vol: 7.544M	Avg Vol: 17.94M
P/E: 10.89	Yield: 2.70%

Live Quote

Distortions

Market efficiency
Minority discount

A simple example

What is the value of a book store with a sales of 200 M HUF if another book shop of 150 M HUF revenue has recently been sold for 160 M HUF?

$$P_1/S_1=160/150=1.07$$

$$P_2=1.07*200=\underline{213.3 \text{ M HUF}}$$

Pros and Cons



Quick
Simple
Reflects the current
market conditions

Implicit assumptions
Quality of data
Comparability
Bubbles
Any value could be justified



Multiple dictionary

In multiples	In valuation
P	Price of one single share
$P/E = \text{Price} / \text{EPS}$	E / Profit after tax
EV (Enterprise value)	$E + D - \text{Cash}$
EV/BV (Book Value)	$(V - \text{Cash}) / \text{IC}$
P/BV	E / Book value of equity
$P/S = \text{Price} / \text{Sales per share}$	E / Sales
EV/S	$(V - \text{Cash}) / \text{Sales}$
CF	Operative CF (usually)

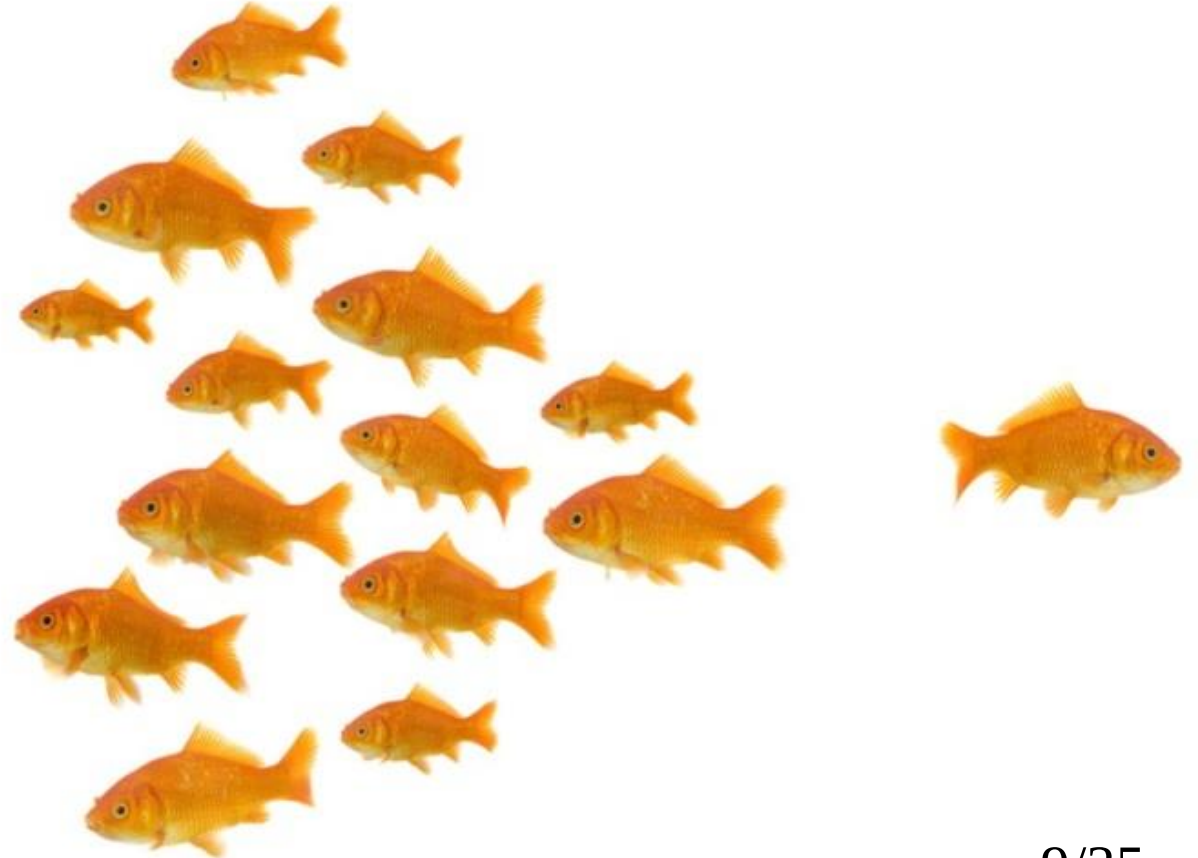
Grouping the multiples

**Profit-
based**

**Revenue-
based**

**Asset-
based**

**Industry
specific**



Defining a multiple

Simple

Easy-to-understand

Clear

Properly defined (last, average, expected, extraordinary)

Accessible

We have the data needed

Choosing the right ratio

Relevance

Either the numerator or the denominator should be some kind of value.

Consistency

The numerator and the denominator should be of the same dimension (V or E).

Uniformity

Same financial year, accounting standard, currency

Empirical tests for applicability

Description power:

What sample size is needed?

What is a high/low value?

Distribution of historical multiple value

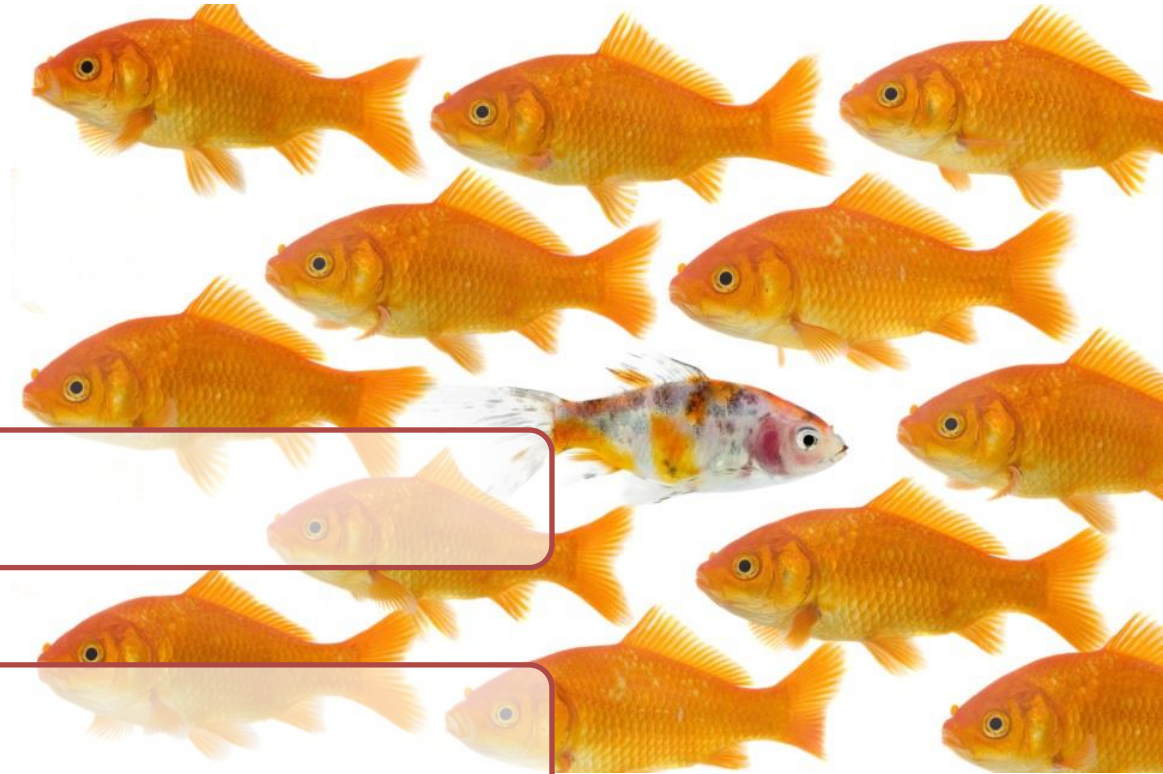
- Is the population homogeneous?

Outliers, averages

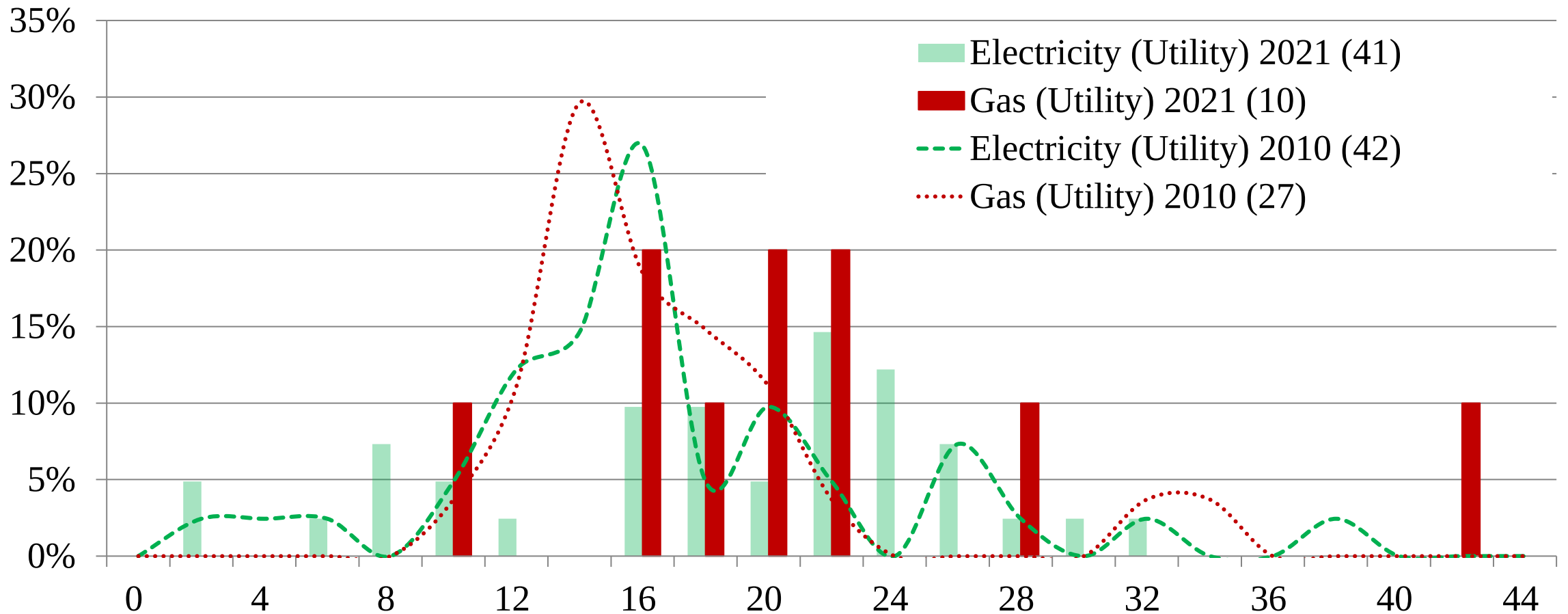
- What needs to be deleted? What would the average show?

Calculation challenges

- Negative profit or equity → distortions



P/E rates (USA, March)



Source: Yahoo! Finance

Content control

What fundamentals influence the value of the multiple?
What are the prerequisites of comparability?

$$P_0 = \frac{DIV_1}{(r_E - g)} = \frac{EPS_1 * div\%}{(r_E - g)}$$

$$\frac{P_0}{EPS_1} = \frac{P}{E} = \frac{div\%}{r_E - g}$$

Dividend pay-off ratio

Growth outlook

Risk of equity

$$\frac{EV}{EBIT} = \frac{E_0 + D_0 - Cash_0}{EBIT_1} = \frac{(1 - t) * (1 - K)}{WACC - g}$$

$$\frac{P}{S} = \frac{P_0}{Sales_1} = \frac{\frac{EPS}{Sales} * div\%}{r_E - g}$$

$$\frac{EV}{Sales} = \frac{E_0 + D_0 - Cash_0}{Sales_1} = \frac{\frac{EBIT}{Sales} * (1 - t) * (1 - K)}{WACC - g}$$

$$\frac{P}{BV} = \frac{P_0}{BV(E)_0} = \frac{ROE * div\%}{r_E - g}$$

$$\frac{EV}{BV} = \frac{EV_0}{BV(E + D)_0} = \frac{ROI * (1 - t) * (1 - K)}{WACC - g}$$

The process of multiple-based valuation

0. Pick a multiple.

1. Choose comparable firms. Similar risk, growth, profit generation
(No industry or size requirement theoretically.)

2. Control for differences. (Similar but not the same.)

Subjective corrections (weighted average)

Modified multiples (corrected for differences)

3. Check reasonability.

Industry regressions

Market regressions

Assumptions of multiple types 1

Profit-based (P/E, EV/EBIT): the same link between the right CF and the given profit.
(g, CF-generation, risk)

Revenue-based (EV/S): from a given amount of revenue any firms generates the same profit.
(g, CF-generation, risk, „margin”: CF/S)



Assumptions of multiple types 2

Asset-based (P/BV, EV/IC): the assets generate the same profit everywhere.

(g, CF-generation, risk, ROI(c)/ROE)

Industry specific (EV/employee): the given specific variable creates the same profit everywhere
(g, CF-generation, risk, „industry”: same technology and efficiency)



P/E effects of differences

$$\frac{P}{E} = \frac{div\%}{r_E - g}$$

Interest rates $\uparrow \rightarrow$ Cost of capital $\uparrow$, P/E $\downarrow$

Risk taking willingness $\uparrow \rightarrow r_E \downarrow$, P/E $\uparrow$

General market outlook $\rightarrow$ average $g \uparrow$, P/E $\uparrow$

Profitability $\uparrow \rightarrow div\% \uparrow$, P/E $\uparrow$

Do not forget that in the long run

$$ROE * (1-div\%) = g^*$$

Interpreting P/E 1

High P/E:

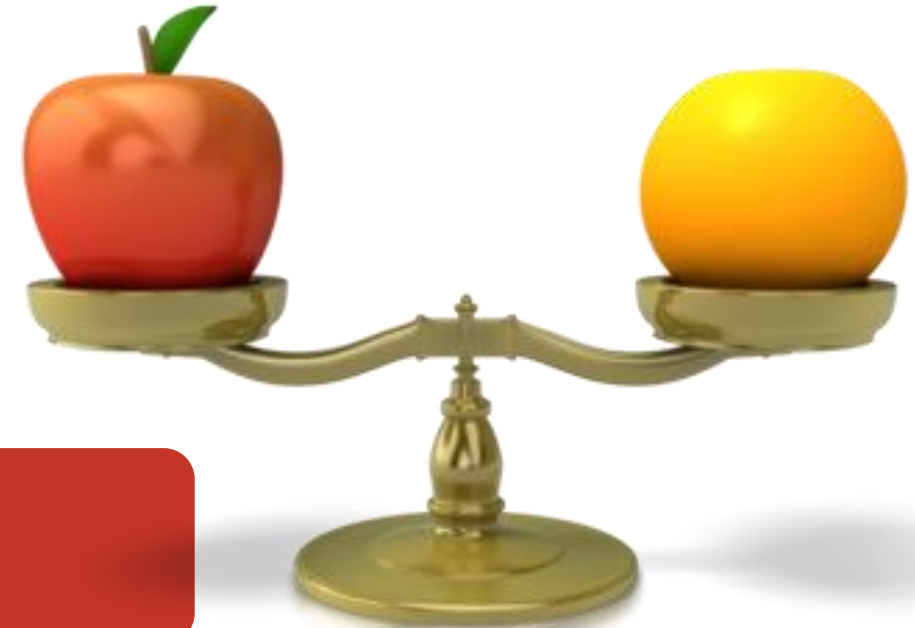
High dividend pay-off

Low r-g

- r big & g huge
- r small & g tiny

P/E/g (PEG)

$$\text{Relative P/E} = (P/E_{\text{Firm}})/(P/E_{\text{Industry}})$$



Interpreting P/E 2

		PEG	
		Low	High
P/E	Low	r high, g low	r high, g very low
	High	r high, g high	r low, g low

Divided payoff ratio?

Only true for firms with the same dividend policy.

Revenue-based multiples

Easy to use
Always positive
No accounting
issues

Very far from the
CF

P/S is inconsistent.

$$\frac{EV}{S} = \frac{\frac{EBIT}{Sales} * (1 - t) * (1 - K)}{WACC - g}$$

Asset-based multiples 1



Stable, easy-to-compare,
loss making firms create no issue



Accounting policy differences,
off-balance sheet items, $BV(E) < 0$

P / BV(E)

$$\begin{aligned} P_0/BV_0 &= ROE * (1+g_n) * OFH / (r-g_n) = \\ &= ROE * P/E \end{aligned}$$

Asset-based multiples 2

$$\underline{EV / BV (D+E) (MV/BV)}$$

$$= ROIC * (1-K) / (WACC - g)$$

$$ROIC = NOPLAT / BV(D+E)$$

$$\text{Stability: } ROIC * K = g \rightarrow$$

$$K = g / ROIC$$

$$EV/BV = (ROIC - g) / (WACC - g)$$



James Tobin
(1918-2002)

Asset-based multiples 3

Tobin's Q

= Market value of Assets / Replacement value of Assets

Original use: When shall we purchase a stock?

NPV (added value) ratio

Importance of off-balance sheet items

Hard to use for valuation (no E or EV in the formula)

Industry specific multiples

Value creation linked to operation

No accounting issues

No further information needed

Same technology and efficiency?

Examples

Number of subscribers, readers, visitors, buyers, patients, quantity produced, owned resource confirmed, sales area, number of rooms

EV/EBITDAR	Retail, airlines	Without rental fees and lease
EV/EBITDAX	Oil and gas, mining	Without exploration expenses
EV/Resource	Oil and gas, mining, lumber	
EV/Capacity	Oil and gas refinery, hotels, restaurants	
EV/Production (Service)	Oil and gas, harbours, airports	
EV/Subscriber	Telecom, media, internet services	
P/tBV	Banks	Tangible book value

A real life example 1

BANKS

	Price (local currency)	2013	P/E 2014F	2015F	P/tBV 2014F	2015F	ROE 2014F	Div. yield 2014F
OTP	4,285	18.7	-11.5	11.1	1.0	1.0	-7.3	3.9
Erste Bank	23.4	158.3	-6.7	12.9	1.2	1.1	-14.2	0.0
FHB	797.0	n.a.	n.a.	95.9	3.1	3.1	-22.2	0.0
PKO BP	32.6	12.6	12.1	11.6	1.6	1.5	12.7	4.5
Pekao	185.55	17.5	16.7	15.5	2.1	2.1	12.4	5.4
Raiffeisen	13.5	4.7	-10.6	8.7	0.5	0.4	-3.9	0.0
Bank Handlowy	106	14.6	15.8	14.3	n.a.	n.a.	11.8	6.2
Komerční Banka	5,304	15.5	15.7	14.9	1.8	1.6	12.4	5.6
Unicredit	5.95	16.1	12.4	10.0	0.7	0.7	5.3	2.7
Intesa	2.98	36.9	16.1	13.0	n.a.	n.a.	6.5	3.8
Peer group's average		34.5	11.2	20.8	1.5	1.4	1.3	3.2
Peer group's average excl. FHB		34.5	11.2	12.4	1.3	1.2	4.0	3.6

Source: Bloomberg, Concorde estimates (OTP, Erste Bank, FHB, PKO BP, Pekao, Raiffeisen)

A real life example 2

UTILITY

	Price (local curr.)	2013	P/E 2014F	2015F	P/BV 2014F	EV/EBITDA 2014F	2015F	Div yield 2014F
CEZ	621.1	9.2	11.2	16.9	1.2	7.2	8.6	5.2
E.ON	14.3	n.a.	16.3	16.5	0.9	5.4	5.7	3.5
RWE	25.0	n.a.	11.5	11.8	2.2	4.5	4.5	3.9
Enel	4.1	13.4	12.9	12.8	1.1	7.0	7.1	3.2
Endesa	18.1	64.9	18.3	17.4	2.2	7.8	7.7	4.9
Iberdrola	6.1	16.5	16.2	15.2	1.1	8.9	8.6	4.5
EdP	3.5	15.9	13.7	14.0	1.5	9.5	9.5	5.4
EdF	24.7	12.4	11.9	12.2	1.3	5.0	5.0	5.1
PGE	20.7	10.6	11.8	12.3	0.9	5.2	5.1	4.3
Tauron	4.9	7.5	7.2	8.3	0.5	4.1	4.2	3.6
Enea	16.2	8.1	8.3	10.6	0.6	4.3	4.5	3.7
Peer group's average		17.6	12.7	13.5	1.2	6.3	6.4	4.3

Source: Bloomberg, Concorde estimates (CEZ)

Source: Concorde, 2 March 2015

A real life example 3

SOTP VALUATION

	EV/Share [HUF]	EV [HUF million]	Multiple	EBITDA [HUF million]	Margin Forecast
Hungarian Fix	257	267,504	4.7x	56,916	33%
Hungarian Mobile	549	572,508	5.5x	104,092	36%
SI/IT	127	132,030	9.0x	14,670	22%
Macedonia	69	71,915	3.8x	18,925	38%
EV	1,001	1,043,957	5.8x	194,603	32%
Net debt	-297	-309,976			
Minority	-47	-49,317			
Dividend	-25	1,043			
Spectrum CAPEX	-91	-95,000			
Fair value	540	590,707			
Upside w dividend	27%				

REGIONAL PEERS

	EV/EBITDA	EBITDA m
Hrvatski Telekom	3.8x	39%
DIGI	4.0x	31%
Telekom Slovenije	3.8x	29%
Telekom Austria	5.5x	32%
Hellenic Telecom	4.6x	32%
Orange Polska	4.6x	25%
Netia	6.3x	27%
Cyfrowy Polsat	6.9x	37%
MegaFon	4.1x	32%
Rostelecom	3.6x	31%
Turk Telekom	4.8x	34%
AVG	4.7x	32%

Source

Source: Concorde, 2 March 2018

Correcting multiples

What if not all similarity requirements are fulfilled?

Recalculated multiples

Our firm differs from the average (g , r , CF , β).
Is that maintainable in the long run?

Discounts & premiums

The peers are different in a given perspective
(other markets, countries, liquidity).

Recalculated multiple

The average forward P/E of European food wholesalers was **15.2** at the beginning of 2019. (Div%=93.07%, g=7.30%*) The firm to value has the same risk, but div%=80%, g=8%.

$$\text{div}\% / \text{P/E} = r - g = 93.07\% / 15.20 = 6.12\%$$

$$r - g_{\text{mod}} = 6.12\% - (8.00\% - 7.30\%) = 5.42\%$$

$$\text{P/E}_{\text{mod}} = 80.00\% / 5.42\% = \mathbf{14.76}$$



*Source: Damodaran Online, 2019

Discounts & Premiums

The operator of a 20-room hotel with unique traits in its region is to value. Similar quality hotels quoted at the stock exchange at 8 M HUF/room.

$$20 * 8 = 160 \text{ million HUF}$$

Long-run competitive advantage: + 10%

Illiquidity of the investment: – 30%

$$160 * 1.1 * 0.7 = \underline{123.2 \text{ million HUF}}$$



Typical mistakes when using multiples

- 1 Description power of multiples not checked.
- 2 Calculation of multiples is not standardised.
- 3 Peers are competitors rather than similar firms.
- 4 Necessary discounts not applied.

Asset-based valuation

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What we still have to cover ...

$$V = V_{OP} + V_{Other}$$

Typical „other” assets

Surplus cash

„Cash not needed for the operation”

Securities

Stocks, bonds, options. Use the market price.

Investments

Strategic: synergies, consolidation

Unused assets

Assets not used by own operation

Over- and underfinanced pension plans

What makes an asset? - Accounting

Owned by the
firm

Expected
incomes exceed
expected costs

Acceptable
uncertainty

What makes an asset? - Finance

Assets generating cash flows

DCF

Assets not generating cash flow

Replacement value, comparable assets

Option-like assets

Real options

Quasi assets

Asset manageable but not ownable

Asset valuation methods

Type of value	Time orientation of data used		
	Past	Present	Future
Entry	Purchase cost	Substitution value (repurchase, recreation) Replacement value	—
Exit	—	Net realisable value, Liquidation value	DCF, Capitalisation, Real options

Based on Wilson (1986)

What is my warehouse worth?

We have bought it
five years ago for
5 M HUF.

It was renovated
last year for
2 M HUF.

Current book
value is
5.6 M HUF.

A similar one can
be bought for
5.9 M HUF in the
neighbourhood.

Last week,
6 M HUF has been
offered for it.

We could build a
similar one over
18 month for
7.1 M HUF.

The present value
of future rent
income is
7.4 M HUF.

Valuation issues 1

1. Current assets

- a) Non-operating cash (missing from the DCF value)
- b) Securities (revaluation)
- c) Receivables (payment risk)
- d) Other lending (will we get that back ever?)
- e) Inventory (FIFO, LIFO, guaranteed purchase price)

Value of cash and securities

Generally, these investment have an NPV of zero, but those could be worth less

Double taxation

Investments below required rate (effective market guarantee zero NPV only at the time of the purchase)

Investor anxiety about poor M&As (e.g. take-over protection)

Valuation issues 2

2. Invested assets

Intangibles: R&D, brand name, franchise, achieves, lists, transferable contracts (purchased vs created, going concern vs liquidation)

Financial assets (futures, forwards, options, warrants, convertible bonds, convertible shares, swaps)

Interests in other firms

Valuation could be challenging (no data):

- market price

- relative pricing

- simplified DCF (target firm cost of capital needed)

May be valued below their market value ...

Information gaps (structure not transparent)

Market scepticism

Is the mother company able to manage the cross-owned portfolio optimally? (The holding discount may even amount to 5-10%.)

Valuation issues 3

2. Invested assets (cont.)

c) Machinery

Real obsolescence (vs D&A):

- 1, economic (market situation),
- 2, operational (wear & tear),
- 3, functional (technology)

d) Real estates (long useful lifetime, obsolescence, market need, inflation vs D&A)

Consider tax effect if asset sale is assumed.

Valuation issues 4

3. Liabilities

Payables (for each BU)

Lease (maintenance cost as expense)

Non-consolidated affiliates

Liabilities to employees

Shareholder loans

Valuation issues 5

3. Liabilities (cont.)

Pension funds (asset or liability)

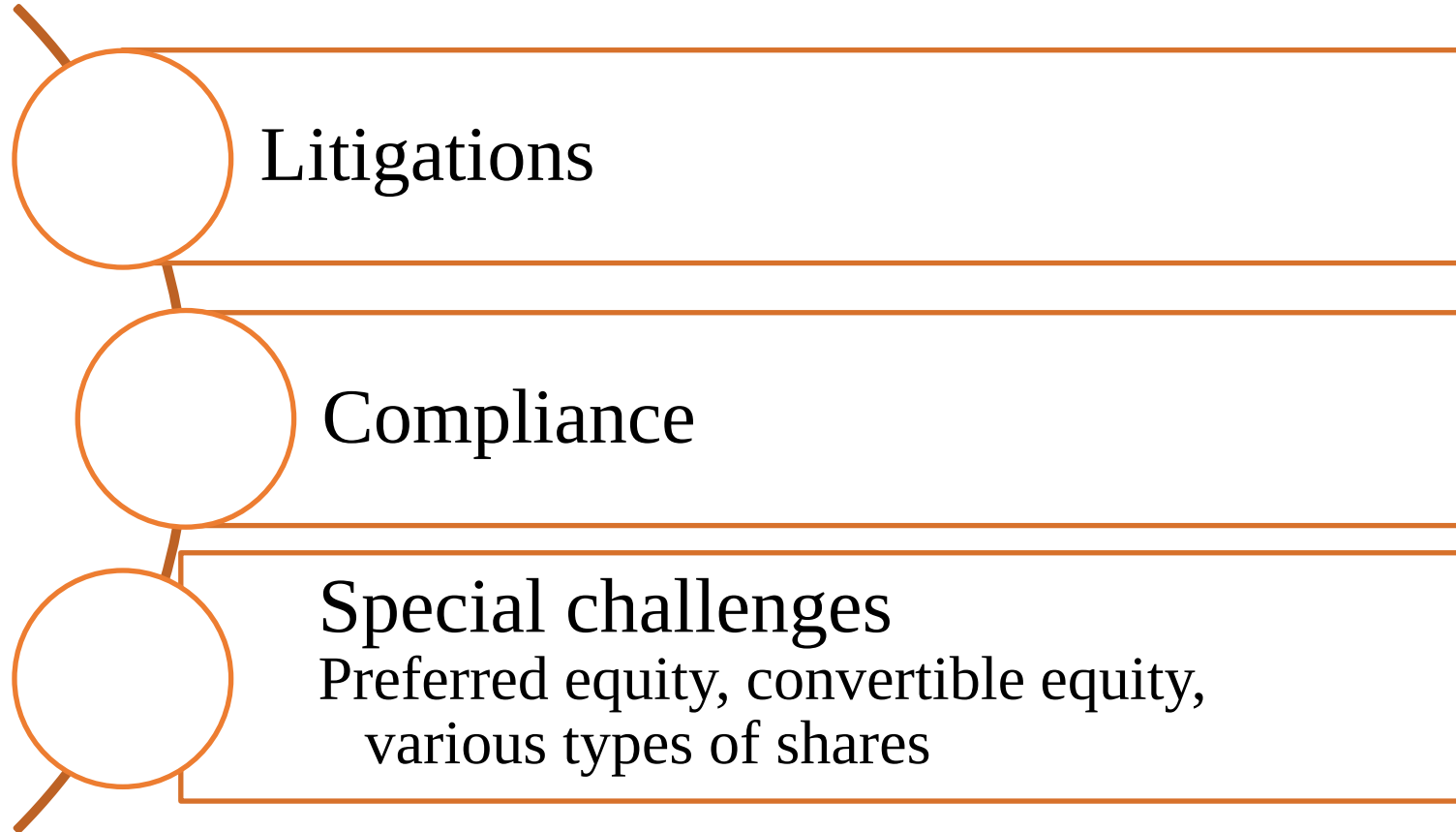
Postponed tax payments (equity or liability?)

Guarantee, warranty, suretyship

Environmental obligations

Valuation issues 6

3. Liabilities (cont.)



Empirical experience

The aim of asset valuation is to estimate the amount of the invested capital.

Valuation methods used to estimate capital employed by business units

Net book value	87%
Replacement value	4%
Gross book value	3%
Market value	1%
Other	5%

Source: Arnold-Davies (2000, p. 159.)

Off balance sheet items

**Proportion of book value in % of the market value
for US listed firms**

1978	95%
1988	28%
2002	< 20%

Source: Personnel Today, 2002

Perceptions of value

Accounting value

Past oriented

Based on documents

Rule of prudence

Aims to inform
outside stakeholders

Business value

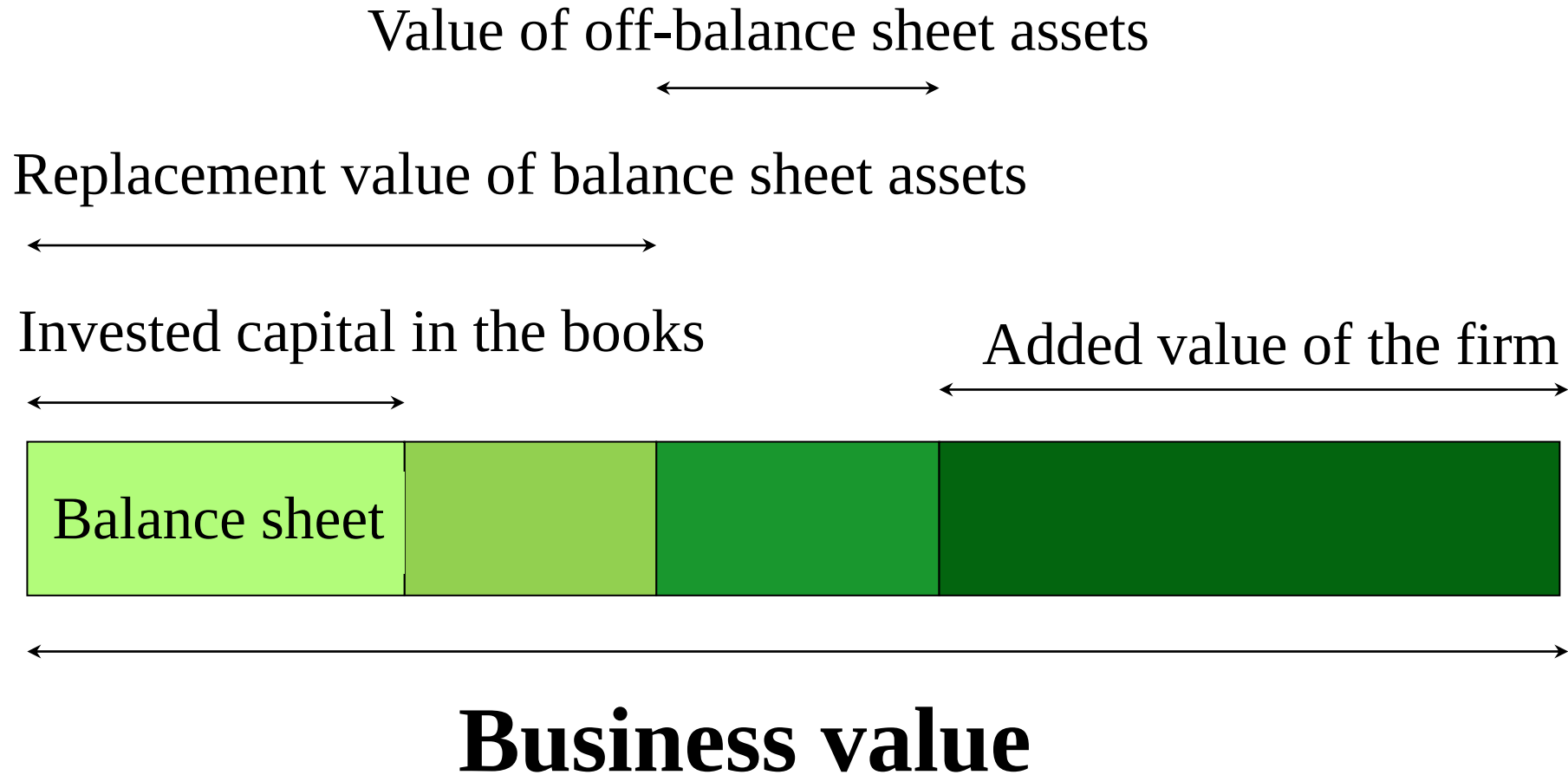
Present and future focused

Predictions, estimations

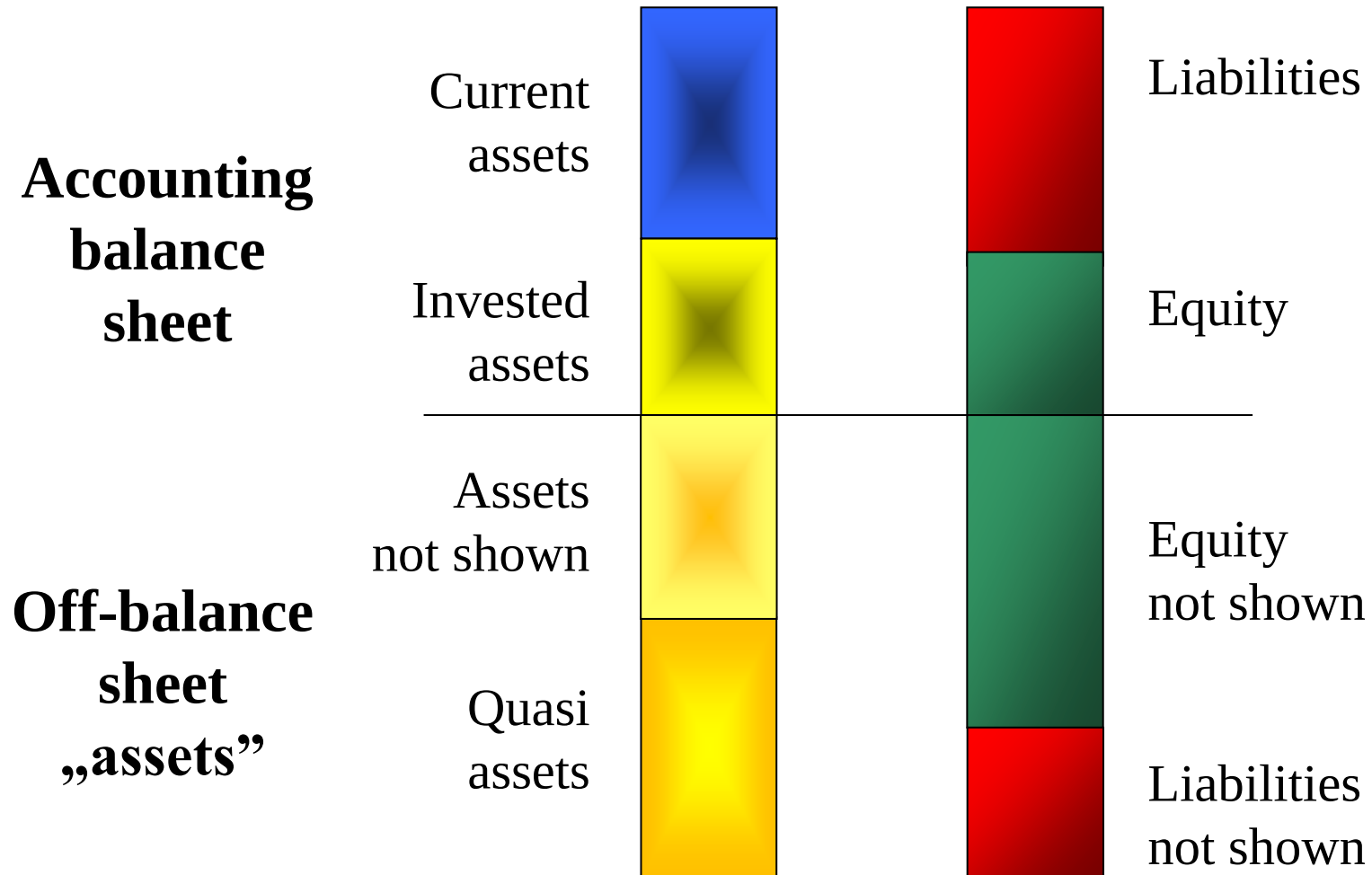
Expected value

Made for owners and
managers

Types of off-balance sheet items



Extended balance sheet



Grouping off-balance sheet items based on type and connectedness

	Ownable	Manageable only
Organisation linked	Brands, transferable contracts	Supply contracts, documented knowledge
Employee linked	— (Playing rights)	Work connections, non-recorded knowledge

Stakeholder	Advantage of good reputation
Investors	Lower price volatility, higher quotes
Employees, management	Lower recruitment cost, higher loyalty (less fluctuation), lower wages
Creditors	Lower rates, more attractive credit terms
Suppliers	Reduced prices, longer payment terms
Retail chain	Lower fees, preferential treatment, wider spatial coverage, more flexible connection
Customers	Premium price, more frequent purchases, higher volume
Competitors	Entry barrier, comparative advantage, openness to cooperation

Do we measure anything? What? How often?

What is the aim of measurement?

Use of money-based
measurement

Tracking changes
(numbers)

Value comparison
(measurement unit)

The value is in the eye of the beholder.
(strategy, synergy)

Quasi assets can not be sold on their own.

ScoreCards 1

1985 Skandia (Sweden)

Five dimensions: finance, customers, employees, processes, renewal

73 traditional + 91 new (intellectual capital) ratios (164 ratios)

1992 Kaplan and Norton: Balanced Scorecard

finance, customers, operational processes, learning-development

1993 Dow Chemical

First in the USA

ScoreCards 2

Mid 1990s PWC

ValueReporting: market, strategy, value creating activities, financial performance

1999 Edvinsson and Malone

Skandia system corrected for redundancies
112 ratios remained even afterwards

Mergers and Acquisitions

Péter Juhász, PhD, CFA

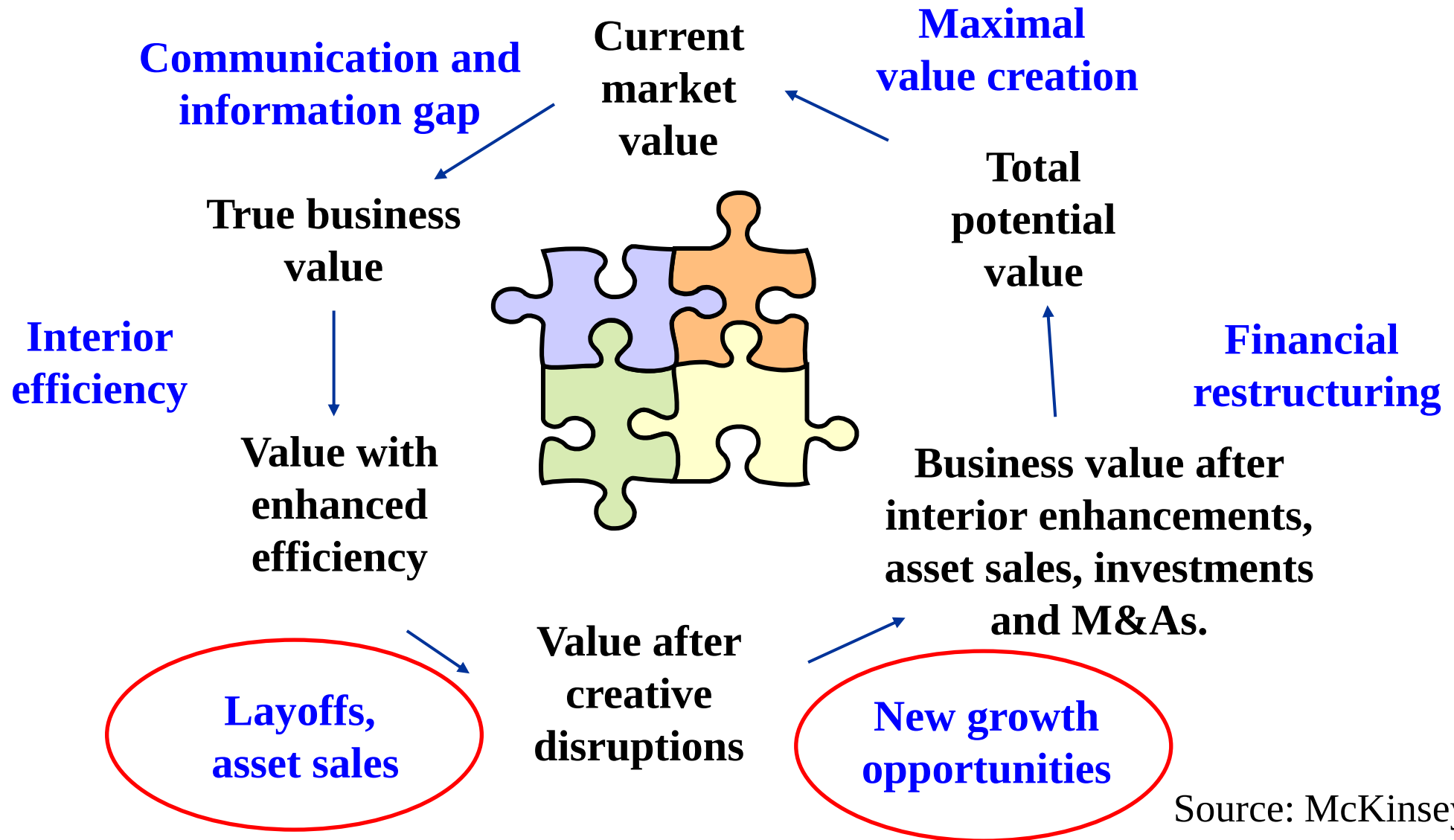


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Value creation areas

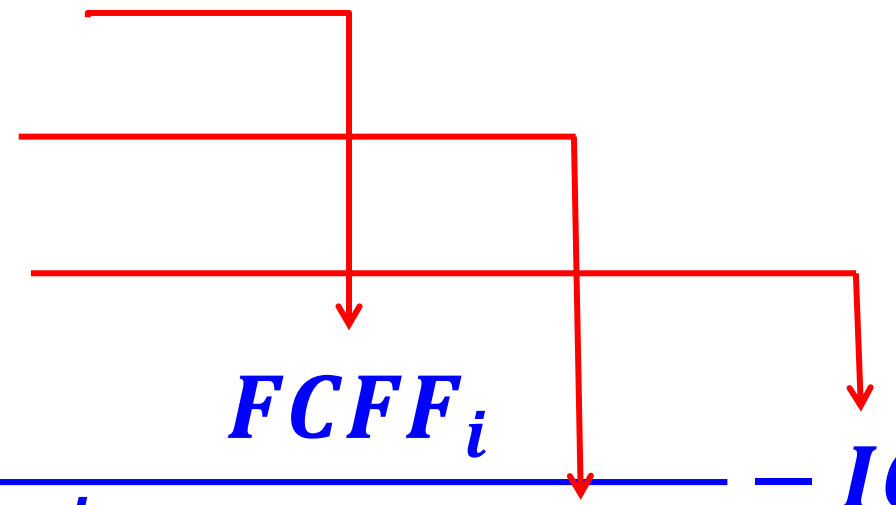


Ways of creating value

Increase cash flows

Reduce risks

Cut back on capital need


$$MVA = \sum_{i=1}^{\infty} \frac{FCFF_i}{\prod_{j=1}^i (1 + WACC_j)} - IC_0$$

Tools:

New, positive NPV projects

Growth,

Sell-off of poor projects

Sell-side motives

Management
issues
(inheritance)

Liquidity issues,
lack of capital
(bankruptcy,
growth)

Stress among
owners
(divorce)

Strategic
opportunities
(better investments
available)

exit
(venture capital)

Buy-side motives

Extensive growth

Efficiency of scale

Market force and
marketing considerations

Vertical integration

Diversification

Taxation

Access cash

EPS increase

Prestige

Complementary resources

Acquiring crown jewels

Undervaluation

Management considerations

Do M&As create value?

For the sellers, absolutely.
The average takeover
premium is 30% in the US.

There is a huge deviation
across buyer effects –
usually they lose value.

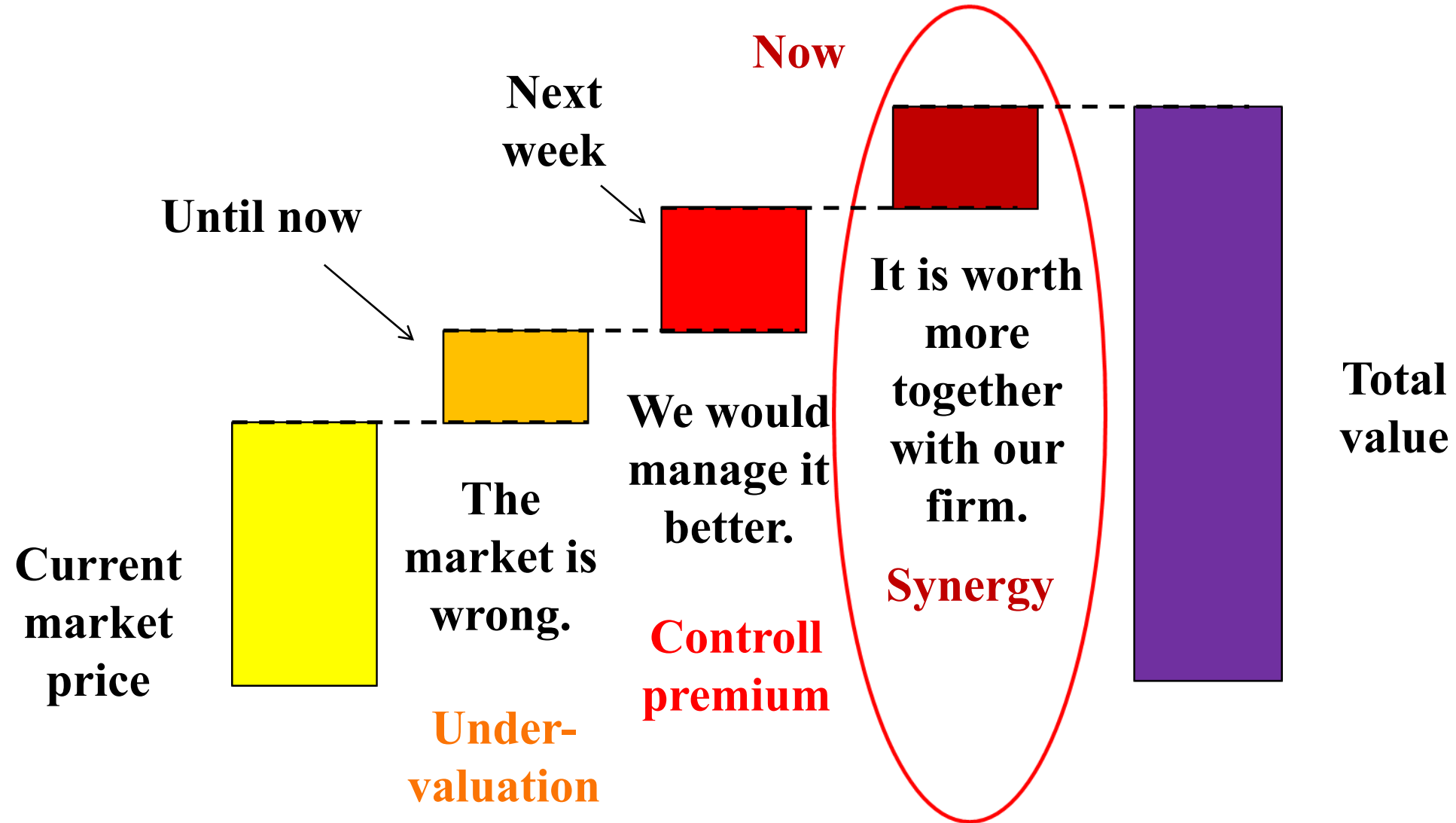
Why paying more than the current price?

Synergy: Two firms together add up for more: $2+2=5$, thus we pay a maximum of 3 for the firm.

Control premium: Should we control the firm, the value will climb from 2 to 3. No need for owning another firm.

Undervaluation „premium”: The fair firm value is 3 today, but the market price is only 2. We pay a maximum of 3, no need to own another firm.

The structure of the total value



Synergy

The effect due to which parts together are more valuable than separately.

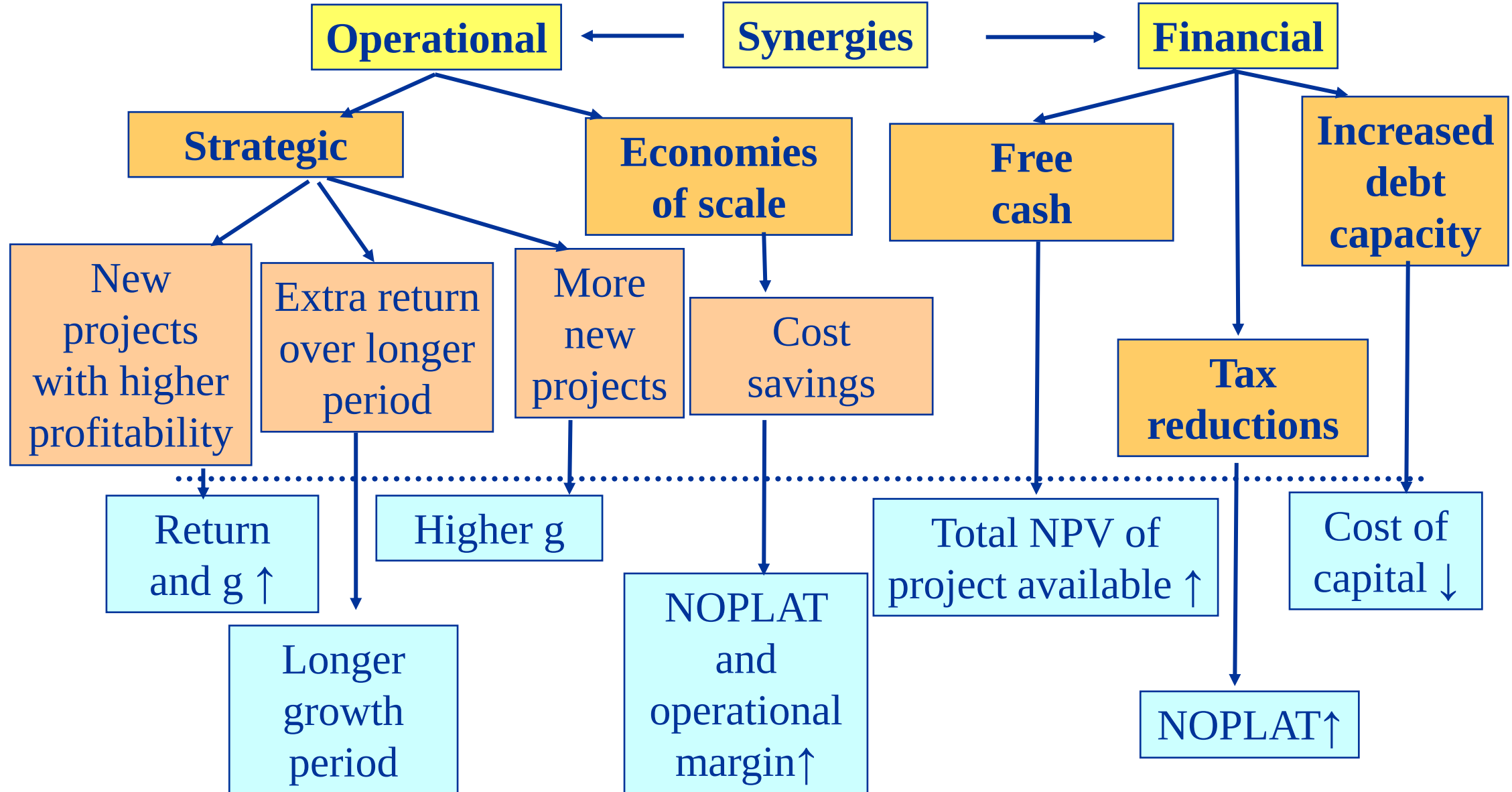
Car – ignition key

Cell phone – SIM card

Computer – software



Typology of synergies



Assessment of synergies

What form do those show?

How do those affect value drivers?

When do those start having an impact?

Modelling operational synergies (higher FCFF):

Cost efficiency – naturally limited

Sales growth synergies – only limited by common sense
(Country and market growth? Maintainable?)

Doubtful synergies

EPS growth – if a firm with a $P/E = 30$ buy another with $P/E = 20$, EPS would grow (And what then? – this is not a value focused approach)

Faster growth – it is of no value alone. We need to review:

- $WACC <> ROIC, RONIC?$
- $Premium <> Synergy?$

The maximal amount of premium 1

Undervaluation „premium”

- + Intrinsic value of the Target
- Market price of the Target

Who pays it?

- The buyer is ready to pay this amount on the top of the market price.

The maximal amount of premium 2

Control premium

- + Intrinsic value of the Target when optimally managed
- Market price of the target with the current management

Who pays it?

- A buyer with better management skills is ready to pay that above the market value.

The maximal amount of premium 3

Synergy premium

- + Intrinsic value of the merged firm
- Stand alone values of both firms when optimally managed

Who pays it?

- A buyer owning another company that could generate synergies.
- Usually just shared. There is no need to pay this completely.

The premium actually paid depends on the bargaining power of the parties.

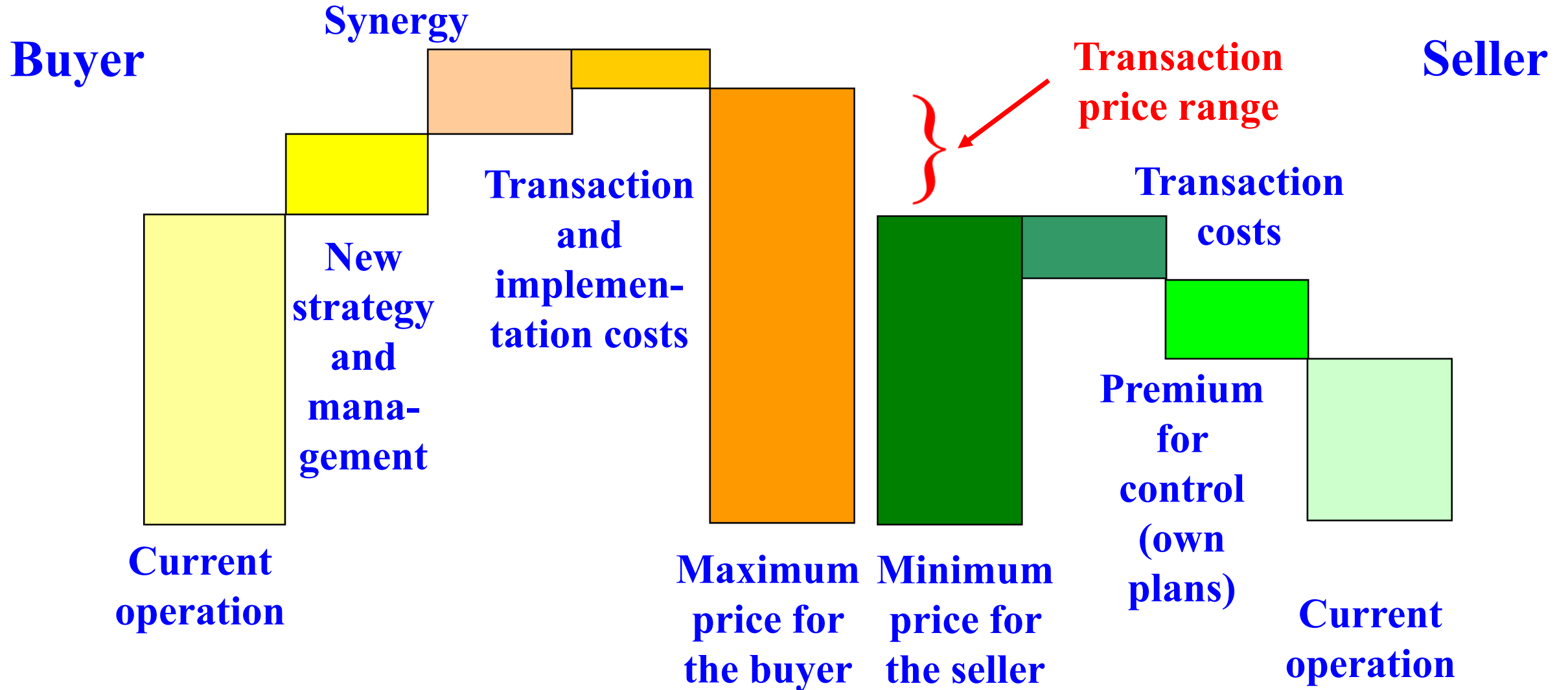
Valid motives for acquisitions 1

Motive	Explanation	Limit price	Comment
Undervalued stocks	$P < E_0$	P (no premium)	Undervaluation at the stock exchange or forced sale and liquidation.
Diversification	More stable CF, less expected risk	E_0 (no premium)	Could not be done on the stock exchange.
Operational synergies	$g \uparrow$, return to scale, market share $\uparrow$, complementary activities	$P + \text{Synergy}$	$V(A+B) > V(A) + V(B)$ Added value from growth in sales or margins

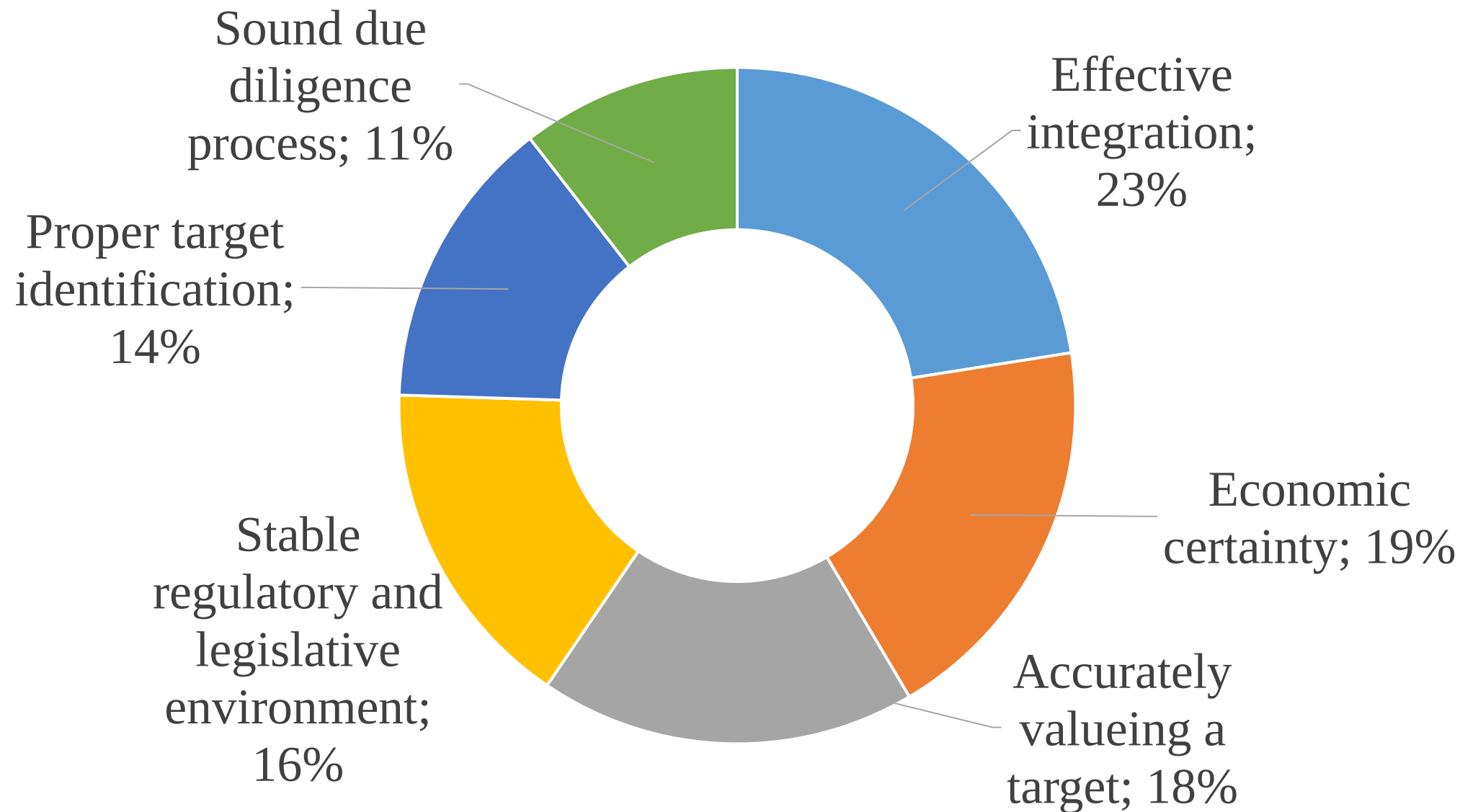
Valid motives for acquisitions 2

Motive	Explanation	Limit price	Comment
Financial synergies	Free cash, leverage opportunity, potential tax savings	P + Synergy	Higher CF, lower WACC or added TS.
Poor management	Resources not optimally used	Value under optimal management: P + control premium	Management issues reflected in price. (Mostly private firms.)

Setting the price



Success factors of M&As



Why would mergers fail?

1. Overpayment (extreme optimism)

- Overoptimism on market opportunities

- Overestimating synergies

- Incomplete due diligence, hidden or unhandled issues

- Overbidding other offers, management ego („the winners’ curse”)

2. Poor integration management

- Lack of setting a new business model

- Uncertainties and conflicts remain unsolved

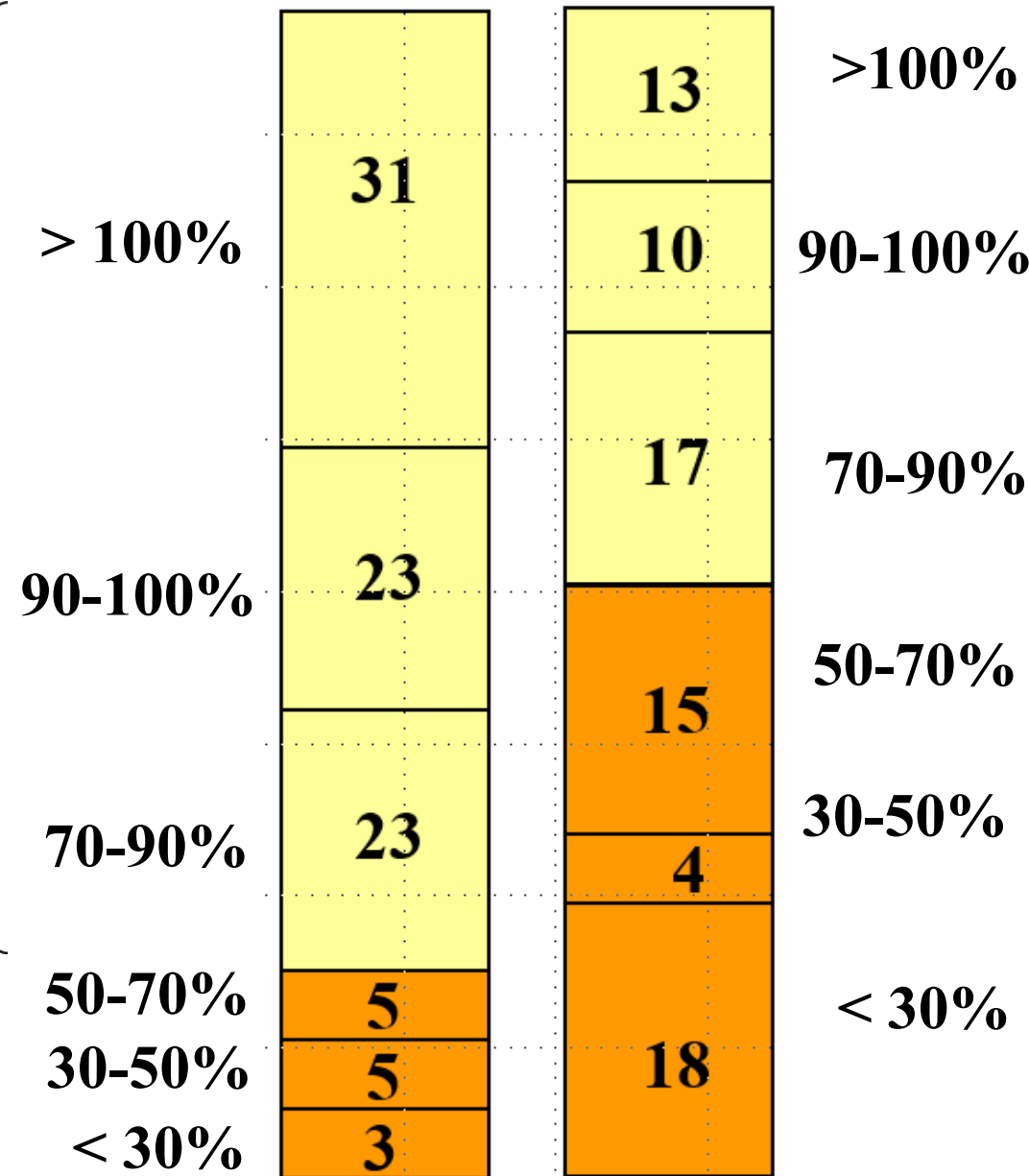
- Late reactions to market pressures

Plans vs Reality

N=90

**Cost
synergy**

In **85 %** of the transactions at least 70% of cost synergies were achieved.



In **52%** of the transactions at least 70% of the sales synergies were realised.

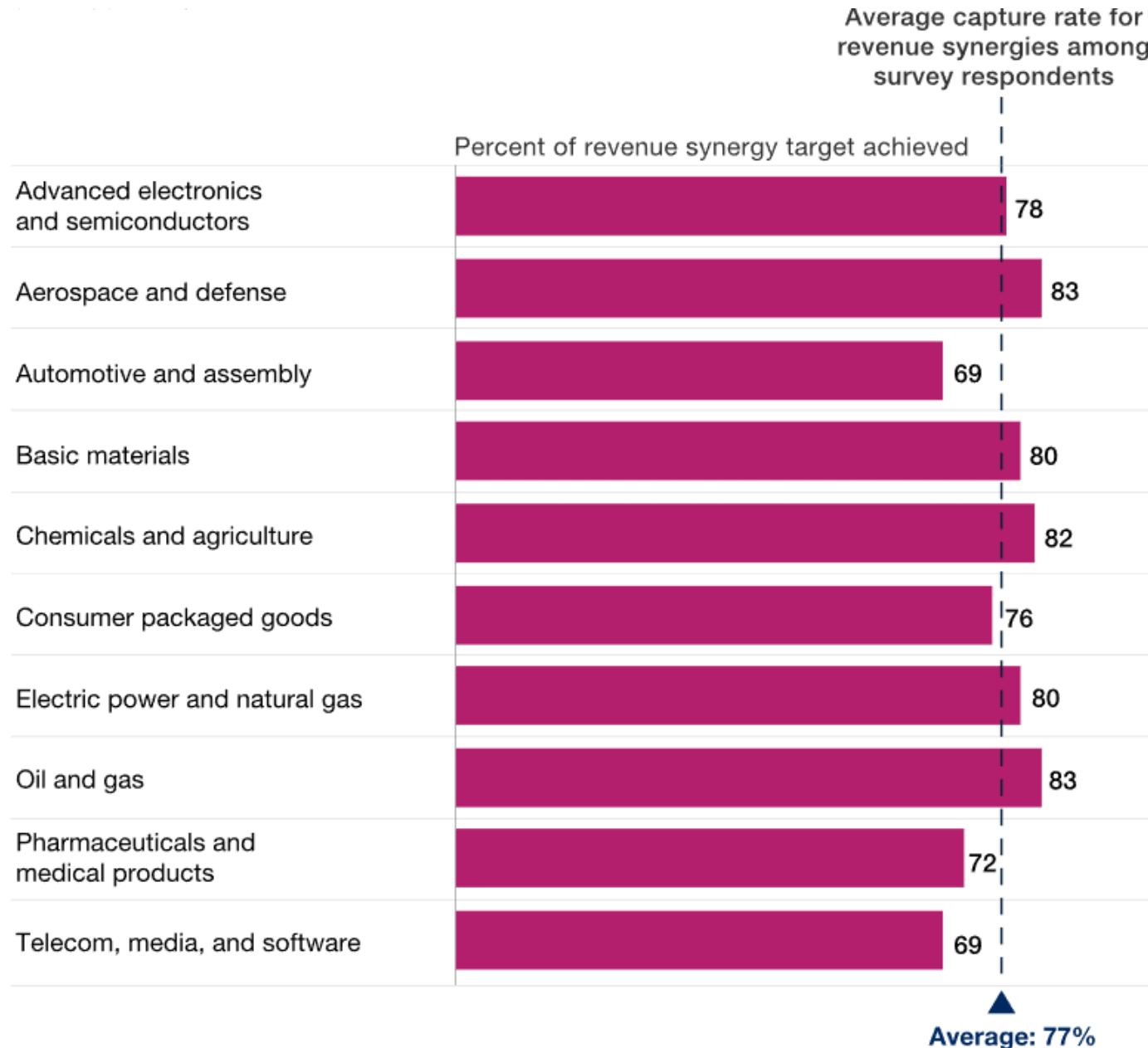
N=77

**Sales
synergy**

Source: McKinsey

Sales synergies realised

Source: McKinsey, 2018



The risk of synergies 1

The likelihood of realising different type of synergies is not the same.

The joint sales will not start growing immediately as

1) the transaction process jeopardises the *customer connections*.

2) competitors aim immediately at our *best staff*,

3) *customers buying from both firms* hit take some of their purchase to a competitor,

4) the *remaining customers* ask for a reduction as thanks to their loyalty

The risk of synergies 2

Costs of realisation

lay-off expenses, restructuring, repositioning, change of names, IT integration
These can add-up to more than one year of synergy

Time need of realisation

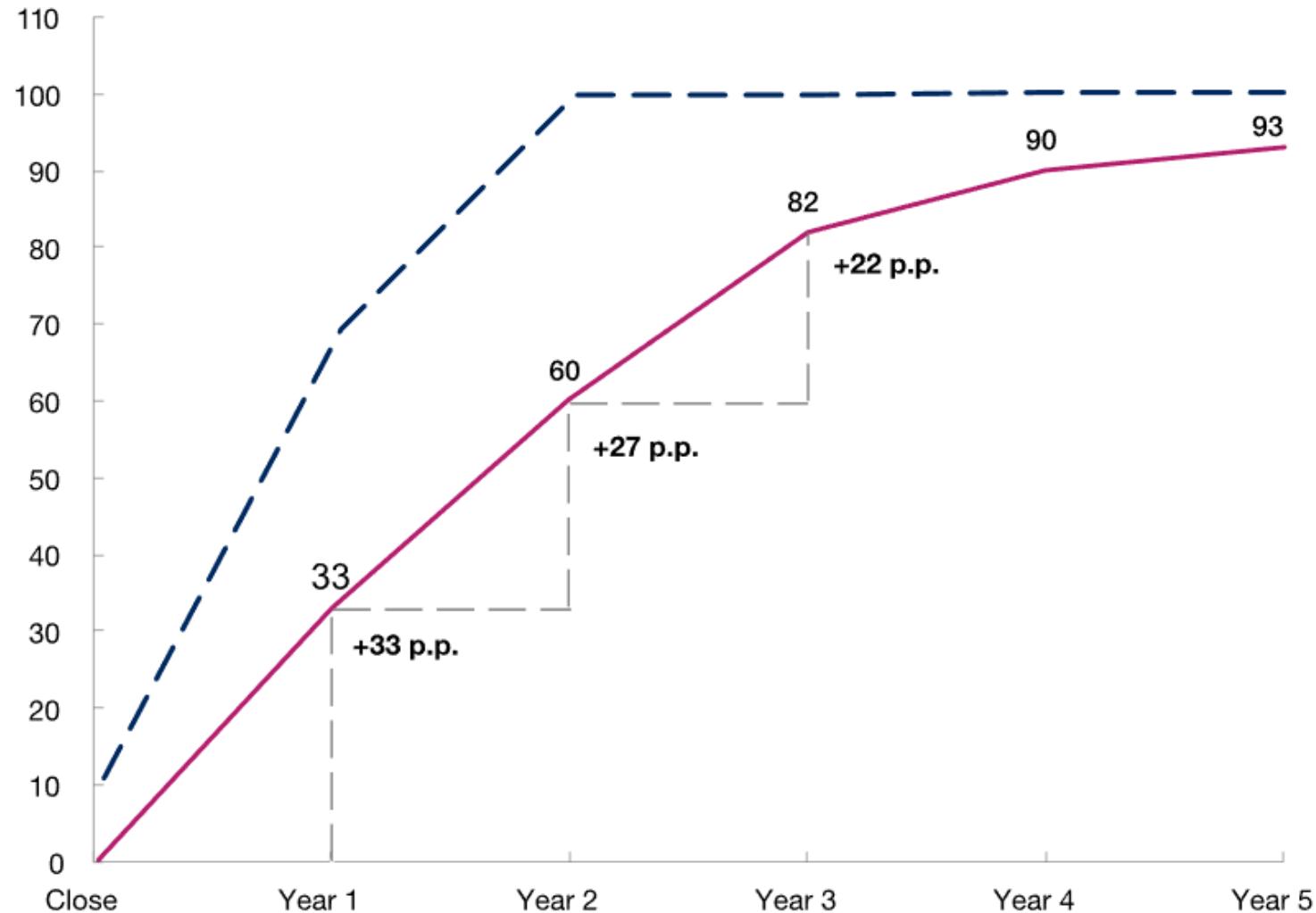
Different customer management systems, trainings, IT integration

**Synergies that could not be realised
during the first complete business year after the transaction
are usually completely lost.**

The time need of realising the synergy

Percent of total synergy target captured by the end of each year after deal close

- Cost synergies (illustrative)
- Average revenue synergies



Source: McKinsey, 2018

How to pay for a firm?



Cash

- Liquidity issues
- All risk remains with the buyer
- All profit stays with the seller



Shares

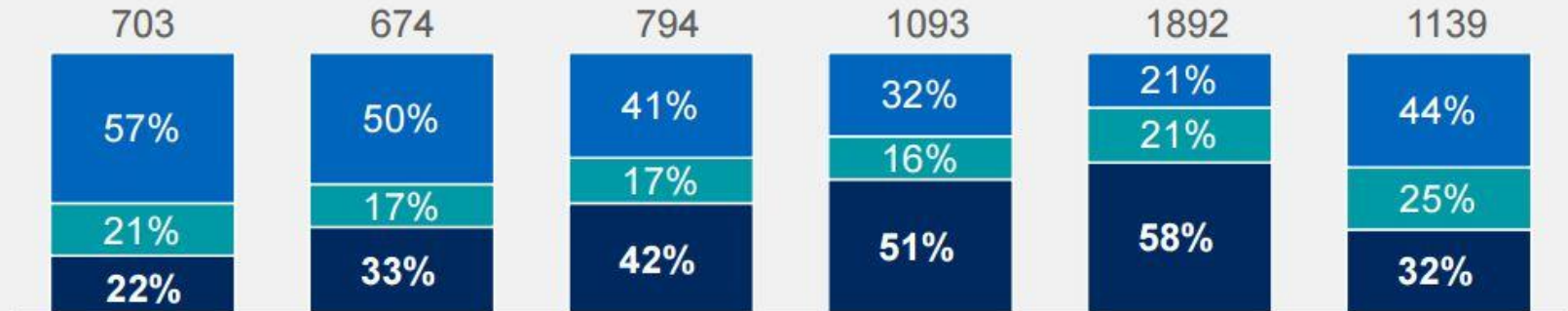
- The seller can not exit
- Shared risks
- No liquidity shock

Cash
Shares
Mixed

Payment methods worldwide

Source: McKinsey, 2017

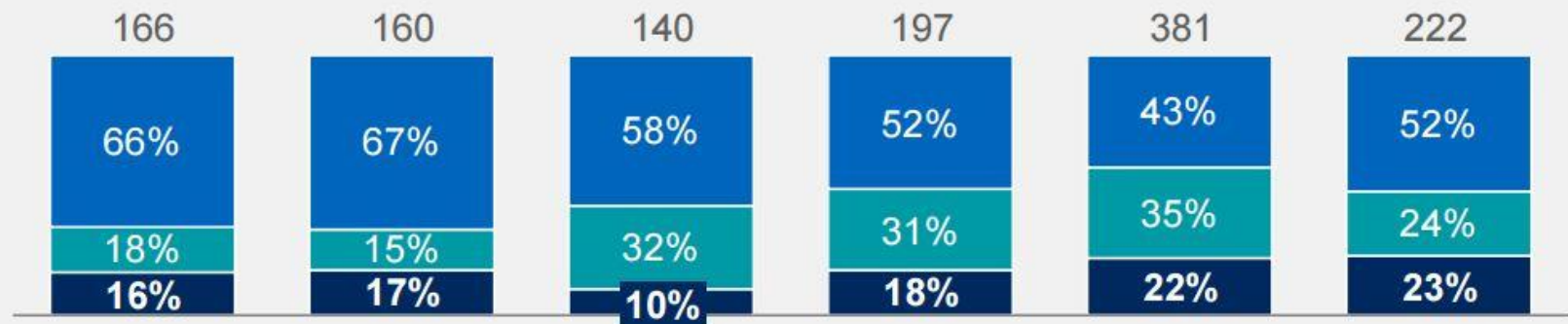
Top 100 global deals by financing means available²



Top 50 EMEA deals by financing means available²



Top 50 APAC deals by financing means available²



2011

2012

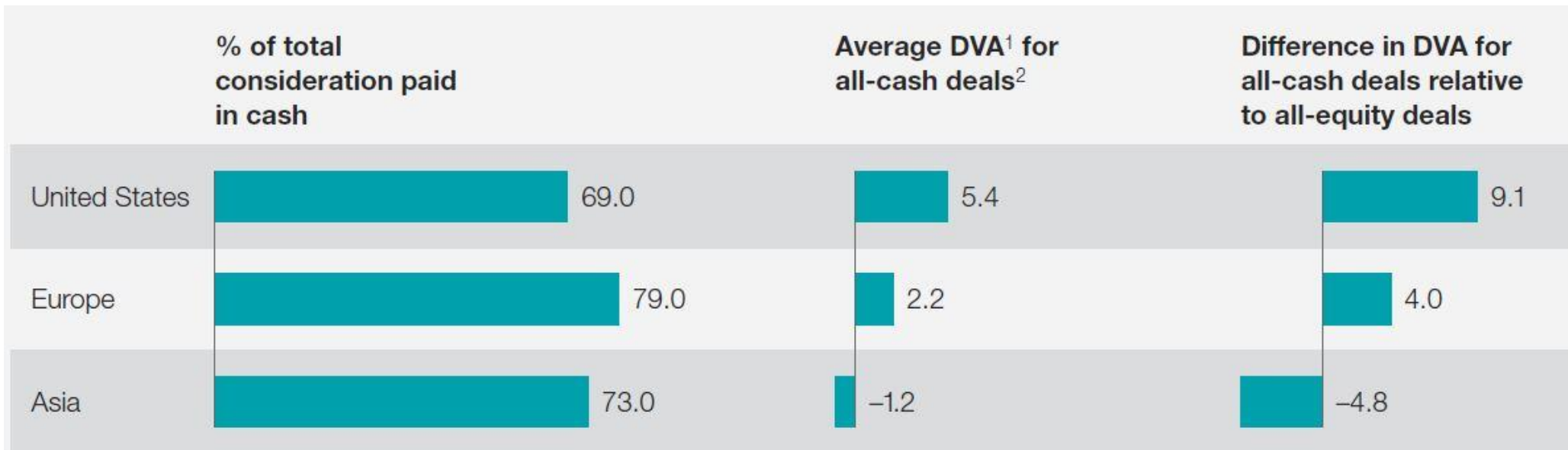
2013

2014

2015

2016

Market view of payment methods



¹ Deal value added.

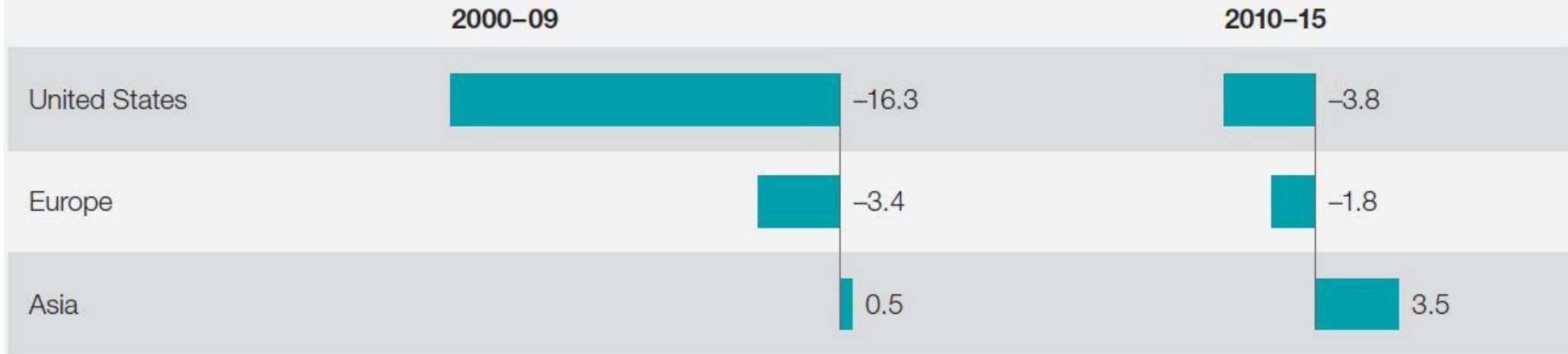
² For M&A involving publicly traded companies; defined as combined (acquirer and target) change in market capitalization, adjusted for market movements, from 2 days prior to 2 days after announcement, as % of transaction value.

Source: Datastream; Dealogic; McKinsey analysis

Source: McKinsey, 2016

Market view of shares-only payment

Average DVA¹ index for all-stock deals,² %



¹ Deal value added.

² For M&A involving publicly traded companies; defined as combined (acquirer and target) change in market capitalization, adjusted for market movements, from 2 days prior to 2 days after announcement, as % of transaction value.

Source: Datastream; Dealogic; McKinsey analysis

Source: McKinsey, 2016

Example

ABC Co. purchases BCD Co. They pass on 100 unit of synergy, one third of the price is payed cash.

How will the price of ABC shares change due to the transaction, if there are 2000 pieces issued?

Data

ABC current value: $V=1000$ $E=800$ $D=200$

BCD current value: $V=300$ $E=200$ $D=100$

Synergy=300

Joint value (before payment): $V=1600$ $E=1300$ $D=300$

BCD shares purchase price: $200+100=300$

Cash: $300 \times \frac{1}{3} = 100$

Shares: $300 \times \frac{2}{3} = 200$

Together (after the payment): $V=1500$ $E=1200$ $D=300$

Results

ABC current quote: $800/2000=0,4$

Synergy to current owners: $300-100=200$

Required values of ABC shares after the transaction:

$$800+200=1000$$

Share price: $1000/2000=0,5$

To be paid for BCD: 200

Number of shares to issue: $200/0.5=400$

Summary

Number of shares after the transaction: $2000+400=2400$

Price quote: $1200/2400=0.5$

Attention! If during the transaction the amount of debt changes or extraordinary dividend is paid, you also need to consider those effects before calculating the share prices and number of shares to issue!

Lessons from the history 1

1. The fusion of **similar-sized firms** is less likely to be successful.
2. Synergies from **immediate cost reductions** (e.g. downsizing) are easier to capture than those linked to future growth, new products, or joint R&D.

Lessons from the history 2

3. Acquisitions focused on **smaller private firms** are more likely to be successful. The harder the competition, the more likely the loss in value is.
4. **Hostile takeovers** promise better management and results than friendly transactions.

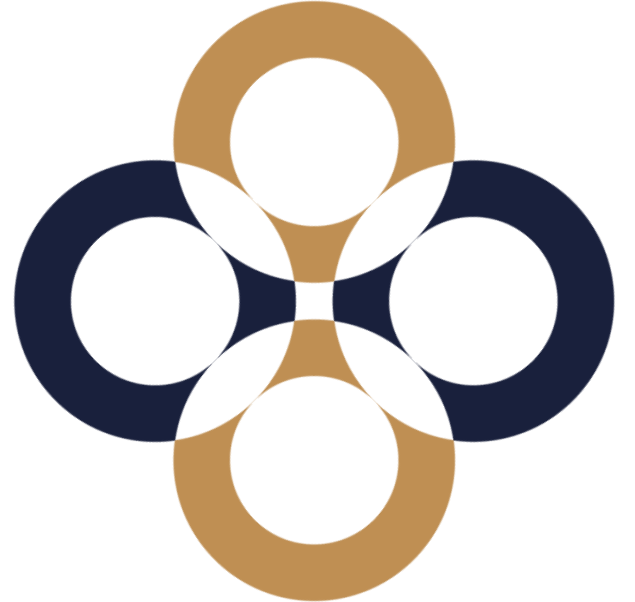
Lessons from the history 3

5. **Companies with outstanding efficiency** are more likely to be able to create value by acquiring firms.
6. **Firms carefully considering** synergies and M&As are more likely to be successful.



Discounts and Premiums

Péter Juhász, PhD, CFA



Why do we need D&Ps?

To include effects into our valuation that the given method has not yet (fully) considered.

The use and extent of discounts and premiums vary across valuation methods even within the same valuation task.

Inclusion of key effects

Effects	DCF	Multiples	Asset-based
Business outlook	CF plan	Industry average	–
Market judgement	Discount rate	Market ratios	Market prices
Liquidity	Premiums and discounts		Different!
Ownership structure			None!
Off-balance sheet items, others			Should be already included

What is the base of correction?

Each approach calls for a different correction.

DCF: Theoretical intrinsic value under perfect conditions (liquid markets, no information asymmetry)

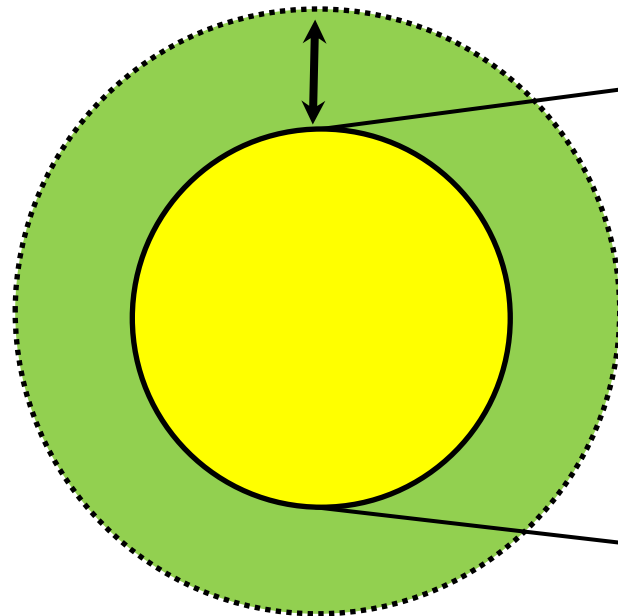
Market multiples: Average value of similar firms (typical, listed, liquid firm)

Comparable transactions: We are exactly like the average of the peers (same country, markets, size, liquidity)

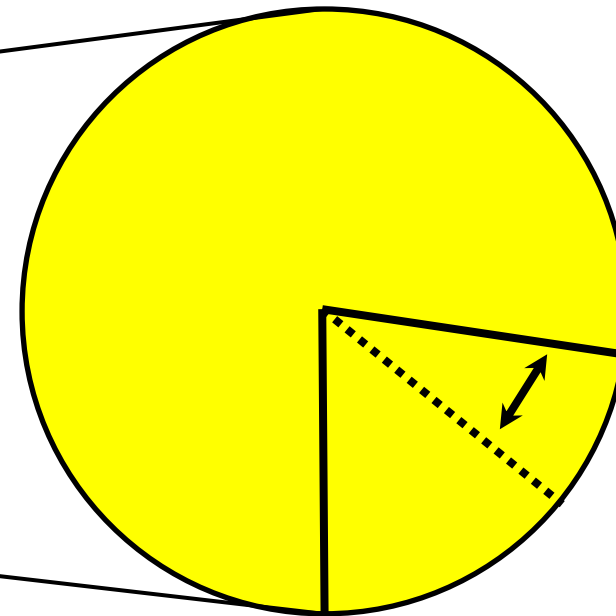
Types of corrections

Corrections should only hit the value of operation.

Corrections for
equity value



Corrections for
a given stake



Discounts & Premiums 1

D&Ps for total equity

Liquidity („illiquidity”) discount (Several meanings: assets, size, financial power, chance of going public.)

Key person discount

Complex portfolio without synergy

Environmental, legal, and dependent liabilities

Long term liabilities (contracts, warranties, guaranties, warrants)

Discounts & Premiums 2

D&Ps for an equity stake

Liquidity discount (no buyer e.g. due to the ownership structure)

Control rights (minority, blocking minority, majority, strategic majority)

Additional voting right (preferred shares)

Big package discount (if sold current price would fall)

The valuation process

1. Determining the value of operation (V_{op})
2. Valuing non-operative assets (NOA)
3. Calculating the total firm value ($V = V_{op} + NOA$)
4. Valuing the foreign capital (D)
5. Determining the market value of equity ($E = V - D$)
- 6. Finding the price of a share or stake**

Valuing equity stakes

The value of a firm is 1200, its debt amounts to 200 and there are 2000 shares issues. What is the market price of a single share?

$$MV(D+E)-MV(D)= MV(E)$$

$$1200-200=1000$$

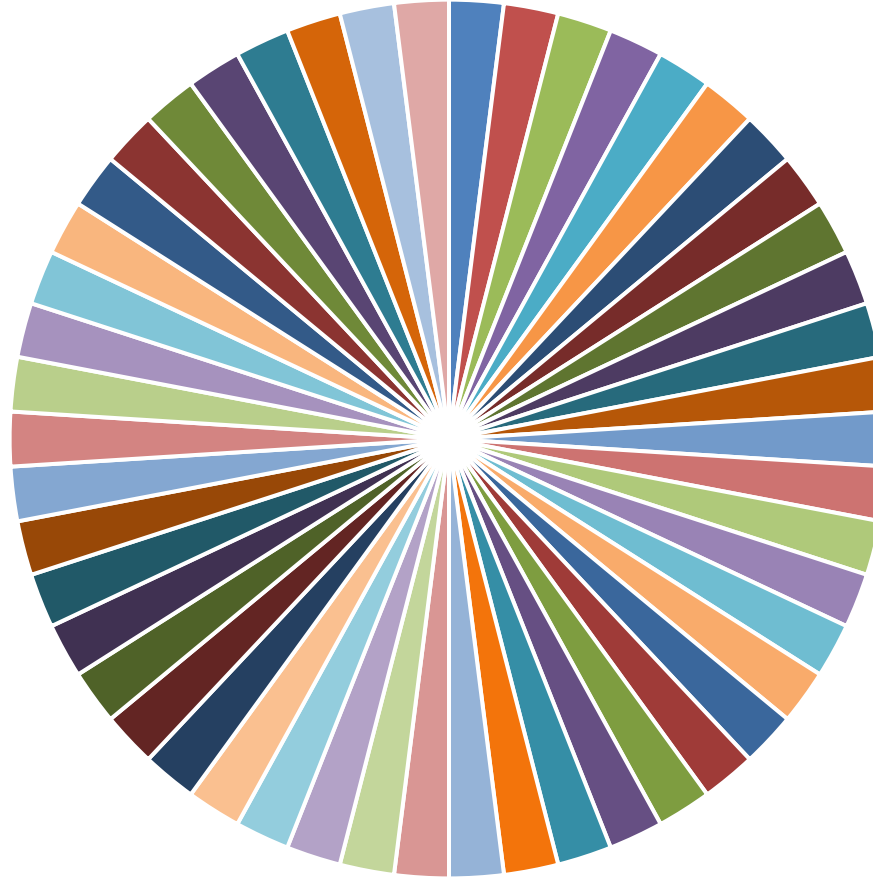
?

$$P=1000/2000=0,5$$

?

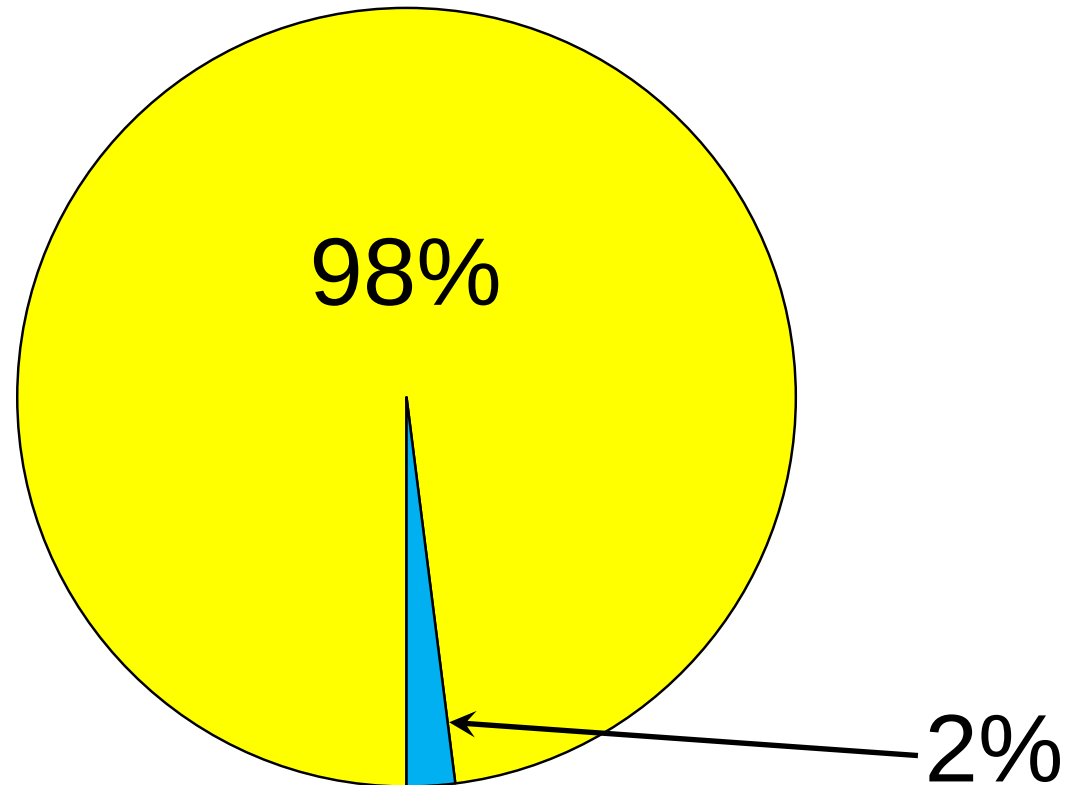
$$MV(E)*q <?> MV(E*q)$$

The value of single share 1



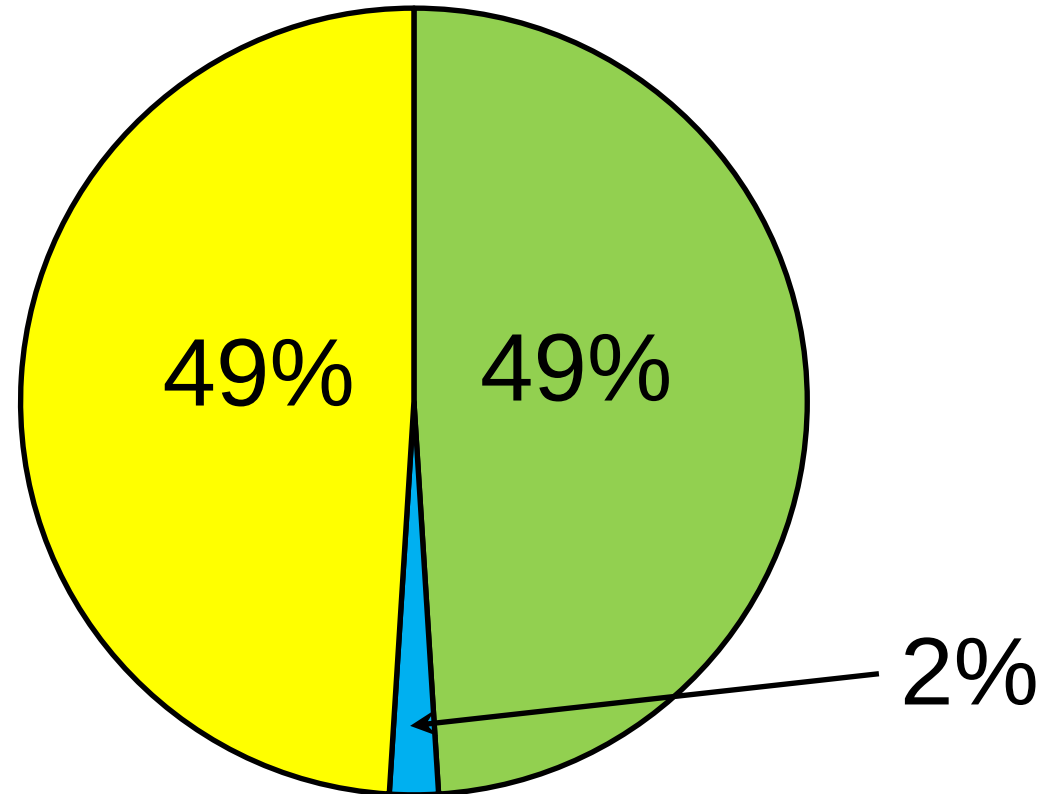
$$MV(E)*q = MV(E*q)$$

The value of single share 2



$$MV(E)*q \gg MV(E*q)$$

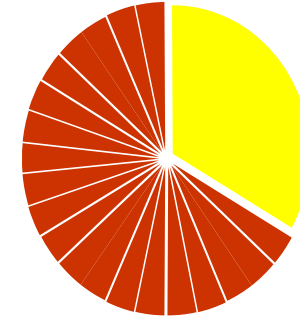
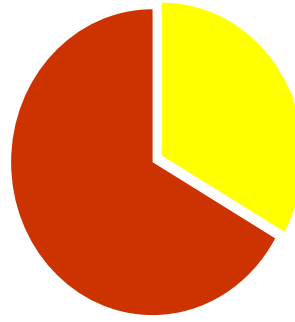
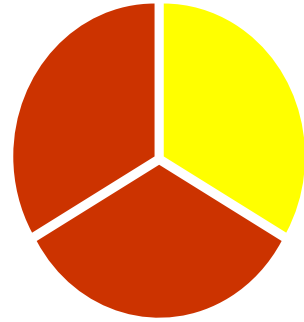
The value of single share 3



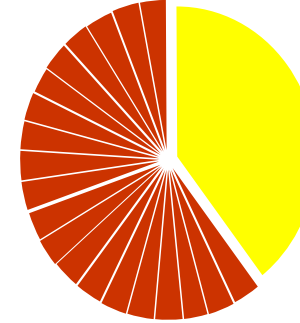
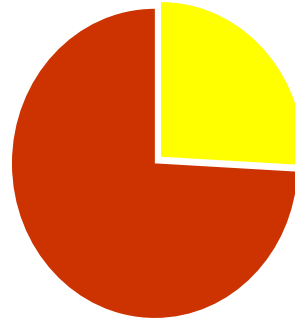
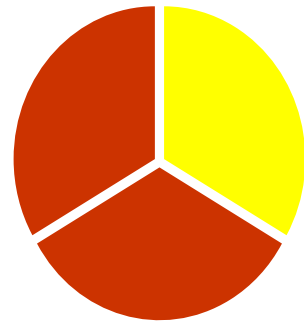
$$MV(E)*q \ll MV(E*q)$$

The value of a 33% stake

**Ownership
stake**



**Value
stake**



-30%
(or -22.1%)
discount

+30%
(or +18.2%)
premium

The sum is always 100%

60%-40% division of the ownership rights

Control premium (%)		Minority discount (%)
10.0	$60 \times 0.10 / 40 =$	15.0
15.0	$60 \times 0.15 / 40 =$	22.5
20.0	$60 \times 0.20 / 40 =$	30.0
25.0	$60 \times 0.25 / 40 =$	37.5

Ownership-structure effect

What kind of opportunity to influence the operation of the firm is offered by the given stake? (Why you pay for the control is that you can make the firm more valuable using it.)

The exact size of the control premium/discount can only be estimated on a case-by-case basis.

The value of the control

1. The likelihood of a change in the control of the firm
2. The value added by the change in the control:
quality of the current management,
difficulty of exercising the control



When to look for a control premium? 1

Poor asset management

Too conservative investment policy

Extreme desire for investment

Missed strategic opportunities

M&As without synergies



When to look for a control premium? 2



Too conservative financing structure

Extreme indebtedness

Poor choice of financing forms and sources

No trust of spending the accumulated cash correctly

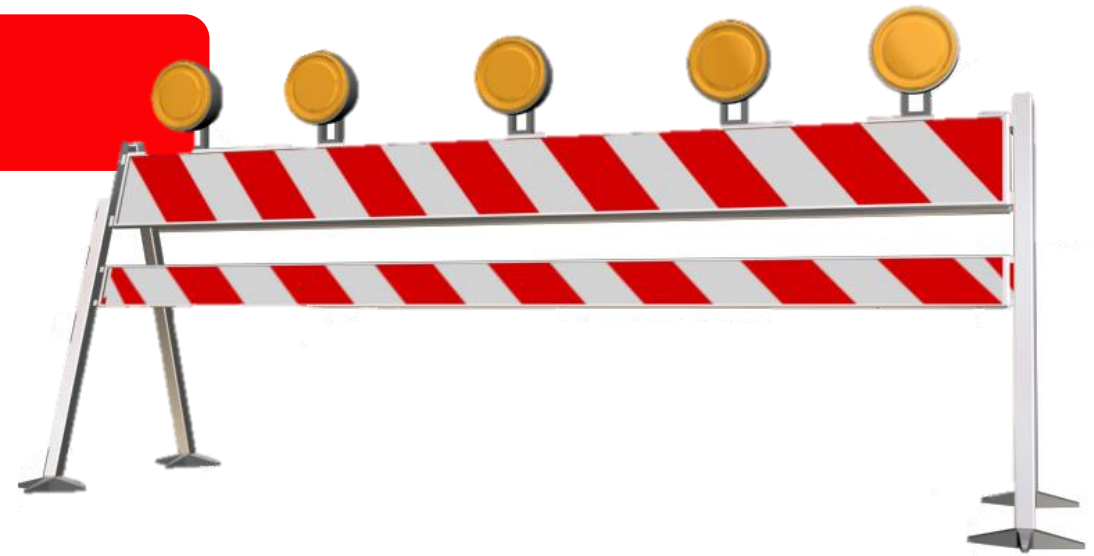
Likelihood of a change in control

Institutional barriers

- Takeover regulations
- Limitations of ownership
- Evading conflicts of interest

Firm-specific barriers

- Voting preferred shares
- Cross-ownership holding structure
- Significant management ownership
- Very diversified ownership
- Base rules limiting ownership

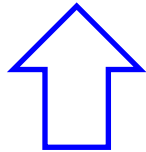


Value of control for listed firms

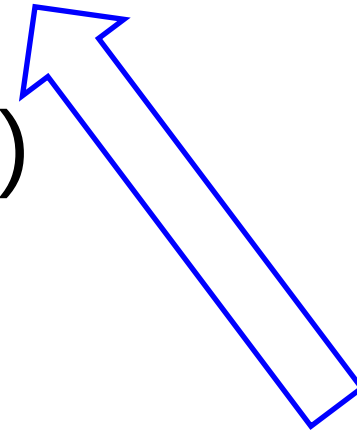
Market value = current value +

+ probability in change of control *

* (optimal value – current value)



We could do it better,



but it is not for sure that we may.

Control premium for public firms

A premium may be paid if a subjective valuation dictates a higher than current market price. (Information on mistakes)

Quotes may move significantly if the estimated likelihood of a change in control would modify. (News on possible M&As)

Weaker investor protection leads to lower share prices.

Private firms: Control premium 1

For private firms, control may have a significant value as usually management would only change if ownership changes.

Minority discount depends on the ratio of the optimal and the current equity value.

Control premium 2

Example on the difference between the valuation of stakes of 49.9%:

Current equity value= 100

Optimal equity value= 150

optimal value 49.9% = 74.85

current value 49.9 % = 49.90

$$\text{Minority discount} = 1 - \frac{49.90}{74.85} = 33\%$$

Control premium 3

Minority discounts are inversely linked to management abilities.

You do not always need a 50%+1 stake for the control.

The value of any stake depends heavily on whether the owner is capable and willing to be involved into the management of the firm .

Deviations across the world

More efficient investment protection: 2-5% (e.g. US, UK)

Less strict: 60-65% (e.g. Brazil or Czechia)

Median: 10-12%.

What is liquidity? 1

Everything can be sold at a low-enough price.

The level of liquidity is different across assets.

Price effect (timing, information effect,
continuity of prices)

Alternative cost of waiting

Transaction costs



What is liquidity? 2

High transaction fees:

real estate agents: 1-4%

action houses for paintings: 15-30%

Reasons for higher fees:

fewer agents,

non-standardised products,

more personal knowledge needed



Marketability discount

Dividend
payment

Availability
of
information

Accessibility
of potential
buyers

Size
of the stake

Put
option

Limitations
on sale

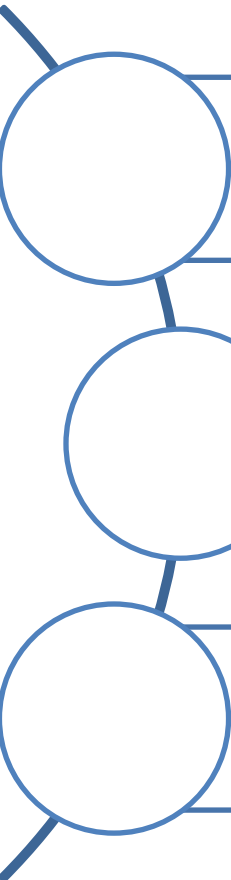
Measuring liquidity discount 1

Non-liquide investments have no well-organised markets – higher selling costs, less stable sales price



Measuring liquidity discount 2

How to quantify illiquidity discount?



Liquidity discount in the stake or share price at the end

Increasing required yield for liquidity risks

Multiple-based valuation based on firms with similar liquidity

Use one method only at a time.

Setting the liquidity discount 1

Fixed discount (range) – strongly relying on former studies that have set 25-35% for shares with limited liquidity.

Many have shown that the samples were distorted and the realistic value is far smaller.

Setting the liquidity discount 2

Firm-specific discount – depends on the possible number of buyers and the difficulty of the sale. (Real estates are easier, collections are harder to sell.)

The liquidity of private firms may change from firm to firm and buyer to buyer.

Lower liquidity discount

More liquid
assets

Stable, positive
cash flow

Going public
is near

Majority stake

Bigger firm
(Costs fixed)

Operational
and asset value
are similar

Liquidity discount in the yields

Liquidity discount appears as a premium in required yield. **The premium desired changes over time.**

Liquidity discount may increase during times the liquidity is needed. (When market liquidity is poor asset liquidity becomes more important.)

In the long run everything is liquid. How long may you add a liquidity correction?

Liquidity discount in the yields

Returns of liquid and less liquid shares show a historical average difference of 3.0-3.5%.

Smaller firms on average had a 4.0% higher yield than S&P companies (15.7% vs 11.7%, 1984-2004).

Life cycle matters: younger companies yielded 19.9% while more mature firms offered 13.7% (vs 11.7% S&P500) (1984-2004)

Links among discounts

More liquid markets have lower
lack-of-control discounts

It is easy to sell if the majority owner does not do a good job.

Discounts and premiums

Liquidity discount (assets, size, financial power, chance of going public)

Control rights (blocking minority)

Additional voting rights

Big package discount

Key person discount

Discount on portfolios without synergy

Environmental, legal, dependent liabilities discounts

Long term liabilities (contracts, warranties, guaranties, warrants)



Market experience

Court decisions and market transaction,
US (1980-1997)

Minority discount:

21-31%

Marketability discount:

(10) 20-60 (80) %

Big stake discount:

0-20%



Example

The HardWork Co. has three owners (40%-35%-25%). What is the value of the 35% stake, if the DCF value of the total equity amounts to 100 under the current strategy? (All parties agree on the valuation.)

**If investors only focus on their
own intention to buy...**

Outside buyer

Own stake after the deal:

$$100 * 35\% = 35.0$$

Private firm (total of equity)

(liquidity, transparency)

-10%

Strategic minority

+ 5%

$$35 * 0.90 * 1.05 = 33.1$$

Premium used

$$33.1 / 35.0 - 1 = -5.4\%$$

25% owner

Own stake until now:	$100 * 25\% =$	25.0
Illiquidity of the stake:		-10%
	$25 * 0.90 =$	22.5
Own stake after the deal:		
	$100 * 0.60 =$	60.0
Control premium		+17%
	$60 * 1.166 =$	70.0
Value of 35% stake	$70 - 22.5 =$	47.5
Premium used	$47.5 / 35 - 1 =$	35.7%

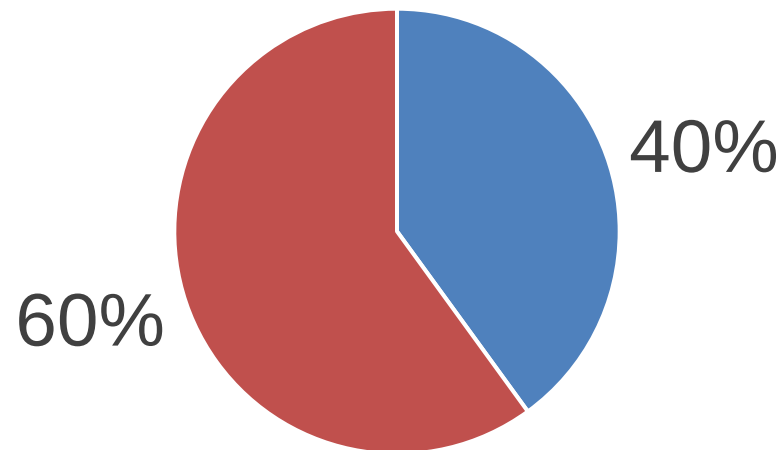
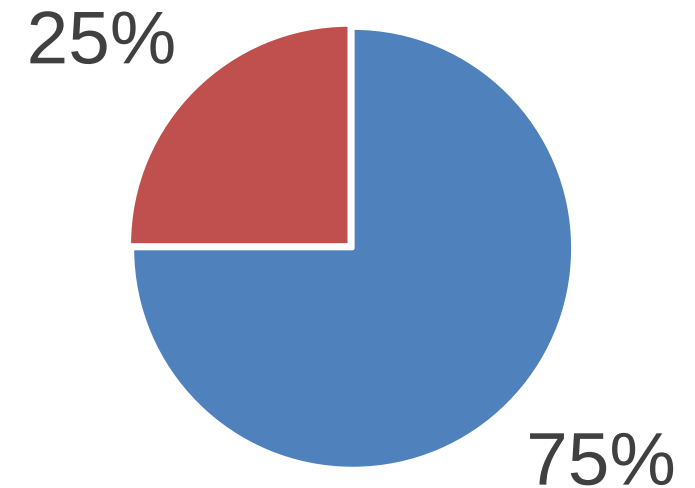
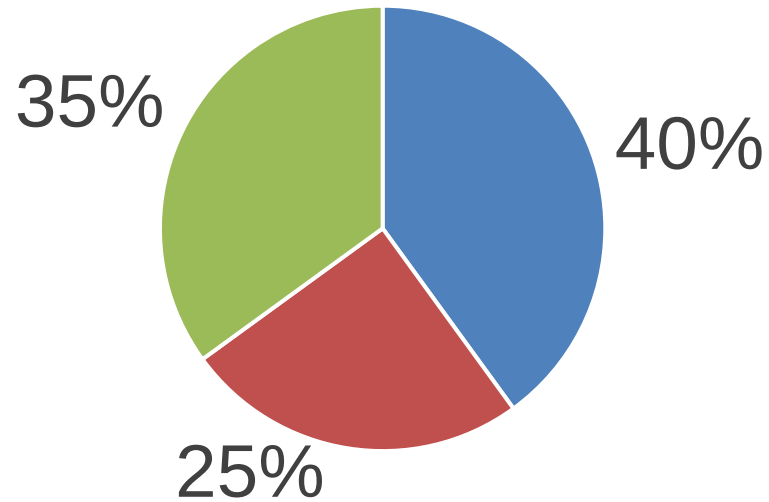
40% owner

Own stake until now:	$100 * 40\% =$	40.0
Own stake after the deal:		
	$100 * 0.75 =$	75.0
Strategic control premium		+13%
	$60 * 1.1333 =$	85.0
Value of 35% stake	$85.0 - 40.0 =$	45.0
Premium used	$45 / 35 - 1 =$	28.6%

What if everyone is interested?

Owners do not contrast to the current status rather than to the future status they will be in if not purchasing.

Possible outcomes



For the 25% owner

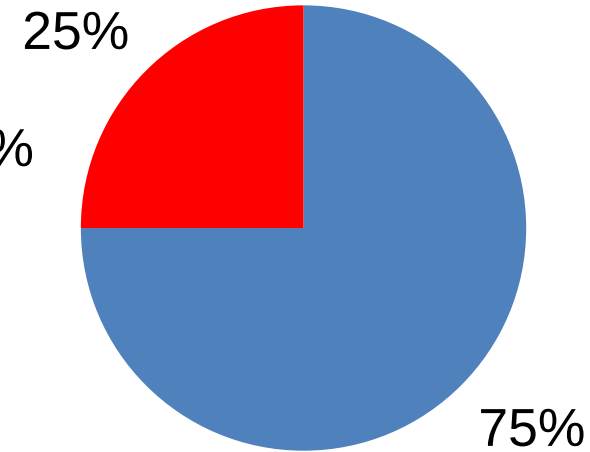
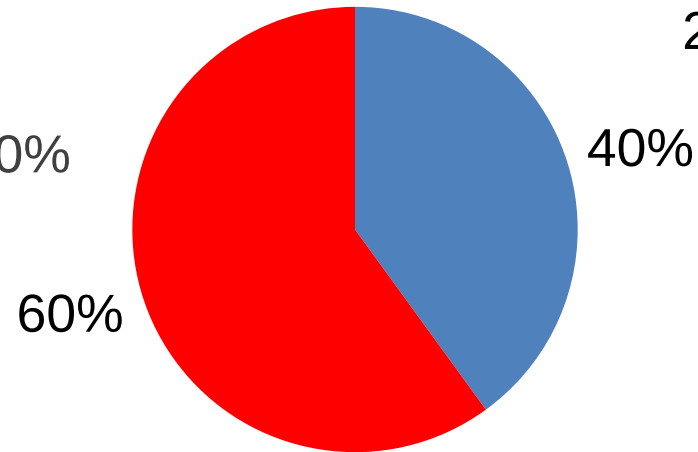
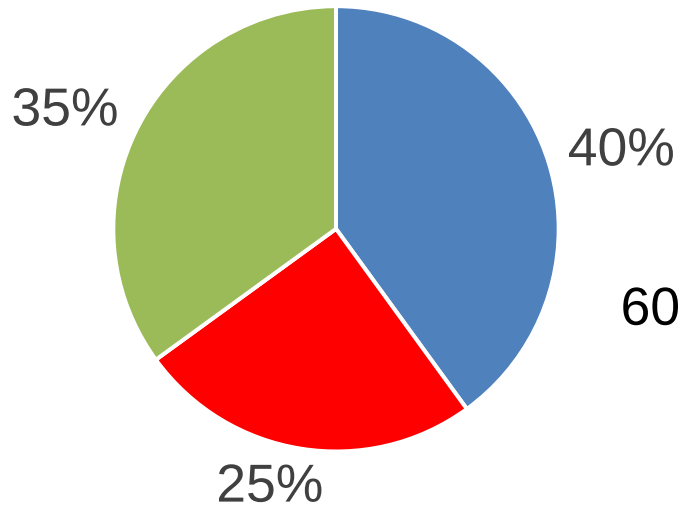
Own stake if 40% owner buys:	25.0
Liquidity and minority discount	-40%
$25 * 0.60 =$	15.0
Own stake (25%+35%) after the deal:	
$100 * 0.60 =$	60.0
Control premium	+17%
$60 * 1.166 =$	70.0
Value of 35% stake	$70.0 - 15.0 =$ 55.0
Premium used	$55 / 35 - 1 =$ 57.1%

For the 40% owner

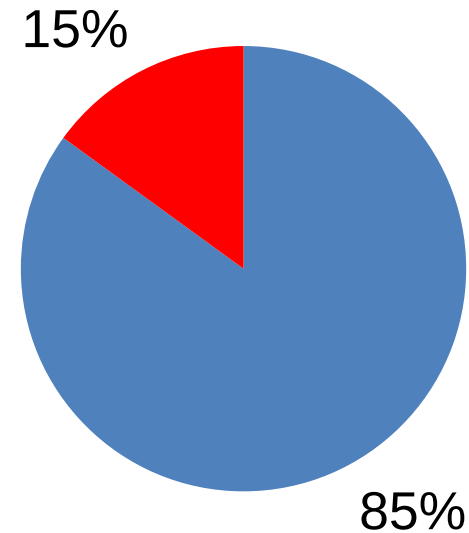
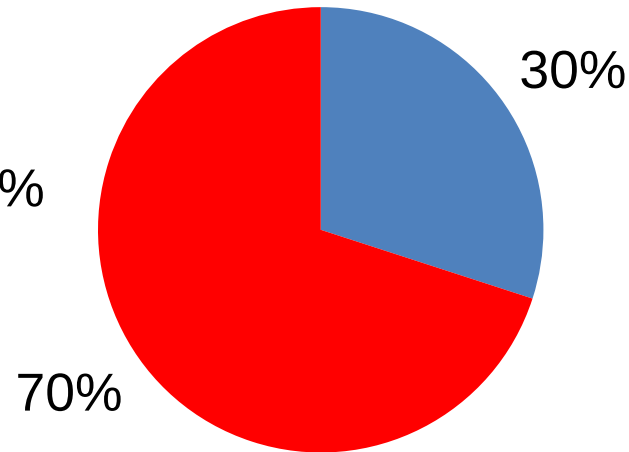
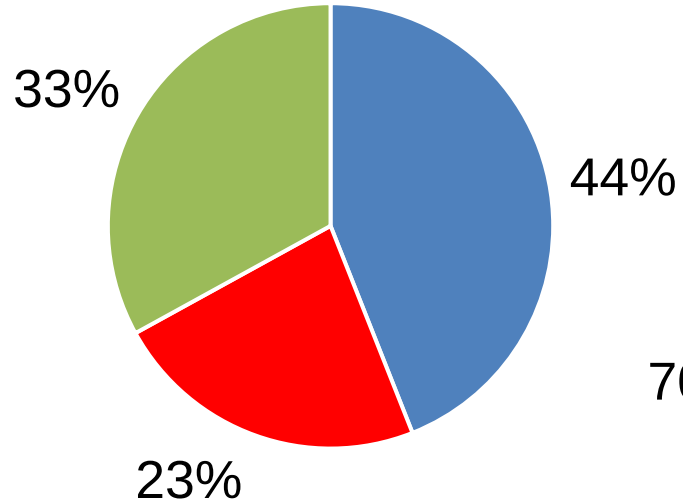
Own stake if 25% owner buys :	40.0
Liquidity and minority discount	-25%
$40 * 0.75 =$	30.0
Own stake (40%+35%) after the deal :	
$100 * 0.75 =$	75.0
Control premium	+13%
$60 * 1.20 =$	85.0
Value of 35% stake	$85.0 - 30.0 =$ 55.0
Premium used	$55 / 35 - 1 =$ 57.1%

Dividing value across stakes

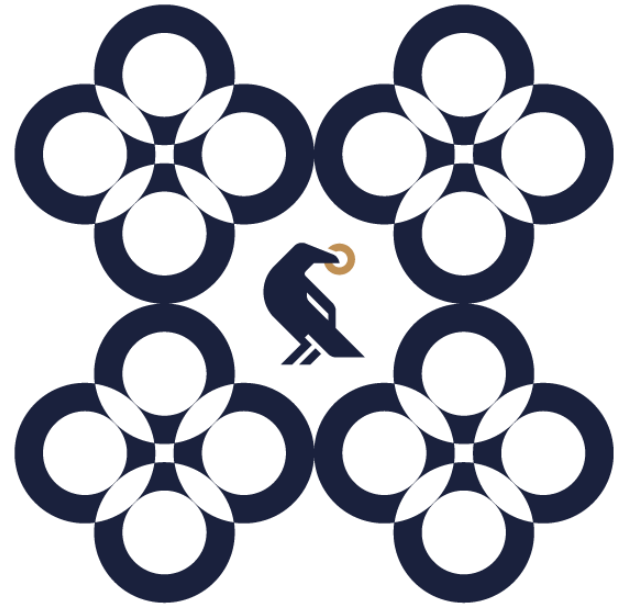
Ownership



Value



**How is it done
in the industry?**



What premium is added to the discount rate to account for the size of the entity being valued?

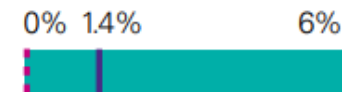
Equity value: <\$50 million



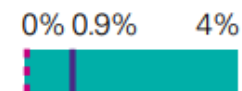
Equity value: \$51 – \$100 million



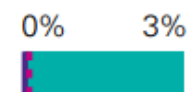
Equity value: \$101 – \$250 million



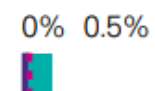
Equity value: \$251 – \$500 million



Equity value: \$501 – \$1,000 million



Equity value: >\$1,001 million

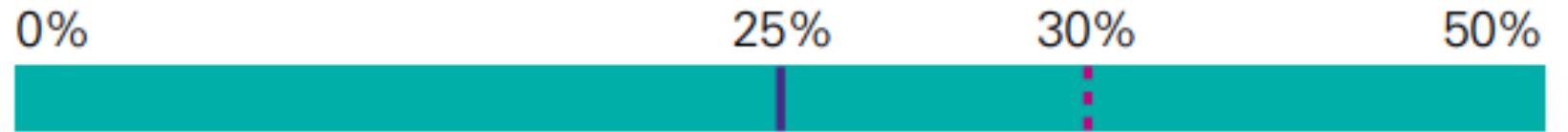


— Median - - - Mode

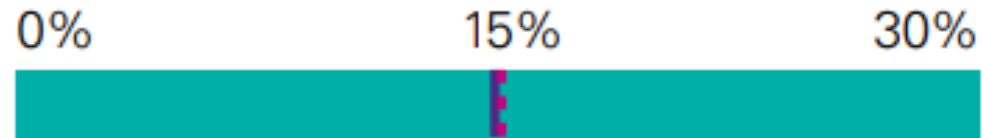
Source: For all it's worth –
KPMG Valuation Practices
Survey 2017 Australia
<https://assets.kpmg/content/dam/kpmg/au/pdf/2017/valuation-practices-survey-2017.pdf>

What discount do you apply in valuing a minority interest?

Size of stake: 1% – 24%



Size of stake: 25% – 49%



Size of stake 50:50 JV

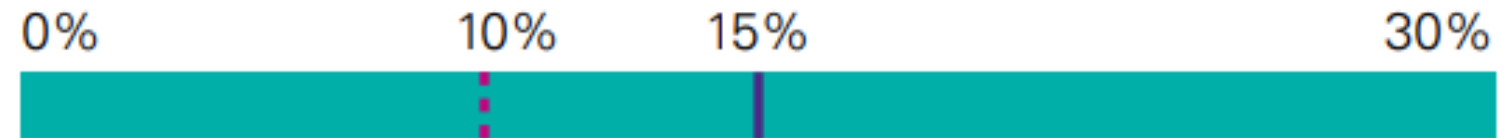


— Median - - - Mode

Source: For all it's worth –
KPMG Valuation
Practices Survey
2017 Australia

What premium do you apply to reflect the level of control offered by the size of the stake being valued?

Size of stake: 51 % – 74 %



Size of stake: 75 % – 99 %



Size of stake: 100 %



— Median - - - Mode

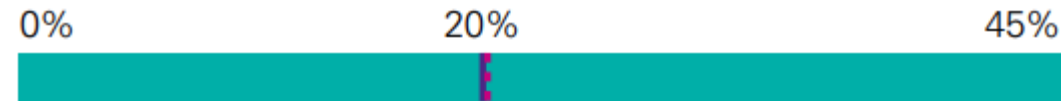
Source: For all it's
worth –
KPMG Valuation
Practices Survey 2017
Australia

When a marketability discount is required, what sized discount do you apply?

Size of stake: 1 % – 24 %



Size of stake: 25 % – 49 %



Size of stake: 50 %



Size of stake: 51 % – 74 %



Size of stake: 75 % – 99 %

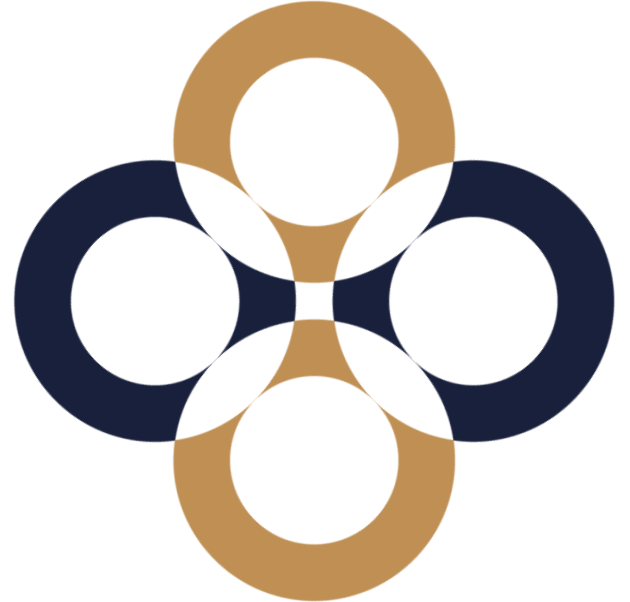


Size of stake: 100 %



Private Equity

Péter Juhász, PhD, CFA



Sources of financing

Internal

Profit
reinvestment

Unneeded
assets

Debt

Bank loan

Bonds

Equity

Current
investors

New owners

What is private equity?

Specialised investor targeting private companies, willing to obtain control over the firms and exit within some time specified in advance.

How is private equity different?

Projects are considered on a portfolio basis not individually.

Successful projects compensate for capital and return lost in failure projects.

Investor provides knowledge next to capital.

The exit is pre-planned.

Benefits

Management
knowhow

Strategic
view

Efficiency
increase

Synergies

M&A
opportunities

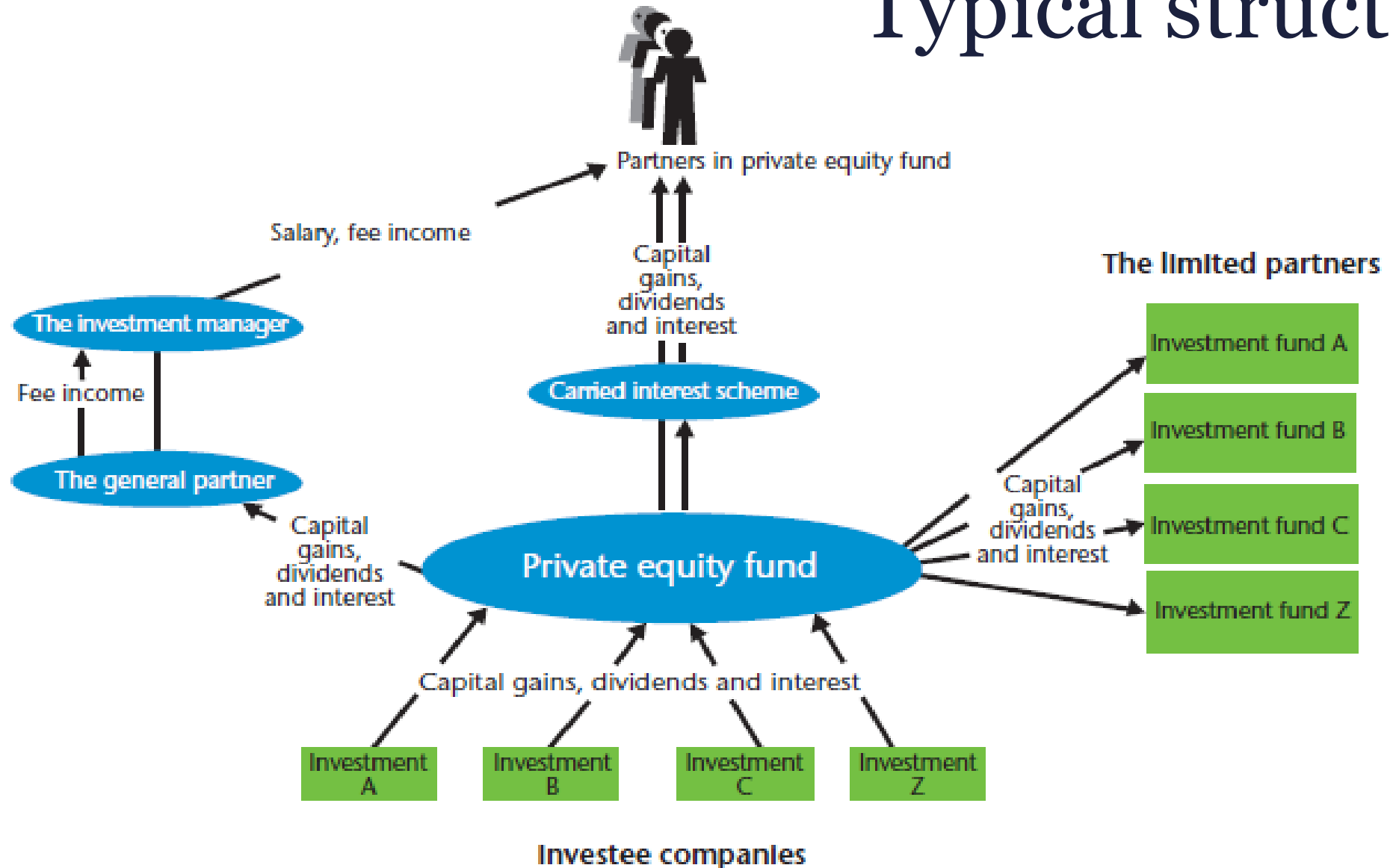
Limit on
agency
problem

Value based
compensation

Where does it come from?



Typical structure



Forms of private equity

Direct

Investor
owns the
shares
directly

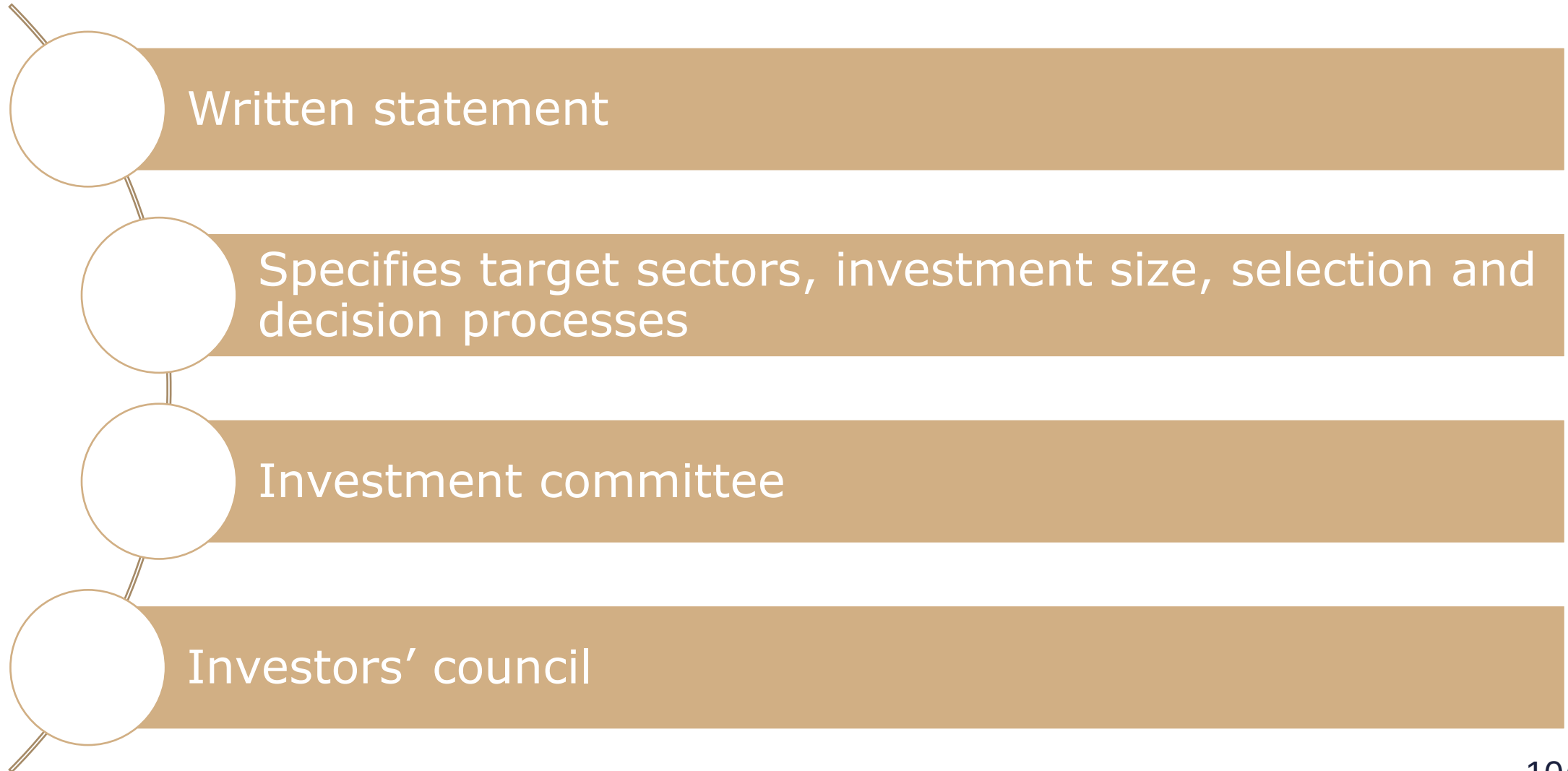
Indirect

Investor and
fund
manager
are different
entities

Typical investors

	Direct	Indirect
Individuals	✓	✓
Companies	✓	✓
Banks		✓
Insurance companies		✓
Investment funds		✓
Pension funds		✓
Governmental institutions		✓

Investment policy



Categorisation

Sector focus

Sector knowledge

High exposure to a specific sector

Duration of investment and cycle of sector to be harmonised

Sectors before expansion or consolidation

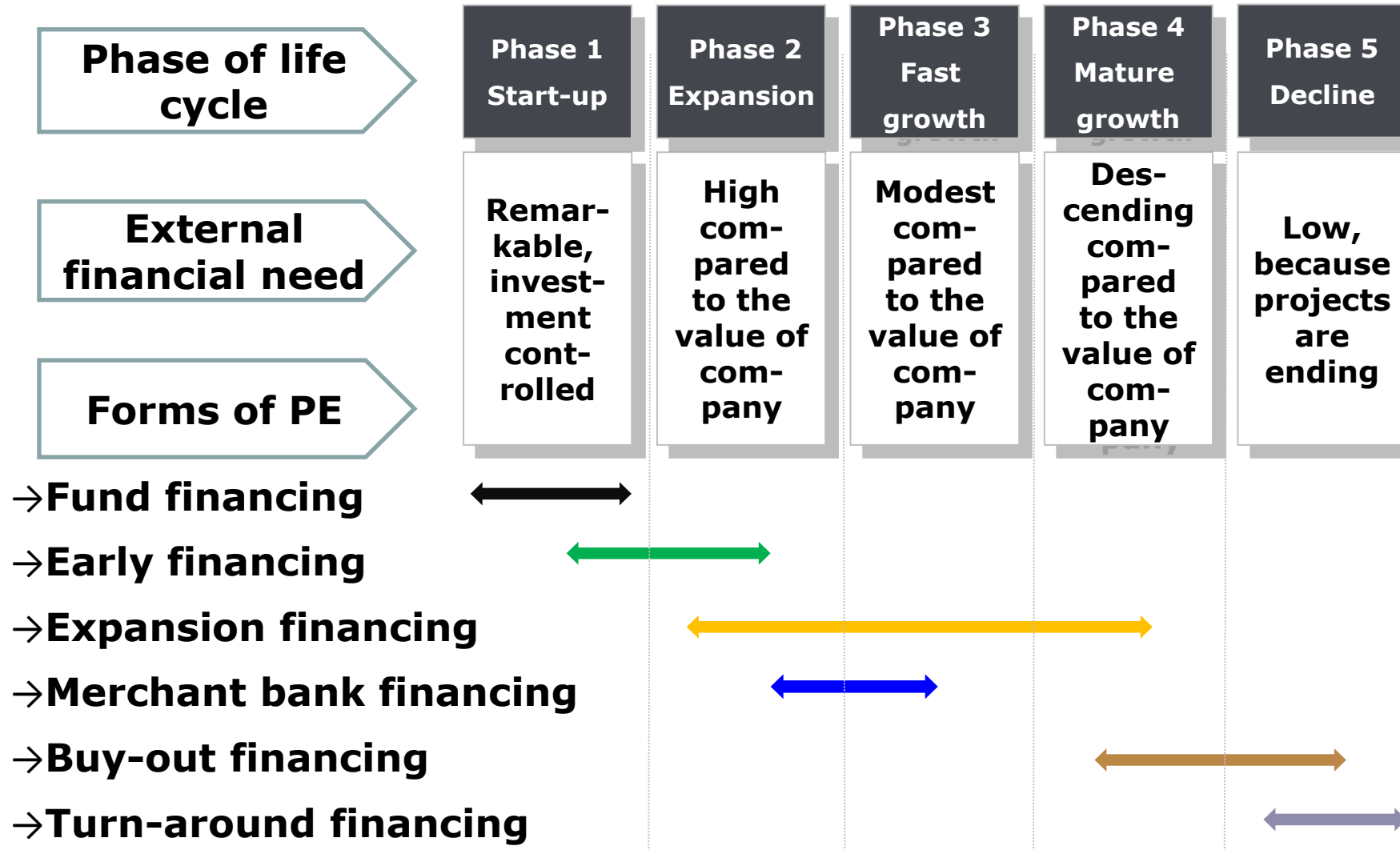
Region focus

Countries in similar economic cycle

Size focus

1/5/20/100/500 M euros

Life cycle focus

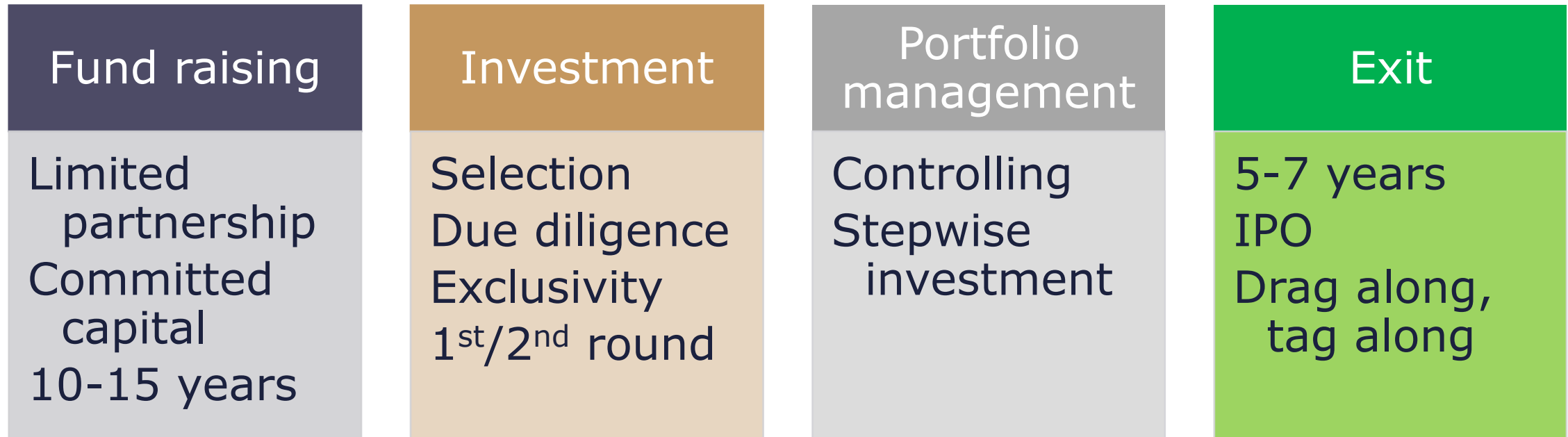


Other categorisations

Leverage

Acquired stake

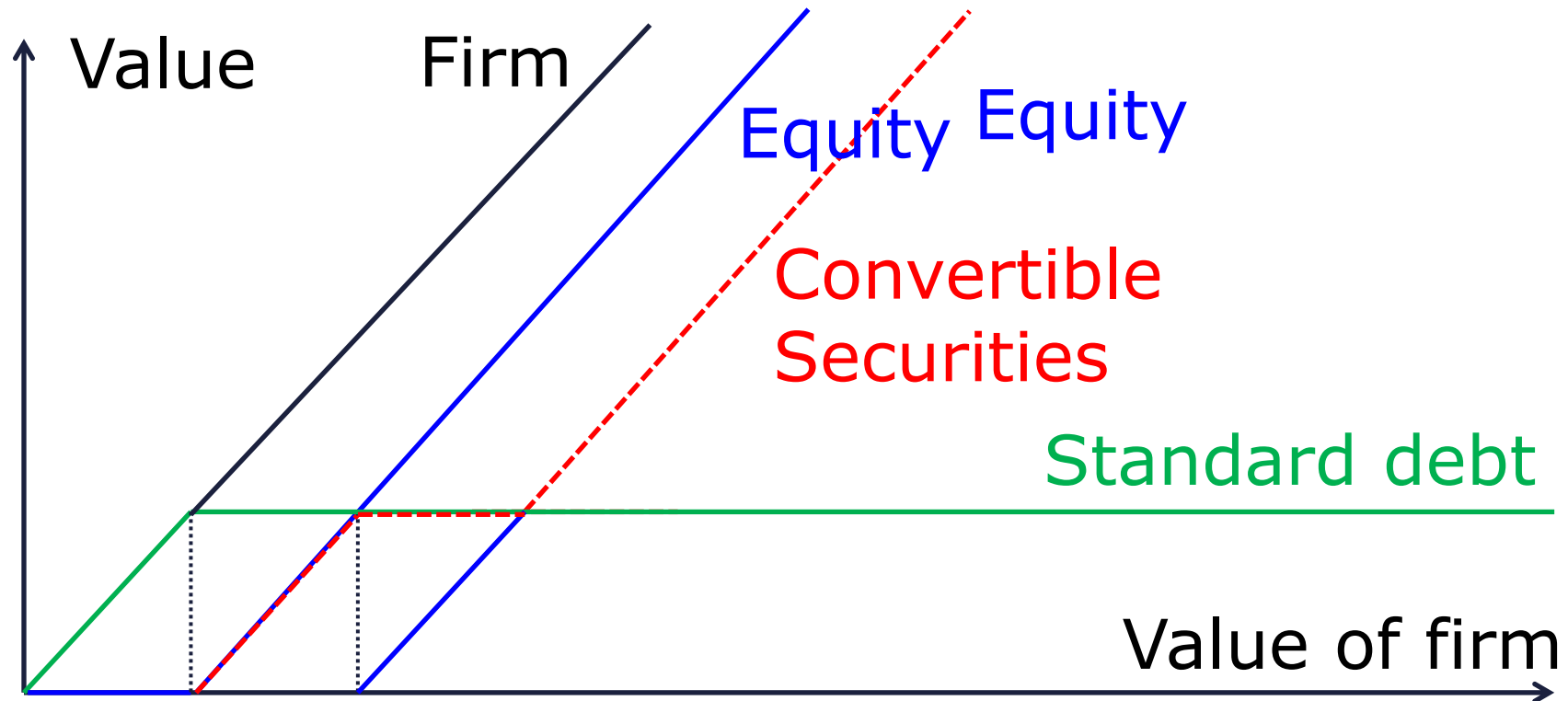
Steps of PE investments



Convertible securities

Convertible loans, bonds + warrants

Convertible preferred shares



Venture capital

Private Equity that invests in start-ups and early stage deals.

Capital for Start-ups

FFF

Family, Friends, and Fools

Business Angels

Wealthy individuals who invest in start-ups and early stage deals.

Equity crowdfunding

A group of individuals who invest in start-ups and early stage deals.

Venture Capital

Private Equity that invest in start-ups and early stage deals.

Types of Venture capital

Early stage

Seed, start-up, first-stage

Expansion

Later stage: Acquisition, Buy-out



Dark side of private equity

Growth pressure

Efficiency pressure

Exit pressure

Ethical issues

Conflicts of interest	Buyout/exit not always in the interest of all owners
Vulnerability of employees	Restructuring may mean lay-offs
Capital markets	Public-to-private deals, Insider info
Tax avoidance	PE firms and management usually very good at tax optimising

Required return for VC

At a VC fund with an average investment horizon of 4 years 70% of similar projects fail. In case of default, 110% of initial investment can be collected at the end. Cost of financing is 20%.

Required return for projects surviving:

$$70\% * 110\% + 30\% * (1+r)^4 = 1.2^4$$

$$30\% * (1+r)^4 = 1.3036$$

$$\mathbf{r = 44.38\%}$$

Drivers of investment success

Better
selection
(specialisation)

Close control

Professional
assistance

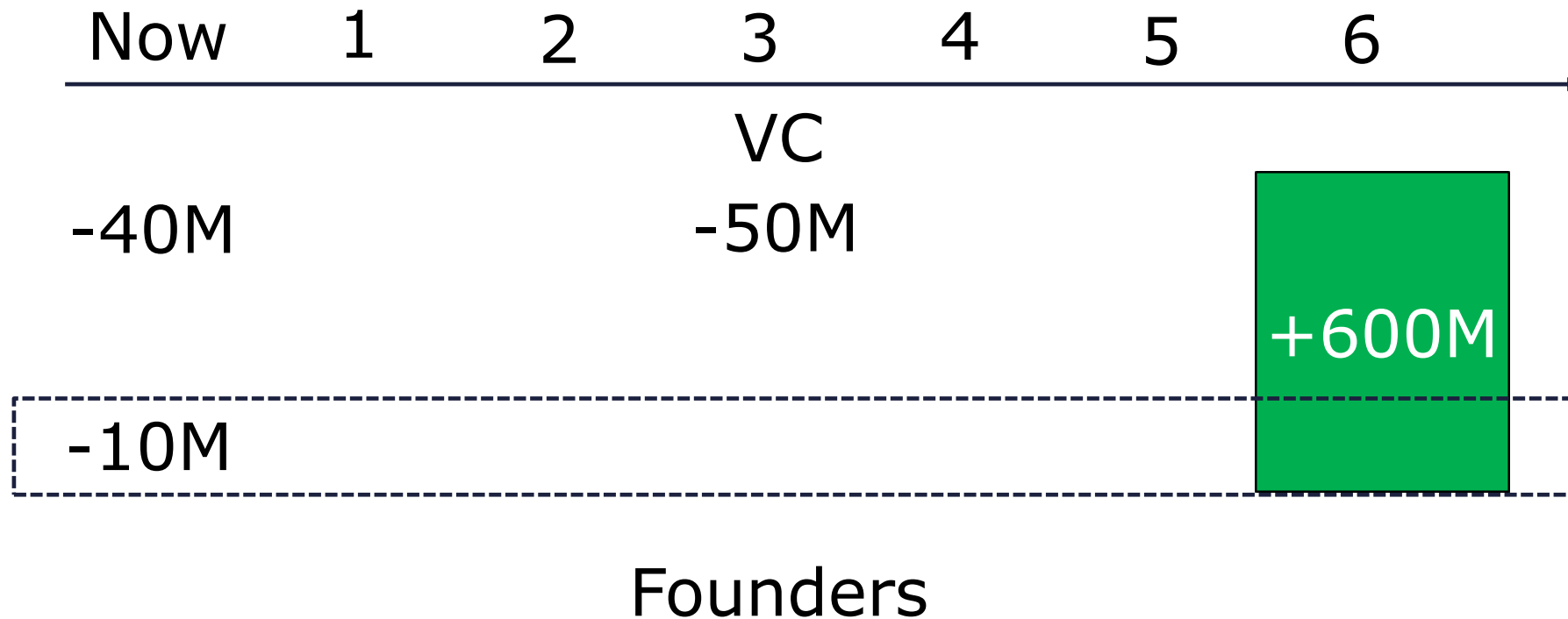
Covenants
and collaterals

Fair price and stake 1.

A firm with a capital of 10 M needs an immediate investment of 40 M. In case of success after 3 years another 50 M will be needed and after 6 years the total of equity may be sold for 600 M.

What stake should VC receive for the equity increases, if required rate of return is 40%?

Investment plan



Fair price and stake 2.

Equity value (E) in 6 years: 600.00 M

In 3 years, after capital: $600.00/1.4^3=218.66$

In 3 years, before capital:

$$218.66-50.00=168.66$$

Stake for investment of 50 M in 3 years:

$$50.00/218.66=22.87\%$$

Calculations

Now

1

2

3

4

5

6

Post-Money 219M

VC -50M

Pre-Money 169M

23%

+600M

VC -40M

Founders -10M

Fair price and stake 3.

In 3 years, before capital: $218.66 - 50.00 = 168.66$

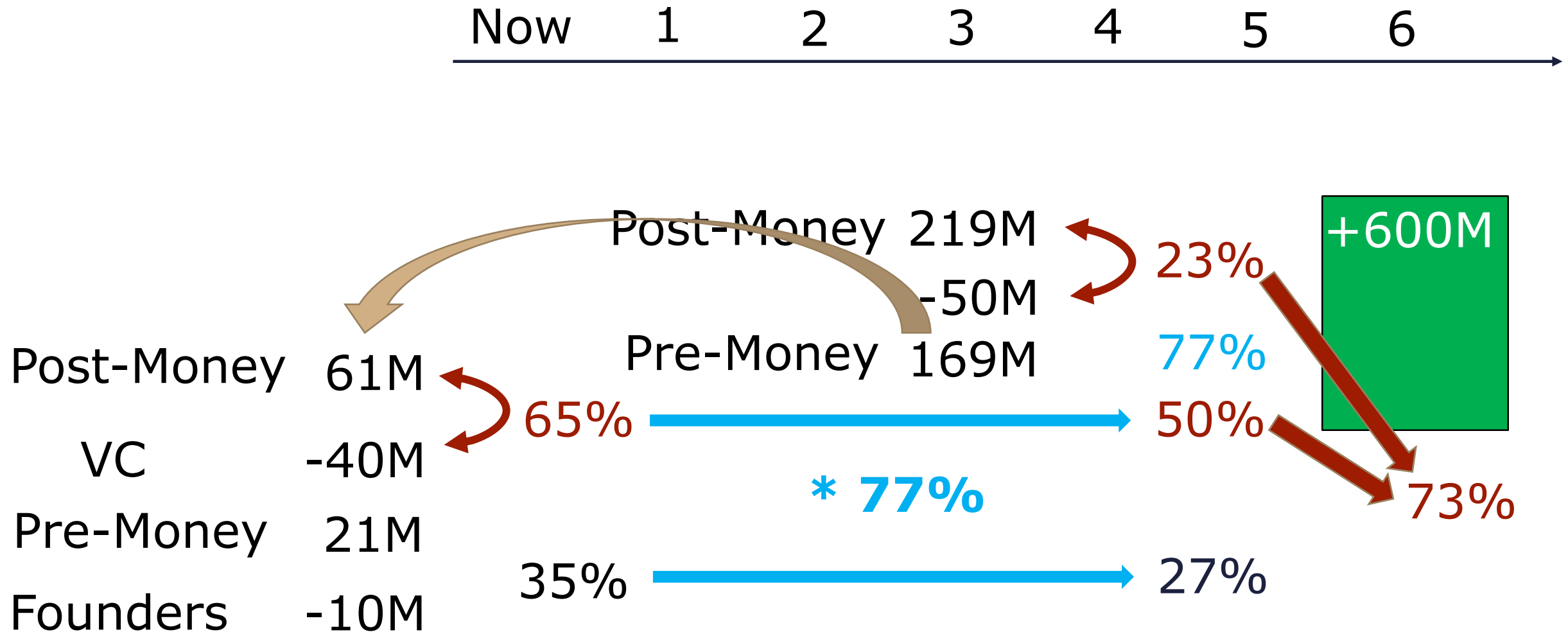
Today, after capital: $168.66 / 1.4^3 = 61.46$

Today, before capital: $61.46 - 40.00 = 21.46$

Stake for current investment:

$$40.00 / 61.46 = 65.08\%$$

Calculations



Fair price and stake 4.

	VC	Founders
Now	0.00%	100.00%
After investment	65.08%	34.92%
In 3 years, after investment	73.07%	26.93%

Stake for capital in year 3: 22.87%

VC total:

$$65.08\% * (1 - 22.87\%) + 22.87\% = 73.07\%$$

Stake of founders:

$$34.92\% * (1 - 22.87\%) = 26.93\%$$

Fair price and stake 5.

Return to founders:

$$600.00 * 26.93\% = 161.58$$

$$161.58 / 21.46 = (1+r)^6$$

$$\mathbf{r = 40.00\%}$$

Measuring value created

EVA:
 $IC \cdot (ROIC - WACC)$

Residual income:
 $PAT - St \cdot r_E$

ΔMVA :
 $(V_1 - IC_1) - (V_0 - IC_0)$

SVA:
 $\Delta E \pm CF$

CFROI:
IRR on FCFE

TSR:
$$\frac{P_1 + Div_1}{P_0}$$

**Balanced
Scorecard**

The secret of successful VBM

Top
management
support

Including
operational
decision

Ability
development

Practice focus

Individual KPIs

Provide data
access

Standardised
format for
reports and
analysis

Bonuses linked
to value creation

Obligatory
value-based
justification of
resource needs

Source:
McKinsey, 1994